

Summary Sheet

Summary

Large national parks often face large protection problems related to understaffing. Etosha National Park (ENP) is a prime example, as it covers 22,935 km² in northern Namibia while only hosting 295 staff ([Republic of Namibia, Ministry of Environment and Tourism, nd](#)). That ratio is about one ranger per 77.7 km², far below a common planning benchmark of 1 ranger per 5 km². In addition, poaching has worsened in recent years, and wildfires and droughts continue to hurt the environment and its animals ([Save the Rhino International, 2023](#); [Associated Press, 2025](#)). Because of these numerous issues, we were tasked with creating a model that shows where protection is needed, gives a plan for how resources should be used under realistic conditions, and explains how those resources should be used when unexpected conditions occur.

Rather than relying on one single model per requirement, our solution is built as a sequence of connected modules. In each stage, the park's data passes through several smaller models, each of which improves one part of the final result before passing it to the next module. This modular pipeline allows the model to handle a more realistic conservation problem than any one method could capture alone.

In the first stage, we evaluated risk across the park by dividing it into cells and using a four-part scoring stack: ecological importance scoring, combined threat scoring, ranger coverage scoring, and priority ranking. Together, these modules measured how important each cell is, how exposed it is to danger, and how well current resources can protect it. A cell only qualifies as protected if it meets our detection-and-response standard, and we then combined all cell results into the spatially defined, park-wide Species Protection Rate (SPR), which measures the share of ecologically weighted land that is truly protected.

Second, we developed a resource-allocation pipeline that combined a Circuit-Theory Infrastructure Model with a Two-Stage Stochastic Program. We first used circuit theory to identify places that are hard for the current protection system to reach and support, which let us look past only where threats were high and also consider where the park is structurally weak. We then used a two-stage allocation model to first choose where to allocate long-term resources under budget and staffing limits and to then decide how to re-allocate patrol, drone, and monitoring effort across numerous disruption scenarios.

In the third stage, we used a forecasting pipeline built from historical monthly reconstruction, spatial persistence testing, time-pattern regression, a Hilbert Space Gaussian Process, and an Ensemble Kalman Filter. These modules were used so that the model could reflect repeating seasonal patterns and future uncertainty instead of assuming that the park will stay static. This allowed us to project SPR forward for five years under many scenarios and estimate how many rangers would be needed to keep protection above target.

When applied to Etosha, our model found that the park operates near its protection target under baseline conditions but falls meaningfully short under severe drought, which concentrates both animals and threats while degrading ranger mobility. The western interior corridor near Galton Gate emerged as the highest-priority weak region in both wet and dry seasons, so our resource-allocation model concentrated new infrastructure there. Sensitivity analysis showed that capital budget is the strongest driver of results, followed by patrol mobility and staffing, while smaller tactical parameters such as shift length have much less effect.

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Letter to the Decision Makers

Dear Members of the International Mission for Monitoring Conservation,

Etosha National Park in Namibia is one of Africa's most important nature reserves. It covers nearly 23,000 km² and is home to countless mammals and birds, some of which are endangered. However, the park is in trouble. Poachers hunt illegally inside it, wildfires burn large areas, and droughts dry up the water that animals need to survive (SOURCE, SOURCE, SOURCE). On top of all that, only 295 workers are available to guard this enormous area. Our team built a math-based system to determine exactly where protection is needed most, what resources to send, and whether those choices will still work five years from now.

How Our System Works

We split the problem into three layers, similar to 3 assembly lines in a factory, that work together to ensure an efficient process from start to end.

Layer 1: Diagnosis. We divided a map of the entire park into smaller 120.8 km² rectangular zones to evaluate if each area was considered “protected” or not. For every square, we measured four kinds of danger: poaching, fire, animals crowding around scarce water, and land wearing out over time. Then we checked how many of each type of animal, such as predators or herbivores, live in each square and how well rangers can currently watch over it. Historical incident pressure & Park incident logs / model-ready covariates & Monthly panel & Spatial kernel smoothing and normalization for risk channel input. A square only counts as “protected” if rangers can actually spot a threat and get there fast enough to stop it. Just being inside the park's fence isn't enough.

Layer 2: Deployment. Once we know where the trouble is, we need to figure out what to do. We used a concept called circuit theory to picture the park like an electrical circuit: ranger stations are the batteries, roads are the wires, and rough terrain acts like resistance that slows the flow. Places where the “current” of protection barely reaches are what we call “dark zones.” Our model then picks the best combination of new guard posts, drones, and cameras the park can afford, and checks whether that plan still works if something goes wrong, such as a wildfire blocking a road or some equipment breaking down.

Layer 3: Forecasting. Using six years of weather and nature data, we ran a simulation of the next five years. We created a single score called the Species Protection Rate, or SPR, that goes from 0, where nothing is protected, to 1, where every important part of the park is fully safe. Our model found that under normal weather, the SPR should climb to about 0.91 in five years. But if a bad drought hits, it will drop to about 0.82, which tells managers they'd need more people or need to move their equipment to newly optimized locations.

What We Found

1. Guard the edges first. The southern and western edges of the park are the least protected because poachers can reach them easily, but ranger patrols struggle to cover them. Directing resources there will make the biggest improvement in protection per dollar spent.
2. Move teams with the seasons. In the dry months, animals gather around waterholes, which makes them easy targets for poachers. In the wet months, tall grass dries out and causes more fires. A plan that keeps rangers in the same spots year-round will always leave the park under-protected in one season. Our model adjusts patrol and drone effort based on the time of year.
3. Fix access before buying technology. Some areas are at a high risk just because there's no road to get there. We found that building one small outpost or upgrading one road can improve protection more than placing expensive surveillance equipment in areas that are already accessible.
4. Drought Awareness. Droughts are the single biggest long-term danger to the nature conservation area. Droughts force animals into small areas, kill off plant cover, and make it harder for rangers to move around. We recommend keeping a flexible reserve of extra staff and setting up water-point monitoring in advance.
5. The framework works at other parks. We adapted our three-step approach to Bandhavgarh National Park

in India (a similar reserve) and Te Wahipounamu in New Zealand (a very different landscape). The 3 layer structure stays the same and just has to be adjusted with local animal data, threats, and maps.

Final Management Takeaway

Most conservation plans stop after making a map of where the problems are. Ours goes further, detailing what to buy, where to put it, and whether the plan can survive realistic budgets and worst-case scenarios. We start by finding the most at-risk and hardest-to-reach zones, simulate how the plan is executed in different scenarios, and then decide how many rangers are needed to keep the park protected.

With our system and your support, we can help protect Etosha National Park and its over 700 species, along with countless other reserves around the world.

Sincerely,
Team #17289

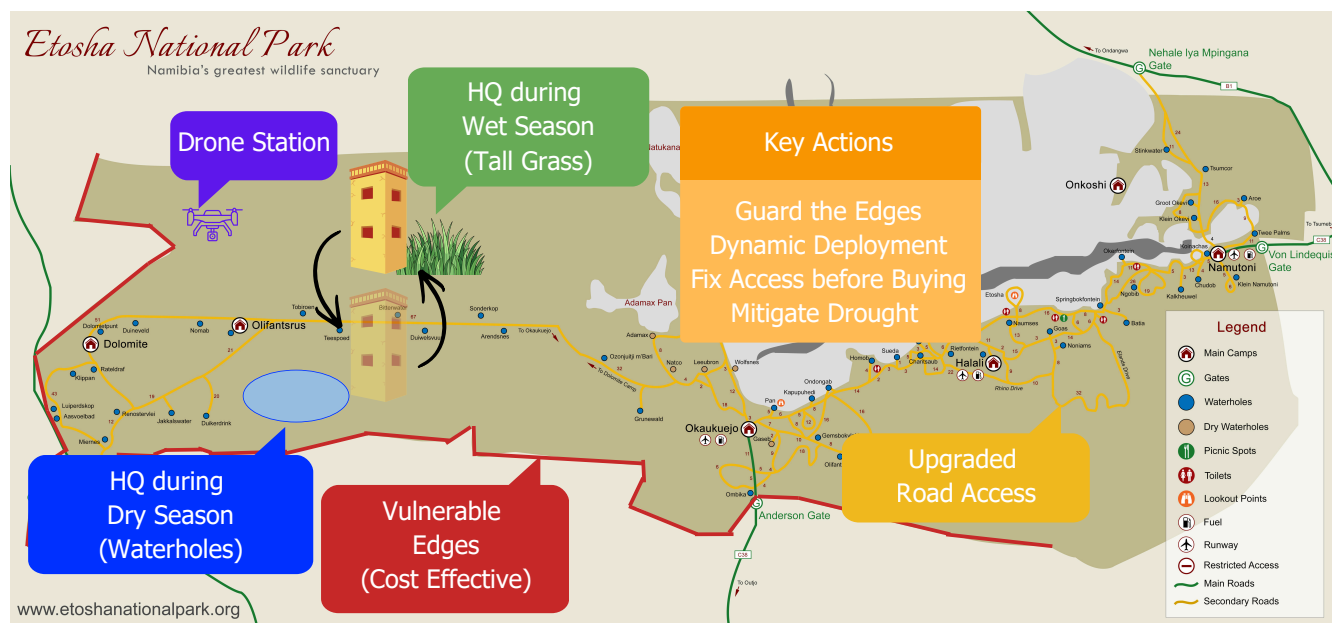


Figure L1. Executive summary map of Etosha National Park overlaid with the four key management recommendations from the three-layer framework. Blue box (west): the Dolomite sector is identified as the highest-priority intervention zone due to structural isolation from ranger bases. Orange box (center-north): seasonal HQ redeployment — waterhole-adjacent in dry season, tall-grass-edge in wet season — addresses the shifting risk geography. Yellow box (center-south): vulnerable park edges are the most cost-effective targets for camera and sensor placement. Red box (south): road access upgrades unlock patrol coverage in the Anderson Gate corridor. Map base from <https://www.etoshanationalpark.org/>.

1 Introduction and Full-Model Overview

1.1 Background

Etosha National Park (ENP) spans about 23,000 km² across savanna, salt-pan systems, and mopane woodland in northern Namibia. The park supports 114 mammal species and one of the world's largest free-ranging black rhinoceros populations (*Encyclopaedia Britannica*, 2019; Mendelsohn et al., 2010; Republic of Namibia, Ministry of Environment and Tourism, nd), so it is a core asset in Namibia's biodiversity strategy.

Protection pressure is increasing. Reported rhino poaching losses in Namibia rose sharply in recent years, with major incidents linked to Etosha's protection system (*Save the Rhino International*, 2023; *Africa Geographic*, 2023). In September 2025, a wildfire that started outside ENP burned through a large portion of the park (*Associated Press*, 2025). Seasonal drought then concentrates wildlife around fewer reliable water points, which raises local vulnerability.

Operational capacity remains limited. ENP has 295 personnel over 22,935 km², which is about 77.7 km² per ranger ($22,935/295 \approx 77.7$), far below widely discussed protected-area staffing benchmarks (*Appleton et al.*, 2022). This gap motivates a quantitative framework that links ecological priority, deployment feasibility, and time-varying risk. Because losses are irreversible, "mostly covered" is not enough. Management needs a system that explains where failures occur, why they occur, and which interventions reduce risk fastest.

1.2 Problem Restatement

The IMMC task asks for a mathematical framework that improves wildlife protection decisions in large national parks, with Etosha National Park (ENP) as the primary case.

- **Identify and rank threats.** The model must identify the dominant threats in ENP and rank them across species groups, habitat types, geography, and more.
- **Define protection quantitatively.** The framework must give an explicit mathematical definition of *protection* so outcomes are measurable and comparable.
- **Allocate scarce resources under constraints.** The system must allocate personnel, routes, and monitoring technology under budget, terrain, and operational limits.
- **Evaluate performance through time.** The model must measure conservation effectiveness dynamically and estimate the staffing needed to sustain a target protection level over a period of time.

Our restatement keeps the structure intact and makes each requirement explicit in the three-layer pipeline: Layer I diagnoses failures, Layer II describes deployments, and Layer III forecasts whether those deployments are enough under future stress.

1.3 Core Idea

We solve the planning problem with a three-layer decision pipeline (*International Mathematical Modeling Challenge*, 2026). Layer I diagnoses spatial protection deficits. Layer II converts those deficits into budget-feasible deployment decisions. Layer III forecasts protection trajectories and staffing requirements over time.

The three-layer design explicitly separates diagnostic inference (what is happening and why) from prescriptive optimization (what to do) from probabilistic forecasting (what will happen). This separation is crucial in conservation planning because by passing compressed, validated outputs between layers rather than raw uncertainty fields, each layer can be stress-tested independently.

The pipeline separates two metrics by purpose. SPR (the Species Protection Rate) is the weighted fraction of ecologically valuable park area where threat detection and response capacity jointly meet minimum thresholds; a value of 1.0 means full coverage, while 0.5 means roughly 50% of weighted area is not adequately protected.

II measures operational deployment effectiveness under scenario-specific constraints.

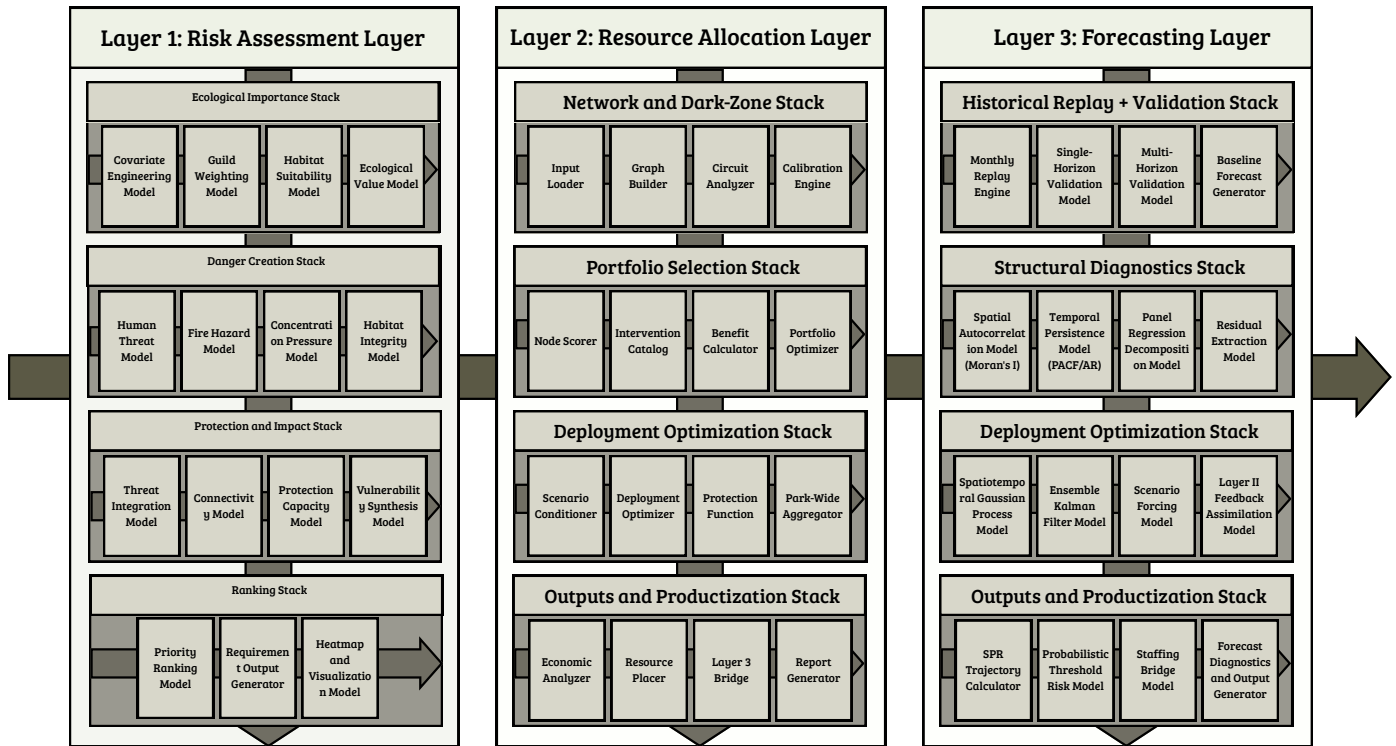


Figure 1: Three-layer pipeline showing the main stages of the model and their substacks.

2 Global Assumptions and Variables

2.1 Global Variables Used in All Three Layers

Table 1: Core variables shared across all three layers, grouped by role to keep notation consistent from diagnosis through allocation and forecasting.

Symbol	Meaning
i, z	Cell or zone index used across all layers.
$g \in G$	Conservation guild index and set.
t	Time index (seasonal in Layer I–II; monthly in Layer III).
ξ	Scenario index for uncertainty analysis.
$E(g, i)$	Guild-specific ecological value.
$T(g, i)$	Guild-specific integrated threat.
$P_{\text{cap}}(g, i)$	Guild-cell protection capacity.
SPR, SPR_t	Species Protection Rate (static or time-indexed).
U_z, S_z	Layer II under-service and deployment priority scores.
$\Pi(y, \xi)$	Layer II operational deployment score under scenario ξ .
W / W_{std}	Spatial relationship structure used in allocation and forecasting.

Global Assumptions

Table 2: Global assumptions (G1–G4) shared across all three layers.

Assumption	Justification	Modeling Role
G1. Planning units are fixed spatial cells on a monthly timestep.	Protection deployment decisions are made at area and season/month scales rather than at point-level real-time control.	Establishes a common spatial-temporal backbone so Layer I outputs can pass directly into Layer II and Layer III.
G2. Multi-source covariates are harmonized to a shared resolution and are jointly comparable after preprocessing.	Environmental, infrastructure, and patrol covariates originate from different products and must be normalized before fusion.	Enables consistent cross-layer feature transfer without re-estimating variable definitions in each layer.
G3. Guild aggregation with fixed weights preserves decision-relevant ecological asymmetry.	Cell-level species abundance is incomplete; guild-level structure retains conservation priority while improving stability.	Keeps SPR and priority scoring interpretable and comparable across diagnosis, optimization, and forecasting outputs.
G4. Management actions tune detection and response, else the system stays stable.	Forecasting requires continuity with recent history, while scenario analysis represents structural departures.	Supports coherent handoff from static diagnosis to scenario recourse and five-year trajectory forecasting.

Full parameter values appear in Appendix Table 31; all modeling choices above are stress-tested in §6.

3 Layer I — Spatial Risk Diagnosis

3.1 Layer I Core

Layer I provides the spatial diagnosis for the full framework. It identifies where protection fails, explains why those failures occur, and ranks where intervention should begin.

Protection performance is measured by Species Protection Rate (SPR), defined in Eq. (6) as the weighted share of ecological value that satisfies minimum protection thresholds.

Layer I shows a stable threat order in failing cells. The dominant driver is **human access pressure**. The secondary drivers are **water-concentration risk** in dry season, **fire pressure** during acute seasonal shocks, and **habitat degradation** as chronic background load.

The protection rule is guild-specific and spatially explicit. Thresholds τ_g^* vary by guild, and SPR is reported both park-wide and by zone. As documented in Table 22 (§3.4), zone-level outcomes are highly asymmetric despite similar aggregate park-wide SPR. The strict all-guild rule (Eq. 39) is deliberately conservative. A cell that passes four guild thresholds but fails for megafauna is counted as unprotected at the park level.

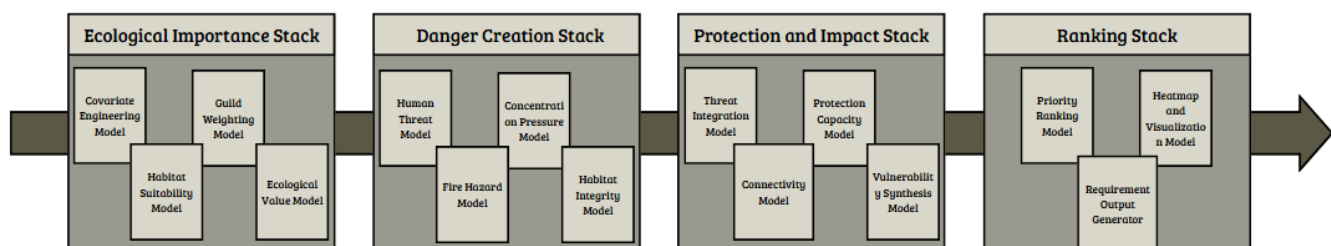


Figure 2: Layer 1 flowchart showing relationship between stacks and submodels.

3.2 Variables, Assumptions, and Justifications

Core Assumptions

Layer I assumptions define how ecological value is scored, how threats are fused, and when a cell is classified as protected.

Table 3: Layer I structural assumptions (L1-1 through L1-4).

Assumption	Justification	Modeling Role
L1-1. Ecological value is represented by a bounded multiplicative form combining habitat suitability, vegetation productivity, water access, and habitat intactness.	Critical ecological factors are complementary: one weak factor should not be fully offset by another strong factor.	Produces interpretable guild-cell ecological value $E(g, i)$ for residual vulnerability and SPR weighting.
L1-2. Threat channels are fused with a Noisy-OR operator under conditional-independence approximation.	Poaching access, fire hazard, water concentration, and degradation can each independently trigger failure risk.	Returns bounded integrated threat $T(g, i)$ while preserving multi-channel escalation behavior.
L1-3. Protection capacity is the joint effect of detection and timely response, with response midpoint calibrated at $\tau^* = 2.0$ h.	Observed anti-poaching outcomes are most sensitive to whether patrol units can intercept incidents before completion.	Links infrastructure and mobility constraints directly to protection probability $P_{\text{cap}}(g, i)$.
L1-4. A cell is classified as protected only when all guild thresholds are satisfied.	High-consequence guild failures must not be hidden by averaging across lower-risk guilds.	Imposes conservative SPR classification aligned with ENP's flagship-species mandate.

Key Variables and Calibration Assumptions

Table 4: Core Layer I variables by stage of the spatial risk pipeline (ecological value, threat fusion, protection capacity, and priority/SPR outputs). Variables are bounded to $[0, 1]$ unless noted.

Variable	Type	Range	Role in Model
$E(g, i)$	Continuous	$[0, 1]$	Ecological value; enters the SPR numerator with guild weighting.
$T(g, i)$	Continuous	$[0, 1]$	Integrated threat from Noisy-OR fusion of four threat channels.
$P_{\text{cap}}(g, i)$	Continuous	$[0, 1]$	Joint detection-response capacity in each guild-cell pair.
$V(g, i)$	Continuous	$[0, 1]$	Residual vulnerability: $E \cdot T \cdot (1 - P_{\text{cap}})$.
$\text{priority_score}(i)$	Continuous	$[0, 0.09]$	Guild-weighted intervention urgency ranking.
I_i^{prot}	Binary	$\{0, 1\}$	Equals 1 only when all guild thresholds are satisfied.
SPR	Continuous	$[0, 1]$	Park-level weighted protected fraction.

Variable	Type	Range	Role in Model
$O(g, i)$	Continuous	$[0, 1]$	Habitat suitability from blended covariate/occurrence information.
$H(g, i), F(g, i), W(g, i), D(g, i)$	Continuous	$[0, 1]$	Human, fire, water, and degradation threat channels.
$p^{\text{det}}(g, i)$	Continuous	$[0, 1]$	Detection probability from patrol and technology effort.
$p^{\text{resp}}(i)$	Continuous	$[0, 1]$	Response probability from travel-time logistic mapping.
$\tau(i)$	Continuous	Hours	Travel time from nearest base to cell i .
$C(i)$	Continuous	$[0, 1]$	Circuit-theory connectivity modifier used in capability scoring.

We map travel time to response probability using a logistic midpoint $\tau^* = 2.0$ h (Appendix Eq. A.10). The 2-hour response threshold ($\tau^* = 2.0$ h) is the central parameter in the response-capacity function. At ENP's primary gravel-road patrol speed of 40 km/h, this threshold implies an approximately 80 km effective response radius from each ranger base. The calibration is consistent with documented anti-poaching response requirements in large southern African reserves (Critchlow et al., 2017; Hilborn et al., 2006). Sensitivity analysis in §6 shows that a 40% reduction in patrol speed (effectively tightening the response window) produces a -18.66% change in Π , confirming this parameter as one of the model's primary drivers.

Guild weights are $\omega_g = (0.35, 0.25, 0.20, 0.10, 0.10)$ for (megafauna, predators, herbivores, birds, habitat fauna). Megafauna receives the largest weight (0.35) for three reasons: ENP's black rhinoceros (*Diceros bicornis*) is Critically Endangered and Etosha holds an estimated 72% of Namibia's population (Africa Travel & Tourism Association, 2024; Save the Rhino International, 2023); large herbivores are keystone species whose loss can disrupt nutrient cycling, vegetation, and trophic structure (Estes et al., 2011; Ripple et al., 2016); and megafauna drive tourism revenue that supports conservation finance (Lindsey et al., 2017). Predators receive the second-highest weight (0.25) because large carnivores — especially lion (*Panthera leo*) in ENP — regulate prey systems through trophic cascades, and their decline is strongly linked to ecosystem instability (Ripple et al., 2014). In ENP, predator pressure is also tightly coupled with prey concentration around waterholes. Herbivores (0.20) are weighted next as the core prey base and seed-dispersal group. Birds (0.10) and habitat fauna (0.10) keep lower but nonzero weights: they remain ecologically important, but are generally more abundant, more dispersive, and under lower direct poaching pressure (McGowan et al., 2020).

This hierarchy is consistent with conservation-prioritization evidence showing that flagship-focused strategies can retain about 79–89% of broader biodiversity representation while preserving financing and public-engagement leverage (Lindsey et al., 2017; McGowan et al., 2020).

Protection thresholds are $\tau_g^* = (0.75, 0.65, 0.55, 0.55, 0.60)$. Higher thresholds for high-consequence guilds prevent a cell from being labeled protected when critical-guild response reliability is still weak. Thresholds were calibrated so that baseline dry-season SPR falls in the 0.85–0.90 range, consistent with the park's operational protection target. A cell with megafauna threshold 0.75 requires substantial simultaneous detection and response probability before counting as protected. Uniform downward perturbation to $\tau_g^* = 0.50$ across all guilds raises dry-season SPR to approximately 0.94; uniform upward perturbation to $\tau_g^* = 0.80$ lowers it to approximately 0.82. The chosen values sit near the center of this range while respecting the asymmetric consequence of megafauna protection failure.

Full parameter values appear in Appendix Table 31; all modeling choices above are stress-tested in §6.

3.3 Architecture and Submodel Logic

How the Layer Works: Four Steps

Step 1: Ecological Value. We compute ecological value for guild g in cell i with a multiplicative form:

$$E(g, i) = S_g \cdot O(g, i) \cdot X_{\text{ndvi}}(i)^{\alpha_g} X_{\text{water}}(i)^{\beta_g} X_{\text{intact}}(i)^{\gamma_g} \quad (1)$$

Here, S_g is guild urgency and $O(g, i)$ is habitat suitability. The three X terms are normalized support variables. Exponents satisfy $\alpha_g + \beta_g + \gamma_g = 1$ (Appendix B; Table 12). We use this multiplicative ecological form to encode factor complimentary when weakness in one habitat input cannot be fully offset by another (Dasgupta, 2021).

Step 2: Threat Integration. We estimate four threat channels per guild and cell: human access $H(g, i)$, fire hazard $F(g, i)$, water-concentration risk $W(g, i)$, and habitat degradation $D(g, i)$. We fuse them with a Noisy-OR operator:

$$T(g, i) = 1 - \left[(1 - H(g, i))(1 - F(g, i))(1 - W(g, i))(1 - D(g, i)) \right] \quad (2)$$

This expression returns the probability that at least one threat channel is active while preserving bounded outputs (Appendix B). The Noisy-OR formulation assumes conditional independence between threat channels given latent risk state, which is a standard practical approximation in probabilistic reasoning (Pearl, 1988).

Step 3: Protection Capacity. Protection capacity is the joint effect of detection and response:

$$P_{\text{cap}}(g, i) = 1 - (1 - p^{\text{det}}(g, i))(1 - p^{\text{resp}}(i)) \quad (3)$$

where $p^{\text{det}}(g, i)$ is detection probability and $p^{\text{resp}}(i)$ is timely response probability (Appendix B and Appendix B).

Step 4: Priority and Protection Classification. Residual vulnerability is

$$V(g, i) = E(g, i) \times T(g, i) \times (1 - P_{\text{cap}}(g, i)) \quad (4)$$

and weighted intervention priority is

$$\text{priority_score}(i) = \sum_g \omega_g \cdot V_{\text{adj}}(g, i) \quad (5)$$

with guild weights $\omega_g = (0.35, 0.25, 0.20, 0.10, 0.10)$.

Here $V_{\text{adj}}(g, i) = V(g, i)(1 - \delta_{\text{conn},g}C(i))$ is connectivity-adjusted vulnerability, which discounts vulnerability by the cell's circuit-theory connectivity score $C(i)$ scaled by guild dispersal capacity $\delta_{\text{conn},g}$, reflecting that well-connected cells have lower effective intervention urgency because wildlife can relocate.

A cell is classified as protected only if all guild thresholds are met. This all-guild rule prevents high-risk failures from being hidden by averaging. Thresholds are guild-specific: $\tau_g^* = (0.75, 0.65, 0.55, 0.55, 0.60)$ for (megafauna, predators, herbivores, birds, habitat fauna).

Finally, park-level SPR is

$$\text{SPR} = \frac{\sum_i I_i^{\text{prot}} \times E_{\text{weighted}}(i)}{\sum_i E_{\text{weighted}}(i)} \quad (6)$$

where $E_{\text{weighted}}(i) = \sum_g \omega_g E(g, i)$.

3.4 Results and Interpretation

What We Found

Layer I indicates moderate park-wide coverage with concentrated failure zones. Dry-season static SPR is 0.890, while wet-season static SPR is 0.856. Of 158 in-park cells, 143 satisfy all-guild thresholds in dry season and 137 in wet season. The failing 15–21 cells cluster in the western interior corridor near Galton Gate (Table 20). This is a localized and persistent failure cluster, not a park-wide collapse. Threat rankings differ by aggregation level. Across all cells, water-concentration risk dominates dry-season means, while habitat degradation dominates wet-season means (Figure 12). In failure cells, however, chronic human-access pressure is the dominant driver in both seasons (60% of failures), followed by habitat degradation (Table 23).

Zone-level results are asymmetric: salt-pan zones achieve perfect SPR in both seasons because low wildlife density keeps residual vulnerability below threshold regardless of moderate protection capacity. Boundary cells ($n = 2$) also achieve perfect SPR due to adequate road access and low ecological value. The priority distribution is strongly right-skewed: 66% of cells fall below priority score 0.005, while the top five western-interior cells score 0.051–0.071 — roughly 7–10 times the park median (Table 21). Targeting fewer than 10 cells (approximately 6% of the park) would address a large share of total protection deficit.

Priority score and challenge score are different outputs. Priority score is the main ranking metric used in Section 3. Challenge score is a separate diagnostic reported alongside the top-priority cells in Table 21 and should not be read on the same numeric scale as priority score.

Layer I answers *where* protection fails and *why*, but it cannot prescribe what to buy or where to deploy it under budget constraints. Layer II takes the priority rankings and structural isolation scores produced here and converts them into an implementable action steps.

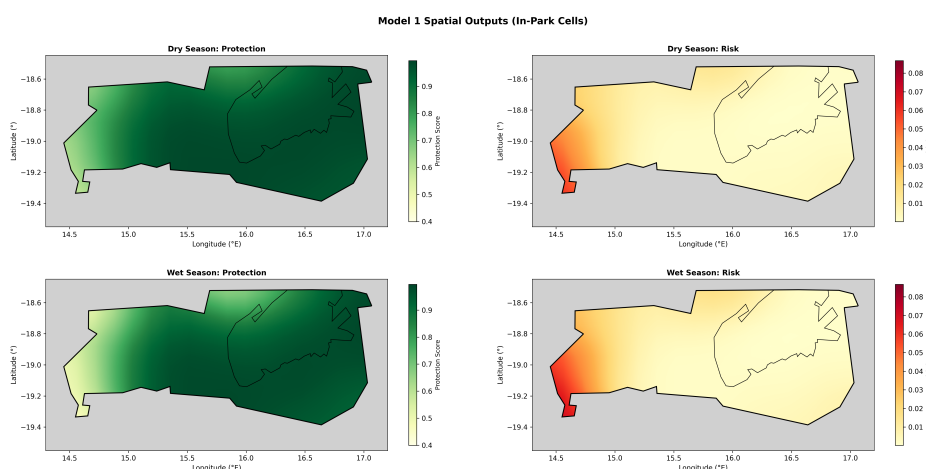


Figure 3: Seasonal spatial outputs from Layer I across the 158 in-park cells comparing wet and dry season through the metrics of protection and risk score.

4 Layer II — Resource Allocation and Protection Design

4.1 Layer-II Core Idea

Layer II converts Layer I diagnosis into implementable action steps by distinguishes **existing resources** (current ranger force and installed infrastructure) from **new additions** selected by optimization under budget constraints. SPR and Π measure different quantities. SPR (Layer I) is an ecological coverage metric: the weighted share of habitat that meets minimum protection thresholds. Π (Layer II) is an operational score: how

effectively a portfolio delivers weighted zone protection under scenario-specific implementation constraints. The numerical gap between these metrics is expected. Dry-season $SPR = 0.890$ means about 89% of ecologically weighted area meets minimum thresholds. $\Pi = 0.310$ measures something different: operational performance in the highest-priority under-served zones, benchmarked against the aspirational $\Pi = 0.70$ level. In short, ENP can be near target overall (high SPR) while still having weak coverage in its hardest-to-protect zones (low Π).

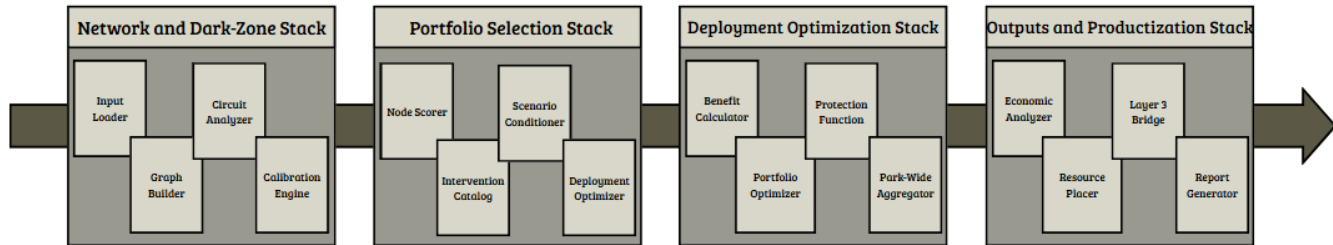


Figure 4: Layer 2 flowchart showing the relationship between stacks and submodels.

4.2 Variables, Assumptions, and Justifications

Layer II translates Layer I diagnosis into constrained deployment decisions. Table 5 lists the core decision, state, and score variables used by the optimization pipeline.

Table 5: Core Layer II variables and their role in the optimization pipeline.

Variable	Type	Range	Role in Model
U_z	Continuous	$[0, 1]$	Under-service score from effective resistance diagnostics.
S_z	Continuous	≥ 0	Zone priority: $(1 - P1_z)(1 + \kappa U_z)$.
y_i	Binary	$\{0, 1\}$	Stage-1 portfolio decision for intervention i .
B_i	Continuous	≥ 0	Strategic benefit of intervention i over affected zones.
$\tilde{B}_i$	Continuous	≥ 0	Lag-discounted intervention benefit, $\tilde{B}_i = \delta_i B_i$.
$\Pi(y, \xi)$	Continuous	$[0, 1]$	Park-wide operational protection score in scenario ξ .
$P_{z,\xi,t}$	Continuous	$[0, 1]$	Zone-time protection from the recourse protection function.
R_z^{eff}	Continuous	≥ 0	Effective resistance from zone z to ranger bases.
$\tilde{W}_z$	Continuous	≥ 0	Dark-zone-amplified zone weight in objective aggregation.
n_r	Integer	≥ 0	Number of patrol teams added to baseline staffing.
$X_{z,\xi,t}$	Continuous	≥ 0	Patrol hours allocated to zone z in time t , scenario ξ .
$d_{z,t,\xi}$	Continuous	≥ 0	Drone hours allocated to zone z in time t , scenario ξ .
$c_{z,t,\xi}$	Continuous	≥ 0	Sensor effort allocated to zone z in time t , scenario ξ .

The most influential structural assumptions governing this layer are as follows; full parameter values appear in Appendix Tables 25–26.

Table 6: Layer II structural assumptions (L2-1 through L2-4).

Assumption	Justification	Modeling Role
L2-1. Patrol mobility varies by road class (25/15/10 km/h).	Travel speed is structurally heterogeneous across ENP road types.	Determines feasible response radius and therefore whether a zone is operationally reachable under each scenario.
L2-2. Marginal protection gain from additional patrol effort is sublinear.	SMART-derived ranger evidence indicates the largest gains occur when expanding from low baseline coverage, with diminishing returns at higher intensity (Critchlow et al., 2017).	Prevents over-allocation of staff in the optimizer and supports balanced camera/sensor/patrol portfolios.
L2-3. Effective resistance is the primary structural-isolation diagnostic.	Straight-line distance misses multi-path bottlenecks created by road topology and terrain accessibility.	Generates under-service score U_z used to prioritize structurally dark zones in Stage 1 and Stage 2.
L2-4. Baseline capital budget is fixed at \$750K for planning-order portfolio design.	The value is broadly consistent with documented protected-area investment scales in sub-Saharan Africa (Bruner et al., 2004; Lindsey et al., 2017).	Anchors the baseline portfolio while preserving budget sensitivity analysis at \$250K and \$1.5M in §6.

4.3 Topsis

TOPSIS Decision

Before finalizing Layer II, we compared candidate allocation frameworks with TOPSIS. We evaluated three allocation frameworks: Pure Integer Optimization (PIO), Circuit-Theory Infrastructure Modeling (CTIM), and a Two-Stage Stochastic Program (TSSP). Evaluation used ten weighted criteria; Uncertainty and Robustness received the highest weight (0.14) given ENP’s compounding shock environment, and Computational Tractability the lowest (0.06) since all candidates are feasible at project scale. Full criteria definitions, weights, normalized scores, and distance computations appear in Tables 39–44. A ± 1 perturbation of any individual score does not change the final ranking.

4.3.1 Selection Decision

PIO underperforms because it treats cells independently and does not represent seasonal or scenario uncertainty. CTIM is strongest as a *diagnostic* method because effective resistance identifies structurally isolated zones. TSSP is strongest as a *prescriptive* method because it optimizes fixed investments and scenario-adaptive recourse. We therefore use a hybrid: CTIM for diagnosis, then TSSP for deployment.

4.4 Architecture and Submodel Logic

4.4.1 Circuit-Theory Diagnosis

Layer I identifies *where* protection is weak. This stage determines whether those zones are structurally reachable. We model ENP as a weighted graph with bases, gates, waterholes, and zones as nodes. Roads define edge conductance C_{ij} by travel quality. Solving $Lv = b$ yields node potential, and edge flows $I_{ij} = C_{ij}(v_i - v_j)$ expose bottleneck corridors. This network interpretation follows ecological circuit-theory connectivity modeling, which represents movement opportunity as effective conductance across heterogeneous landscapes

(McRae et al., 2008).

Zone effective resistance is converted to an under-service score:

$$U_z = \frac{1}{1 + \exp(-k(R_z^{\text{eff}} - R_{\text{median}}))}.$$

Here, R_z^{eff} is the average effective resistance from zone z to all ranger bases (Appendix B, Eq. 44). R_{median} is the median of that value across all zones in the park. We set $k = 2.5$, which controls how quickly the score rises; with this calibration, zones in the highest resistance quartile typically have $U_z > 0.75$, matching our operational definition of structural isolation.

$U_z \approx 1$ marks structurally dark zones that remain hard to serve even if patrol hours increase.

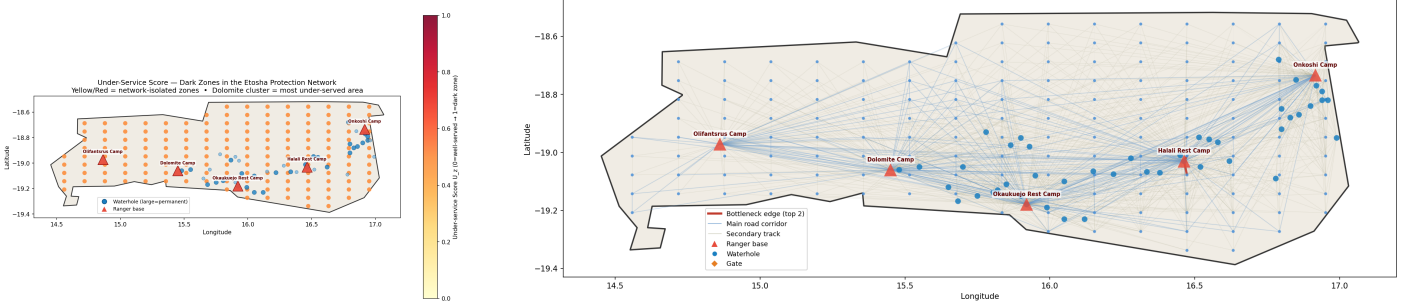


Figure 5: Circuit-theory structural diagnostics for the Etosha patrol network showing underservice score and bottlenecks.

4.4.2 Portfolio Selection

Using U_z and Layer I shortfall $(1 - P1_z)$, each zone receives

$$S_z = (1 - P1_z)(1 + \kappa U_z),$$

which prioritizes zones that are both under-protected and structurally isolated. Candidate interventions (outposts, cameras, sensors, drone bases, road upgrades, patrol teams) receive strategic scores B_i over affected zones. Lag-discounted benefits are optimized by

$$\max_{y_i \in \{0,1\}, n_r \in \mathbb{Z}_+} \sum_i \tilde{B}_i y_i + \gamma n_r$$

where γ is the per-team value weight assigned to additional patrol teams n_r , set equal to the average zone benefit of a single patrol team and ensuring patrol capacity competes for resources on the same scale as fixed infrastructure.

Subject to three binding constraints: capital $\sum_i F_i y_i \leq B_{\text{capital}} = \$750K$, operating cost $\sum_i O_i y_i + 60,000 n_r \leq B_{\text{operating}}$, and staffing $n_r \leq N_{\text{staff,max}}$. Full formulations are given in Appendix Eq. (B.5). We use \$750K as the baseline planning-order capital assumption (Bruner et al., 2004; Lindsey et al., 2017) and test \$250K and \$1.5M alternatives in §6. These constraints keep the optimizer grounded in available budget, staffing, and access conditions. Detailed formulations are in Appendix B.

4.4.3 Recourse Deployment

Portfolio Selection decides what the park owns; Recourse Deployment decides how to use those assets under realized scenarios ξ . The recourse problem allocates mobile resources — patrol hours, drone sorties, and sensor effort — to maximize a weighted park protection score across the fixed asset portfolio selected in Stage 2. Zone protection $P_{z,\xi,t}$ is modeled as a complementary Cobb-Douglas function of patrol effort and combined

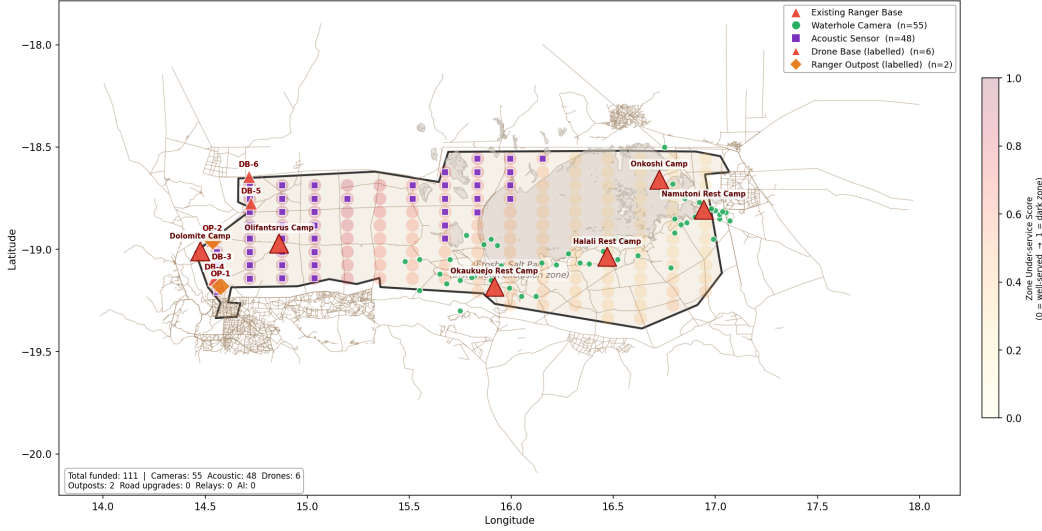


Figure 6: Optimal fixed-asset placement plan selected by the mixed-integer optimizer at the \$750K capital budget.

detection effort: neither channel alone achieves full protection, and both exhibit diminishing returns. The exponential form ensures $P_{z,\xi,t}$ is bounded to $(0, 1)$ by construction, while the permanent boost term $\alpha_b B_z(y)$ adds a structural floor from Stage 2 outposts and road upgrades that persists regardless of which scenario is realized.

$$\max_{x,d,c} \Pi(y, \xi) = \sum_z \tilde{W}_z \bar{P}_{z,\xi}, \quad \tilde{W}_z = W_z(1 + \kappa U_z),$$

with $\bar{P}_{z,\xi} = \frac{1}{|T|} \sum_t P_{z,\xi,t}$ and

$$P_{z,\xi,t} = 1 - \exp\left(-((\alpha_r X_{z,\xi,t})^a (\alpha_d d_{z,t,\xi} + \alpha_c c_{z,t,\xi} + \varepsilon)^b)^{1/(a+b)} - \alpha_b B_z(y)\right).$$

The optimization is re-solved per scenario with operational constraints on shift hours, sorties, and reassignment. Full protection-function derivation is in Appendix B; full recourse constraints are in Appendix B. Stage 2 selects a static intervention plan while Stage 3 reallocates mobile resources after scenario.

4.5 Results and Management Interpretation

4.5.1 What the Placement Map Reveals

At the \$750K planning budget, the optimal portfolio selects 111 additions: 55 waterhole cameras, 48 acoustic sensors, 6 drone bases, and 2 ranger outposts. Four drone bases (DB-3 to DB-6) cluster in the western Dolomite sector, which confirms structural isolation as the dominant allocation driver. The solution prioritizes chokepoint relief over even spacing.

Road upgrades, communication relays, and AI detection are not selected at this budget because lag-adjusted benefit per cost is lower than fast-deployable camera and sensor assets. This is a budget-level result, not a long-run rejection. In the high-capital sensitivity case (\$1.5M), structural assets become more competitive.

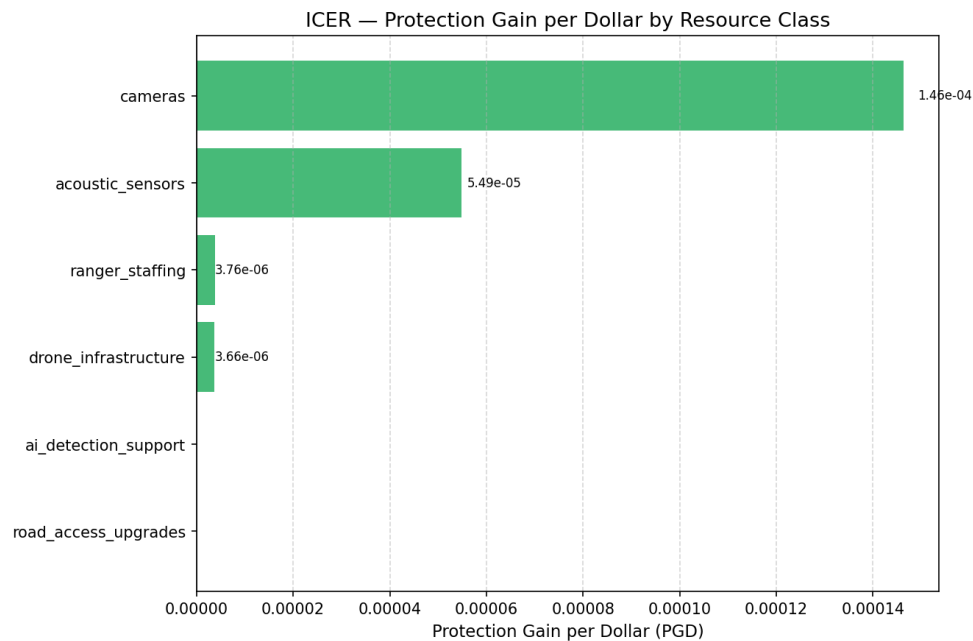
4.5.2 Scenario Performance

Π is an operational effectiveness score, not the same quantity as SPR. We set $\Pi_{\text{target}} = 0.70$ as an aspirational high-coverage operating point where incremental gains begin to saturate. Baseline performance is $\Pi = 0.310$. Wildfire is the most damaging scenario ($\Pi = 0.177$). Drone outage causes only a small drop ($\Pi = 0.307$), which indicates portfolio robustness when aerial capacity is constrained.

Table 7: Park-wide operational protection score Π across eight Layer II scenarios.

Scenario	Prob.	Π	Annual Cost	Staff	Note
Normal conditions	0.35	0.310	\$402,300	20	Baseline
Fewer rangers (65%)	0.10	0.201	\$282,300	10	−35%
Drones break (50%)	0.10	0.307	\$366,300	20	−1%
Poaching surge	0.10	0.311	\$402,300	20	+0.3%
Roads blocked	0.10	0.310	\$402,300	20	0%
Wildfire (30%)	0.10	0.177	\$402,300	20	−43%
Rangers + roads	0.08	0.201	\$282,300	10	−35%
Insider threat	0.07	0.199	\$246,300	10	−36%

4.5.3 ICER: Which Resources Give the Best Value?

**Figure 7:** Protection gain per dollar invested ($PGD_k = \Delta\Pi_k / \Delta Cost_k$) for each resource class at the \$750K budget.

The inverse metric is $ICER_k = \Delta Cost_k / \Delta\Pi_k$. Cameras dominate at this budget because unit cost is low and placement aligns with high-priority waterhole zones. The roughly 40× camera-vs-ranger PGD gap is a *marginal* result at this budget level, not a claim that staffing is unimportant.

This ICER result is marginal and budget-specific. At \$750K, cameras deliver more immediate gain per dollar than added patrol capacity because they relieve the current detection bottleneck while patrol expansion is still constrained by road-network coverage. This does *not* mean staffing is optional: in \$6, a 25% staffing reduction lowers Π by 7.55%, while a 25% increase raises Π by 20.81%. The asymmetry reflects the baseline operating point. Current staffing already supports core coverage, so moderate cuts first affect lower-yield zones, while moderate increases open high-priority zones that were previously underserved.

Layer II identifies what to buy and where to deploy it under budget and scenario constraints, but it produces

a static snapshot of operational effectiveness. Layer III extends this into a dynamic question: given current conditions and plausible future shocks, how is SPR expected to evolve over the next five years, and what staffing is required to keep it above target?

5 Layer III — Forecasting Layer

5.1 Forecasting and State Updating Core

Layer III answers two management questions: how protection is expected to evolve over five years, and what staffing is required to keep SPR above target under stress. This layer extends the Layer I spatial snapshot into monthly trajectories with uncertainty.

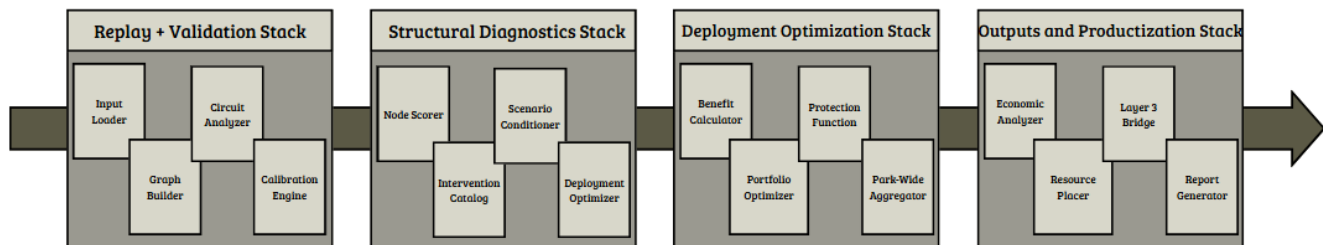


Figure 8: Layer 3 flowchart that shows the relationship between stacks and submodels.

5.2 Variables and Assumptions

Key Variables and Assumptions

Layer III propagates monthly vulnerability states under environmental forcing and scenario uncertainty. Table 8 lists the core variables used in the forecasting pipeline.

Table 8: Core Layer III variables and their role in the forecasting pipeline.

Variable	Type	Range	Role in Model
$x_{i,t}$	Continuous	$[0, 1]$	Monthly cell-level vulnerability state.
$u_{i,t}$	Vector	$\mathbb{R}^k$	Environmental forcing (drought, fire lag, seasonality).
Φ	Matrix	—	Transition matrix combining persistence and neighbor influence.
W_{std}	Matrix	—	Row-normalized spatial weight matrix.
ϕ_i	Continuous	$[0, 1]$	Cell-level persistence weight in AR transition term.
Q	Matrix	—	Process-noise covariance in EnKF forecast step.
R	Matrix	—	Observation-noise covariance in EnKF update step.
SPR_t	Continuous	$[0, 1]$	Monthly park-level Species Protection Rate.
$P(\text{SPR}_t < \tau)$	Continuous	$[0, 1]$	Target-miss probability at month t .
s_t^*	Integer	$\{100, \dots, 400\}$	Minimum staffing level to maintain $\text{SPR}_t \geq 0.90$.

Variable	Type	Range	Role in Model
$\varepsilon(i, t)$	Continuous	—	HSGP spatiotemporal residual field.
k_s, k_t	Functions	—	Matérn-3/2 spatial and exponential temporal kernels.
ℓ_s, ℓ_t	Continuous	km, months	Spatial and temporal length scales ($\ell_s = 28.4$ km, $\ell_t = 3.2$ months).

Table 9: Layer III structural assumptions (L3-1 through L3-4).

Assumption	Justification	Modeling Role
L3-1. Short-horizon vulnerability dynamics are primarily driven by environmental forcing and seasonal structure.	SLX diagnostics show forcing terms explain a majority share of monthly variance.	Supports forcing-driven state transition updates in the EnKF propagation step.
L3-2. Neighbor influence is local and decays with distance under the standardized spatial-weight matrix.	Moran's I and adjacency diagnostics indicate strong short-range spatial dependence.	Justifies the mixed self-persistence + neighbor transition structure in Φ .
L3-3. Cell persistence is moderate, bounded, and shock-sensitive.	Historical replay shows month-to-month carry-over without deterministic locking.	Enables stable multi-year trajectories while allowing scenario shocks to shift forecast paths.
L3-4. Holdout evaluation is an internal-consistency test because ground-truth labels are reconstructed from Layer I on historical covariates.	External field-observed cell-month protection labels are unavailable at required resolution.	Constrains interpretation of validation metrics to pipeline coherence rather than independent external validation.

Full variable and assumption tables appear in the Appendix tables and forecasting appendix sections.

5.3 Architecture and Submodel Logic

How the Layer Works: Four Steps

Step 1: Monthly Panel Construction. We rerun Layer I month by month from January 2019 to December 2024, which yields 72 months across 940 grid cells: $940 \times 72 = 67,680$ cell-month records. The 158 in-park cells are used for all SPR computations, while buffer cells provide spatial context. For validation, months 1–60 are training and months 61–72 are holdout test.

Step 2: Pattern Diagnostics. We test whether spatial persistence supports a spatiotemporal model. Moran's I ranges from 0.9445 to 0.9503. These values are expected for smoothed ecological surfaces, and they confirm that explicit spatial dependence should outperform purely temporal dynamics (Moran, 1950; Cressie and Wikle, 2011). Mean cell persistence is 0.5893, which indicates moderate month-to-month carry over. We then fit a fixed-effects SLX panel to quantify forcing and neighbor effects (Appendix B), following spatial-econometric lag decomposition principles (LeSage and Pace, 2009).

Step 3: Forecasting. We combine an HSGP residual model with an Ensemble Kalman Filter (EnKF). The

EnKF transition is

$$x_{t+1} = \Phi x_t + Bu_t + w_t \quad (7)$$

where x_t is vulnerability state, u_t is forcing, and w_t is process noise. The transition matrix is

$$\Phi = \text{diag}(\phi) + (I - \text{diag}(\phi))W_{\text{std}} \quad (8)$$

so each cell mixes self-persistence (ϕ_i) and neighbor influence.

We run 500 ensemble members for 60 months under five Layer III scenarios: baseline, severe drought, northern boundary breach, fire-season repeat, and combined drought+breach pressure. Layer II scenarios represent short-horizon operational shocks, while Layer III scenarios represent multi-year environmental and security trajectories. An optional Layer II-to-Layer III bridge can inject portfolio-corrected capacity forecasts into the EnKF transition matrix. **The forecasts presented in §5.4 and Figure 9 use baseline bridge mode — they represent pre-investment trajectories without the Layer II portfolio deployed.** This is deliberate: forecasting the return on the Layer II investment requires multi-year deployment assumptions beyond the scope of this analysis. The Layer II portfolio's protection gains are captured in the II metric; the Layer III forecasts quantify what the park faces if current operational conditions continue.

Model diagnostics and forecasting design. Before forecasting, we fit a fixed-effects SLX panel on 67,680 cell-month records to separate cell fixed effects, local forcing, spatial spillovers, and seasonal harmonics. This avoids misreading neighbor influence as temporal persistence and supports a forcing-driven transition structure (Cressie and Wikle, 2011; LeSage and Pace, 2009). The adjusted $R^2 = 0.671$ shows forcing explains most month-to-month variation. The remaining 33% likely reflects stochastic poaching, staffing fluctuations, and legacy composite-input provenance gaps (§7.2), which we absorb with the HSGP residual field and EnKF process noise. The SLX fit uses the full 940-cell panel, including buffer cells, to preserve spatial spillover structure near park edges. SPR reporting still uses only the 158 in-park cells.

We use EnKF because it is practical for high-dimensional ecological state models (Evensen, 2003; Ionides et al., 2006; Wikle and Berliner, 2007). For 940 cells, exact covariance updates scale poorly (roughly $O(n^3)$), while ensemble updates remain tractable and directly represent scenario uncertainty. We apply Gaspari–Cohn localization (50 km; Appendix B) to suppress spurious long-range correlations from finite ensembles.

We set the ensemble size to 500 as an accuracy–runtime compromise: calibration is near 1:1 spread-vs-RMSE (Figure 16), consistent with practical EnKF guidance (Evensen, 2003), and runtime stays under 45 minutes on a 16-core CPU with 64 GB RAM. We pair EnKF with an HSGP residual layer (Matérn-3/2 spatial kernel, $\ell_s = 28.4$ km; exponential temporal kernel, $\ell_t = 3.2$ months) so initialization starts from a realistic spatiotemporal field rather than a flat prior, reducing forecast spin-up.

Step 4: Decision Metrics. We convert ensembles into management outputs. Target-miss probability is

$$P(\text{SPR}_t < \tau) = \frac{1}{N} \sum_j \mathbf{1}\{\text{SPR}_t^{(j)} < \tau\} \quad (9)$$

with threshold τ and ensemble size N . Staffing requirements are computed from a discrete staffing set $\mathcal{S} = \{100, 150, 200, 250, 295, 350, 400\}$:

$$s_t^* = \min\{s \in \mathcal{S} : \text{SPR}_t(s) \geq 0.90\} \quad (10)$$

The exponent 0.7 in this staffing-capacity bridge encodes diminishing patrol returns, consistent with empirical evidence that effectiveness plateaus after early coverage gains (Moore et al., 2018); full details are in Appendix B. Sensitivity tests over exponent 0.5–0.9 yield severe-drought staffing estimates of 310–385, so the 350-staff recommendation is robust with an uncertainty band of about ± 40 staff.

5.4 Staffing and Management Feedback Outputs

Validation note. The holdout metrics in Table 31 measure internal consistency: they test whether the Layer III pipeline reproduces Layer I outputs on held-out months 61–72, not whether the model predicts independently observed real-world protection states. The held-out “ground truth” is itself generated by running Layer I on historical covariates. A 6-month Pearson r of 0.9921 therefore confirms that the forecasting pipeline is internally coherent and that the EnKF transition structure captures the dominant temporal patterns in the synthetic panel — it does not constitute external validation against field-observed poaching or wildlife census data, which were not available at the cell-month resolution required. This is the model’s most significant epistemic limitation and is acknowledged further in §7.2.

Under baseline conditions, SPR starts at 0.874 at month 0, dips slightly in early months reflecting seasonal cycling, and rises to 0.913 by month 60 — above the 0.90 target by year 4. The northern boundary breach and fire-season repeat scenarios both remain within 1–2 percentage points of baseline throughout, confirming that the existing sensor and camera portfolio provides robust perimeter absorption for those shock types. Figure 9 reports month-level values. Figure 10 reports annual means, so early-year averages can be below the month-0 value because they include seasonal lows.

Severe drought ($+2.5\sigma$) is the dominant stress case. Month-60 SPR falls to 0.824 — 8.9 percentage points below target — with a 47% target-miss probability. The combined drought-plus-breach trajectory tracks severe drought closely, confirming drought as the dominant compound-shock driver; the boundary breach adds only approximately 0.7 percentage points of additional SPR loss on top of the drought effect. The 2026–2027 window represents the deepest risk period across both drought scenarios, with annual mean SPR approximately 0.75–0.76 (Figure 10).

The staffing bridge translates these trajectories into minimum headcount requirements: approximately 295 staff sustain $\text{SPR} \geq 0.90$ under baseline conditions, 350 under severe drought, and 400 under combined drought and boundary breach. The baseline target-miss probability of 0.000 reflects that all 500 ensemble members exceed 0.90 at month 60 under baseline conditions — this is a sample lower bound, not a theoretical guarantee.

Management implication: if drought intensifies, the park needs additional staffing capacity.

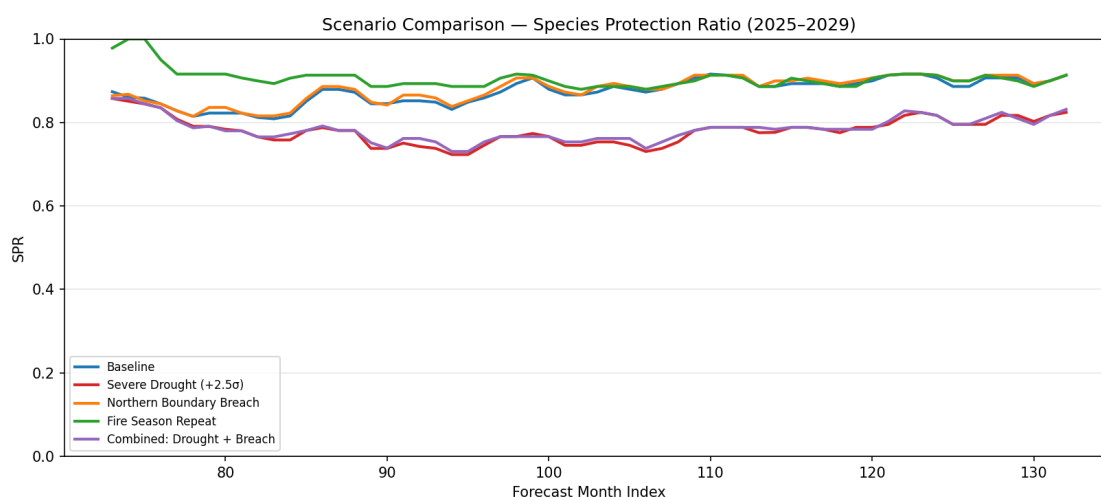


Figure 9: Ensemble mean SPR trajectories over the 60-month forecast horizon (January 2025 – December 2029) under five scenarios.

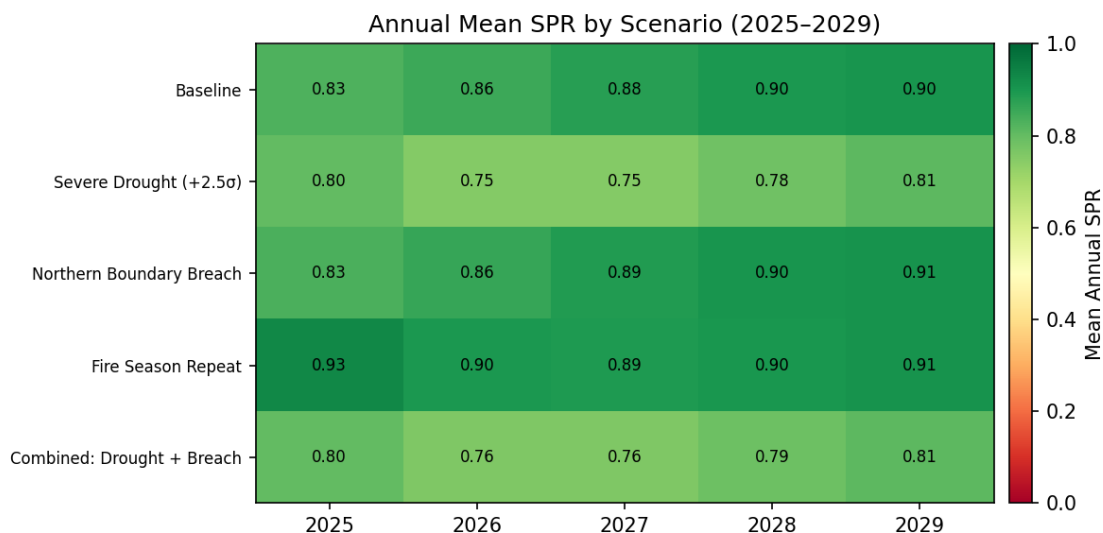


Figure 10: Annual mean SPR heatmap across five forecast scenarios and forecast years 2025–2029. .

6 Sensitivity, Scenario, and Uncertainty Synthesis

6.1 Approach

We run one-at-a-time sensitivity analysis around the Layer II baseline portfolio. Each run changes one assumption or control parameter while all other settings remain fixed. We report effects as $\Delta\Pi$, the percent change in Layer II protection score Π relative to baseline $\Pi = 0.310$.

6.2 Scenario Design and Perturbations

Table 10 lists the tested perturbations and their measured $\Delta\Pi$ values.

Table 10: Sensitivity analysis between 8 scenarios where one factor is being varied.

Scenario	Justification	$\Delta\Pi$ vs. baseline
Weights (all equal)	Tests sensitivity to the baseline multi-criteria weight vector.	−0.0145%
Budget $\pm 25\%$	Tests plausible funding variation around the baseline operating plan.	−18.20%, +15.40%
Natural disaster $\pm 25\%$	Tests uncertainty in wildfire-loss severity.	+1.37%, −1.16%
Staff $\pm 25\%$	Tests realistic changes in ranger availability.	−7.55%, +20.81%
Uniform poaching time	Tests sensitivity to removing day-night concentration.	+7.92%
Drone detection rate (0.2% / 20%)	Tests uncertainty in drone sensing quality under operational limits.	$\approx 0.00\%$
Patrol speed −40%	Tests mobility disruption from degraded road access.	−18.66%
Poaching pressure $\pm 50\%$	Tests large shifts in illegal-harvest intensity.	+34.00% at −50%, −10.20% at +50%
Shift length 8h to 12h	Tests a tactical schedule extension without added personnel.	+0.49%
Capital budget (\$250K / \$750K / \$1.5M)	Tests how fixed-asset purchasing power changes feasible portfolios.	−67.49%, baseline, +64.64%

6.3 Tornado Comparison

Figure 11 visualizes the same $\Delta\Pi$ results. The largest negative bar is low capital (-67.49%). The largest positive bar is high capital ($+64.64\%$). The tornado ordering confirms that investment scale is the dominant first-order driver of Layer II performance.

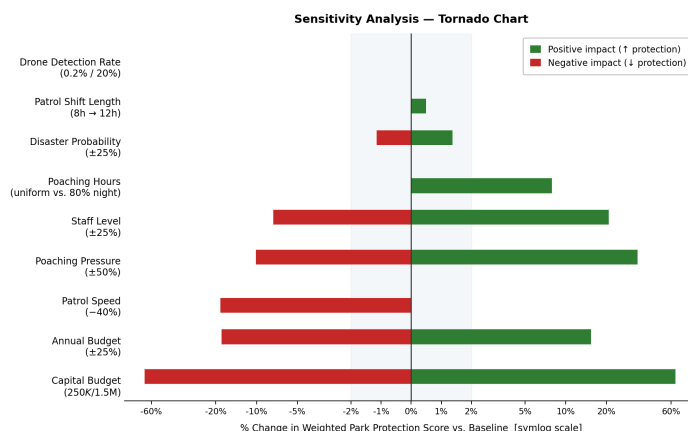


Figure 11: Tornado chart of sensitivity analysis with each bar showing the percentage change in park-wide Layer II protection score.

6.4 Dominant Risk Factors and Interpretation

Capital adequacy is the dominant risk lever. Low capital causes the largest loss in protection effectiveness, and high capital causes the largest gain. Staffing, patrol mobility, and poaching pressure form the second tier of influence. These levers directly change coverage capacity or threat load, so they shift Π more than tactical adjustments. Poaching pressure -50% yields $+34.00\%$ Π . This reflects that Π measures the ratio of protection capacity to threat load — lower threat intensity makes the same capacity more adequate — and does not imply management should discount perceived threat levels. Shift-length changes and drone-detection assumptions produce small effects in this setup. The management implication is direct: prioritize capital stability, staffing resilience, and mobility continuity before tuning minor operational parameters.

7 Strengths, Limitations, and Scope

7.1 Strengths

- **Uncertainty-aware forecasting:** Layer III reports ensemble trajectories, target-miss probabilities, and scenario comparisons rather than a single deterministic trajectory.
- **Manager-facing outputs:** The framework returns explicit decisions (where to deploy, what to purchase, and staffing needed to keep SPR above target), not only abstract diagnostics.
- **Empirically grounded weighting:** Guild weights and ecological-value exponents are anchored to literature on poaching exposure, trophic cascade magnitude, and habitat dependence. If new evidence shifts relative risk, recalibration updates a bounded parameter set rather than requiring full model redesign.
- **Modular reproducibility:** Each layer has explicit inputs, outputs, and parameter tables, which supports component-level auditing and transfer to new parks.

7.2 Limitations

- **Forecast horizon risk:** We test on months 61–72 and get strong agreement with replayed Layer I outputs. We then project 60 months ahead, so uncertainty grows as the horizon grows. When you read year-4 or year-5 SPR, you should treat those values as decision ranges and update plans each season.

- **Reduced-form structure:** We model park-level patterns in threat, response, and coverage. We do not model each poacher move, each ranger route choice, or each animal path. Managers should pair our outputs with ranger field logs before final daily patrol assignments.
- **Cross-layer error propagation:** Layer II and Layer III both use Layer I outputs. If Layer I inputs drift, downstream deployment and forecast outputs drift too. Managers should rerun all three layers after each major data update.
- **Strategic resolution:** We plan at cell scale, so you get clear park strategy. Fine hotspots inside a cell still hide from the model. Managers should use this framework for weekly zone priorities, then use on-ground reports for day-level placement.
- **Regulatory feasibility caveat:** Public reporting lists drone restrictions in ENP ([Africa Travel & Tourism Association, 2024](#); [Windhoek Observer, 2024](#)). Layer II tests show small performance loss under 50% drone outage. Managers should keep a drone-light portfolio and prioritize cameras, sensors, road access, and ranger mobility.
- **Potential non-stationarity:** Climate pressure, enforcement policy, and access routes shift over time. Fixed parameters lose fit when these drivers shift. Managers should recalibrate key threat and response parameters at regular intervals.

8 Model Adaptation

8.1 Selecting the Two Adaptation Parks

We test transferability with one *similar* park (Bandhavgarh, India) and one *structurally different* park (Tē Wahipounamu, New Zealand). Bandhavgarh is a recalibration case because it is a terrestrial reserve with zoned management, road-constrained patrol logic, and strong large-mammal conservation pressure ([Bandhavgarh Tiger Reserve, nd](#); [Ministry of Environment, Forest and Climate Change, 2016](#)).

8.2 Adapting to a Similar Case: Bandhavgarh Tiger Reserve

The three-layer pipeline structure transfers directly. Layer I's Cobb-Douglas ecological value form, Noisy-OR threat integration, Noisy-OR protection capacity, and SPR aggregation all apply without structural modification. Layer II's circuit-theory network diagnosis and two-stage stochastic program carry over unchanged. Layer III's EnKF transition structure and Gaspari-Cohn localization are architecture-level choices independent of the specific park.

Inputs and parameters requiring modification. The guild set and weights must be recentered on Bengal tiger (*Panthera tigris*) as flagship; the human threat calibration shifts from boundary-access poaching pressure to tourism-corridor and forest-fringe village access, which are the primary vectors in Bandhavgarh. Waterhole concentration risk is lower priority; fire hazard and habitat fragmentation replace it as secondary threat channels. Patrol mobility parameters must reflect Bandhavgarh's zoned access system (Core, Buffer, and Multiple-Use zones with different speed assumptions) rather than ENP's open-road network. Response time thresholds may tighten to under 1.5 hours given the smaller managed area.

Data required for calibration. Tiger and prey occurrence records from GBIF and the Wildlife Institute of India; MODIS NDVI and CHIRPS rainfall at comparable resolution; zone boundary shapefiles from the Madhya Pradesh Forest Department eco-sensitive zone notification; road network from OSM; documented poaching incident locations from the reserve's SMART patrol logs.

Expected metric shifts. Because managed area is smaller and access is more tightly zoned, structural isolation (high- U_z zones) should be less prevalent than in ENP. However, tiger-centric weighting concentrates priority-score mass near gate-access corridors, so Layer II deployment should shift toward corridor surveillance and rapid-response siting rather than ENP's western-interior outpost strategy.

8.3 Adapting to a Different Case: Tē Wahipounamu

Components that remain unchanged. The three-layer workflow — diagnosis, allocation, forecasting — applies. SPR as a weighted protected fraction transfers directly. The EnKF transition structure, scenario tree framework, and TOPSIS method-selection logic are park-independent.

Inputs and parameters requiring structural redesign. The threat module requires the most significant changes: the four ENP channels must be replaced by invasive predator pressure (stoats, possums, rats), visitor-stress at high-use sites, and alpine weather disruption. The mobility module must incorporate terrain penalty functions for fjord and alpine topography, seasonal route closures, and weather-dependent response-time multipliers with no ENP analog. Circuit-theory connectivity modeling remains applicable but requires a richer edge-conductance specification encoding vertical terrain transitions, not just road quality.

Data required for calibration. Department of Conservation management plan and track-closure records; LINZ topographic data for terrain penalty calculation; pest monitoring network data (predator-trap density and catch rates) as threat intensity proxies; visitor-use statistics from hut booking systems.

Expected metric shifts. Baseline deployment effectiveness II will likely be lower at first calibration because response access is constrained by fjord and alpine topography. Protection performance will become more dependent on sensor-trap network design and access-node coverage than on drone-centered deterrence, with larger staffing buffers required in closure-heavy seasons.

9 Conclusion

- **Current status:** Layer I reports dry-season static SPR = 0.890 and wet-season static SPR = 0.856. ENP therefore operates near target but has limited resilience in high-risk interior zones.
- **Priority action:** Concentrate investment and adaptive deployment in the western interior corridor near Galton Gate (Dolomite sector), where high-value cells remain structurally hard to reach.
- **Main forecast risk:** Layer III baseline month-60 SPR reaches 0.913, but severe drought reduces month-60 SPR to 0.824 with 47% target-miss probability. So, credible stress requires staffing buffers near 350+.
- **Generalizability:** The pipeline transfers to similar reserves via recalibration and to structurally different systems when threat and mobility modules are redesigned, as demonstrated in §8.
- **Final synthesis:** ENP does not need maximal coverage everywhere. It needs reliable performance at the highest-consequence bottlenecks. Our framework supports that shift from static maps to operational planning.

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A Glossary

Glossary

Term	Definition
Syndicate	Coordinated criminal group that organizes large-scale illegal activities such as rhino poaching.
Species Protection Rate (SPR)	Primary park-level protection metric. It measures the weighted share of ecologically valuable areas that meet defined protection thresholds.
Dark Zones	Areas with persistent structural under-service where patrol and sensing coverage are insufficient for timely detection and response.
Guild	Functional species group with similar ecological role or conservation profile (e.g., megafauna, predators, herbivores).
Cobb-Douglas	Multiplicative value function used to combine habitat-support factors (e.g., water, vegetation, intactness) into ecological value.
Recourse Deployment	Scenario-adaptive reassignment of mobile resources after uncertainty is realized.
Acoustic Sensors	Field devices that detect sound signatures associated with threats, such as gunshots or vehicle activity.
Drone Base	Fixed launch, charging, and maintenance node that supports aerial monitoring operations.
Detection Capacity	Probability that a threat is detected in time by patrol or technology assets.
Response Capacity	Probability that response teams reach a detected threat location within required operational time limits.
Circuit Theory	Network-analytic method that treats movement access as electrical flow over a resistance graph to identify bottlenecks and isolated zones.
Water Concentration Risk	Threat increase caused by wildlife aggregation near limited water points during dry periods.
Boundary Breach	Security failure at the park perimeter that increases unauthorized entry risk.
Ensemble Kalman Filter (EnKF)	Data-assimilation forecasting method that propagates many state trajectories and updates uncertainty through time.
Salt Pan	Large dry lakebed feature in ENP that strongly shapes movement and accessibility.

B Mathematical Derivations

Layer I Mathematical Derivations

A.0 Mathematical Tools, Scope, and Inputs

Pipeline overview. Model 1 runs as a staged pipeline on a fixed 20×47 grid (940 cells), with 158 in-park cells used for protection outcomes. Core stages compute habitat suitability, ecological value, threat channels, connectivity, protection capacity, residual vulnerability, and final park metrics.

Core transforms. We use the logistic function $\sigma(x) = \frac{1}{1+e^{-x}}$, the logit transform $\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$, clipping $\text{clip}(x, a, b) = \max(a, \min(b, x))$, and indicator functions $\mathbf{1}[\cdot]$.

Noisy-OR (probabilistic union). For approximate independent channels with probabilities $p_1, \dots, p_n$, the probability at least one channel activates is $1 - \prod_{k=1}^n (1 - p_k)$. We use this for integrated threat and protection capacity.

A.0.1 Data Inputs, Spatial Assumptions, and Scope

All geodesic distances are computed with the haversine formula ($R = 6371$ km) to avoid planar distortion over Etosha-scale extents. Seasonal means are aggregated as dry (May–October) and wet (November–April). ESA land-cover fractions enter suitability, intactness, and connectivity terms.

The model assumes fixed planning cells, two seasonal states, guild-level species grouping, common $[0, 1]$ scaling for fused channels, and a strict all-guild protection rule to avoid masking failures in high-consequence guilds.

Several composite covariates used by the pipeline (including forage adequacy, water reliability, cover index, boundary access index, poaching-risk raw score, and patrol-effort index) come from legacy preprocessing products whose full generation scripts are not co-located in this snapshot. We therefore treat their provenance as a documented limitation and bound potential impact through sensitivity testing in Section 6.

A.1 Habitat Suitability Blend

Habitat suitability $O(g, i)$ estimates how well cell i matches the ecological requirements of guild g . When occurrence data are available, covariate-based and occurrence-based estimates are blended:

$$O(g, i) = \begin{cases} 0.70 \cdot H_{\text{cov}}(g, i) + 0.30 \cdot H_{\text{occ}}(g, i), & \text{if occurrence data exist,} \\ H_{\text{cov}}(g, i), & \text{otherwise.} \end{cases} \quad (11)$$

The 70/30 split ensures the covariate-based ecological prior dominates while occurrence evidence provides a bounded local correction. The covariate score is:

$$H_{\text{cov}}(g, i) = \left(\sum_k w_k^g x_k(i) \right) [0.9 + 0.1 \cdot LC_{\text{norm}}(g, i)] \cdot \text{seasonal_mod}(g, i) \quad (12)$$

where guild-specific covariate weights w_k^g sum to unity and the bracket term applies a modest land-cover correction (LC_{norm} weight of 0.1 means it refines but does not dominate). The seasonal modifier $\text{seasonal_mod}(g, i)$ is a per-guild, per-season scalar (range 0.85–1.15) adjusting baseline suitability for known seasonal shifts in guild habitat use.

The occurrence term uses a diminishing-returns logistic transform:

$$H_{\text{occ}}(g, i) = \text{remap}(\sigma(-0.5 + 1.5 \log(1 + n_{\text{occ}}(g, i)))) \quad (13)$$

where the remap anchors zero occurrences to a near-zero contribution. Occurrence counts are aggregated within a 14 km radius of each cell centroid, consistent with approximate home-range influence radii of medium-to-large savanna mammals.

A.2 Cobb-Douglas Ecological Value

The ecological value formula integrates three support variables multiplicatively:

$$E(g, i) = S_g \cdot O(g, i) \cdot X_{\text{ndvi}}(i)^{\alpha_g} X_{\text{water}}(i)^{\beta_g} X_{\text{intact}}(i)^{\gamma_g} \quad (14)$$

with the constraint $\alpha_g + \beta_g + \gamma_g = 1$ for each guild.

Why multiplicative form: The Cobb-Douglas structure captures partial complementarity. Water availability, vegetation productivity, and habitat intactness are each necessary for wildlife viability in an arid savanna system. A severe deficit in any one cannot be offset by exceptional performance in others. An additive score would allow excellent vegetation to mask critically limited water access, which is implausible given that water is the primary binding constraint for most Etosha guilds during dry season.

Floor values: Each X term is clipped to a minimum of 0.05 to prevent numerical collapse of the product when any single factor approaches zero.

Table 12: Guild-specific Cobb-Douglas exponents. Each row sums to 1.00.

Guild	α_g (NDVI)	β_g (water)	γ_g (intactness)	Sum
Megafauna	0.30	0.45	0.25	1.00
Predators	0.35	0.30	0.35	1.00
Herbivores	0.45	0.30	0.25	1.00
Birds	0.40	0.20	0.40	1.00
Habitat fauna	0.35	0.25	0.40	1.00

Megafauna and predators are weighted toward water access (large β_g) because predictable waterhole use is the primary mechanism linking water to concentration-based poaching exposure. Birds are weighted toward vegetation productivity (large α_g) because foraging habitat drives their distribution. Habitat fauna are weighted toward intactness (large γ_g) because limited mobility prevents compensatory movement away from degraded patches.

A.3 Environmental Support Variables

The three support terms are defined as:

Vegetation productivity:

$$X_{\text{ndvi}}(i) = \text{clip}\left(\frac{\text{NDVI}(i)}{0.80}, 0.05, 1\right) \quad (15)$$

The NDVI ceiling of 0.80 reflects the upper tail of savanna greenness observed in Etosha.

Water access:

$$X_{\text{water}}(i) = \text{clip} \left(e^{-d_{wh}(i)/20.0} + k_w f_{80}(i), 0.05, 1 \right) \quad (16)$$

where $d_{wh}(i)$ is distance to nearest waterhole (km), $k_w = 2.0$ is the permanent-water bonus coefficient, and $f_{80}(i)$ is the fraction of nearby area with permanent water (ESA land-cover). The decay scale of 20.0 km reflects the approximate range over which water proximity provides ecological benefit at the park scale.

Habitat intactness:

$$X_{\text{intact}}(i) = \text{clip} (1 - \text{bare_frac}(i) - \text{built_frac}(i), 0.05, 1) \quad (17)$$

A.4 Human Threat Model

Human threat is modeled as a logistic function of three spatial predictors:

$$\log \lambda(i) = 1.5 \cdot \text{road}(i) + 1.0 \cdot \text{bnd}(i) + 2.0 \cdot \text{prs}(i) \quad (18)$$

$$H(g, i) = \sigma(\log \lambda(i) + \alpha + \delta_g^H) \quad (19)$$

The coefficient $\beta_{\text{prs}} = 2.0$ is the largest because poaching records exhibit strong spatial persistence: cells with elevated historical risk continue to attract illegal activity. The intercept α is solved analytically so that the park-wide mean megafauna human threat matches a calibration target of 0.35:

$$\alpha = \text{logit}(0.35) - \delta_{\text{meg}}^H - \overline{\log \lambda}|_{\text{in-park}} \approx -2.98 \quad (20)$$

where $\overline{\log \lambda}|_{\text{in-park}}$ is the mean over all 158 in-park cells.

Table 13: Guild log-odds shifts for human threat (δ_g^H). Positive values indicate higher vulnerability.

Guild	δ_g^H	Rationale
Megafauna	+1.2	Rhinoceros is primary poaching target
Predators	+0.3	Some targeted trade in felid parts
Herbivores	−0.2	Lower direct human pressure
Birds	−0.8	Minimal direct poaching threat
Habitat fauna	−0.4	Low poaching exposure

A.5 Fire Hazard Model

The base fire hazard is:

$$F_{\text{base}}(i) = \sigma \left(-1.2 + 2.0 \cdot \text{fire}(i) - 0.8 \cdot \text{ndvi}(i) - 0.012 \cdot \text{rain}(i) + 0.015 \cdot d_{wh}(i) + \gamma_{\text{sea}} \right) \quad (21)$$

where $\gamma_{\text{sea}} = +0.50$ in dry season and -0.80 in wet season. Signs are ecologically motivated: repeated burning and distance from water increase hazard, while greener and wetter conditions reduce it. The seasonal term calibrates so that dry-season fire hazard substantially exceeds wet-season hazard.

The guild-specific fire score is shifted in log-odds space:

$$\text{logit}(F(g, i)) = \text{logit}(F_{\text{base}}(i)) + \log s_g^F \quad (22)$$

avoiding linear-probability distortions near boundaries.

Table 14: Guild fire sensitivity parameters. $s_g^F = 1.00$ (herbivores) is the reference level.

Guild	s_g^F	$\log s_g^F$
Megafauna	0.85	-0.163
Predators	0.70	-0.357
Herbivores	1.00	0.000
Birds	0.60	-0.511
Habitat fauna	1.10	+0.095

A.6 Water-Concentration Risk

Water-concentration risk combines spatial proximity to waterholes with drought stress:

$$M_t(i) = \sigma(0.5 + 1.2(\text{drought_std}(i) + \text{offset}_t)) \quad (23)$$

$$W(g, i, t) = \text{clip} \left(e^{-d_{wh}(i)/\lambda_g} M_t(i) + 2.0 \cdot f_{80}(i), 0.01, 1 \right) \quad (24)$$

where $\text{offset}_t = +1.0$ in dry season and -0.5 in wet season. The exponential decay captures local attraction to water points; the drought multiplier $M_t(i)$ strengthens this concentration under dry conditions and relaxes it when surface water is broadly available.

Table 15: Guild-specific waterhole concentration decay radii λ_g (km).

Guild	λ_g (km)
Megafauna	25.0
Predators	20.0
Herbivores	15.0
Birds	30.0
Habitat fauna	10.0

Larger values indicate wider effective ranging to water sources. Birds and megafauna travel furthest; habitat fauna concentrate only within narrow radii.

A.7 Habitat Degradation

Habitat integrity is a multiplicative function of four structural dimensions:

$$Z_1(i) = 1 - \text{road_norm}(i) \quad (25)$$

$$Z_2(i) = 1 - \text{burn_norm}(i) \quad (26)$$

$$Z_3(i) = \text{bnd_dist_norm}(i) \quad (27)$$

$$Z_4(i) = \text{ndvi_veg}(i) \quad (28)$$

For each guild:

$$I(g, i) = Z_1^{a_{g1}} Z_2^{a_{g2}} Z_3^{a_{g3}} Z_4^{a_{g4}}, \quad \sum_m a_{gm} = 1 \quad (29)$$

and habitat degradation is $D(g, i) = 1 - I(g, i)$. The multiplicative form ensures that chronic structural decline from any single source materially reduces total integrity even when other dimensions are healthy.

Table 16: Habitat integrity exponents per guild. Each row sums to 1.00.

Guild	$a_{g1} (Z_1)$	$a_{g2} (Z_2)$	$a_{g3} (Z_3)$	$a_{g4} (Z_4)$	Sum
Megafauna	0.40	0.25	0.25	0.10	1.00
Predators	0.30	0.30	0.30	0.10	1.00
Herbivores	0.30	0.35	0.20	0.15	1.00
Birds	0.25	0.35	0.15	0.25	1.00
Habitat fauna	0.35	0.30	0.20	0.15	1.00

Z_1 : road integrity; Z_2 : burn recurrence; Z_3 : boundary insulation; Z_4 : vegetation structure. Megafauna and predators weight road integrity most strongly because roads are the primary access vector for poachers.

A.8 Noisy-OR Threat Integration

The Noisy-OR formula integrates threat channels via probabilistic union:

$$T(g, i) = 1 - (1 - H)(1 - F)(1 - W)(1 - D)|_{g,i} \quad (30)$$

This is a standard probabilistic fusion for settings where any channel can independently elevate risk. It preserves bounded outputs in $[0, 1]$, captures compounding from concurrent moderate threats, and avoids the dilution of simple averaging.

Independence assumption: The channels are treated as conditionally independent first-order contributors to risk. Noisy-OR is therefore a transparent and policy-interpretable approximation rather than a claim of exact mechanistic independence. Positive residual correlation (roads amplify both H and D ; drought amplifies both W and F) means the formula slightly overestimates true integrated threat, which is the conservative direction of error for a protection-planning application.

A.9 Detection Capacity

Weighted surveillance effort combines patrol, camera, and drone contributions:

$$\mathcal{E}_{\text{tot}}(i) = m_t (\theta_{\text{pat}} e_{\text{pat}}(i) + \theta_{\text{cam}} e_{\text{cam}} + \theta_{\text{drone}} e_{\text{drone}}) \quad (31)$$

where $m_t = 1.00$ in dry season and $m_t = 0.88$ in wet season, reflecting reduced patrol coverage during wet months. Default camera effort $e_{\text{cam}} = 0.30$ and drone effort $e_{\text{drone}} = 0.10$ are treated as fixed infrastructure baselines.

Guild-specific exponential detection probability:

$$p^{\text{det}}(g, i) = 1 - e^{-s_g^{\text{det}} \mathcal{E}_{\text{tot}}(i)} \quad (32)$$

The exponential form gives diminishing returns to additional surveillance effort.

Table 17: Guild detection sensitivity s_g^{det} . Higher values reflect larger body size and more predictable movement.

Guild	s_g^{det}
Megafauna	1.20
Predators	1.00
Herbivores	0.90
Birds	0.60
Habitat fauna	0.70

A.10 Response Capacity

Travel time uses a seasonal response-time multiplier r_t ($r_t = 1.00$ dry, $r_t = 1.18$ wet):

$$\tau(i) = r_t \cdot \frac{d_{\text{base}}(i)}{40.0 \text{ km/h}} \quad (33)$$

Response capacity:

$$p^{\text{resp}}(i) = \sigma(-2.5(\tau(i) - 2.0)) \quad (34)$$

The logistic form transitions smoothly around the 2-hour target, reflecting gradual degradation of response effectiveness with delay. The steepness parameter $k = 2.5$ controls how sharply capacity drops beyond the threshold.

Protection capacity is modeled as a Noisy-OR union of detection and response channels:

$$P_{\text{cap}}(g, i) = 1 - (1 - p^{\text{det}}(g, i))(1 - p^{\text{resp}}(i)) \quad (35)$$

This formulation captures the probability that at least one protection channel succeeds, with maximum protection when both channels are strong.

A.11 Circuit-Theory Connectivity

Connectivity is computed using circuit theory so that movement potential aggregates across many plausible pathways rather than a single least-cost corridor.

Node conductance:

$$\kappa(i) = e^{-\text{road_norm}(i)} \cdot \text{ndvi_component}(i) \cdot \text{bare_component}(i) \quad (36)$$

Queen-adjacent edge conductance:

$$w_{ij} = \frac{\kappa(i) + \kappa(j)}{2} \quad (37)$$

The weighted graph Laplacian $\mathbf{L}\mathbf{v} = \mathbf{b}$ is solved with source injections at park gate nodes and sink withdrawals at pan-interior nodes. The resulting potential vector is min-max normalized to produce $C(i) \in [0, 1]$.

Connectivity-adjusted vulnerability:

$$V_{\text{adj}}(g, i) = V(g, i)(1 - \delta_{\text{conn},g}C(i)) \quad (38)$$

where $\delta_{\text{conn},g}$ varies by guild dispersal capacity (birds = 0.22, habitat fauna = 0.10).

A.12 Final Aggregation, Thresholds, and Park Metrics

We aggregate guild-level vulnerability into a cell priority score using guild weights ω_g , and classify protection with a strict all-guild threshold rule:

$$I_i^{\text{prot}} = \mathbf{1}[P_{\text{cap}}(g, i) \geq \tau_g^* \text{ for all } g] \quad (39)$$

The park-level Species Protection Rate (SPR) is the weighted ecological fraction that satisfies this rule:

$$\text{SPR} = \frac{\sum_i I_i^{\text{prot}} \left(\sum_g \omega_g E(g, i) \right)}{\sum_i \sum_g \omega_g E(g, i)} \quad (40)$$

The challenge score used for intervention ranking is:

$$\text{challenge}(i) = \sum_g \omega_g \mathbf{1}[P_{\text{cap}}(g, i) < \tau_g^*] E(g, i) T(g, i) \quad (41)$$

Table 18: Guild weights, urgency scalars, protection thresholds, and connectivity discounts.

Guild	ω_g	S_g	τ_g^*	$\delta_{\text{conn},g}$
Megafauna	0.35	0.80	0.75	0.20
Predators	0.25	0.70	0.65	0.18
Herbivores	0.20	0.40	0.55	0.15
Birds	0.10	0.75	0.55	0.22
Habitat fauna	0.10	0.45	0.60	0.10

Layer II Mathematical Derivations

B.1 Graph Laplacian Formulation

The park operational graph $G = (V, E)$ consists of n nodes (ranger bases, gates, waterholes, patrol zones) connected by edges (roads). Each edge $(i, j) \in E$ has conductance C_{ij} representing road quality. The weighted graph Laplacian L is an $n \times n$ matrix defined by:

$$L_{ij} = \begin{cases} \sum_k C_{ik} & \text{if } i = j \\ -C_{ij} & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

The Laplacian system $Lv = b$ balances supply and demand at each node, where v is the vector of node potentials and b is the source/sink vector. Supply nodes (ranger bases) have positive entries in b ; demand nodes (high-threat zones) have negative entries.

Properties: - L is symmetric and positive semi-definite - The null space of L is spanned by the constant vector $\mathbf{1}$ - For connected graphs, L has rank $n - 1$

To solve uniquely, we fix one node potential (typically ground one demand node at $v = 0$) or use the Moore-Penrose pseudoinverse.

B.2 Effective Resistance Calculation

The effective resistance R_{ij}^{eff} between nodes i and j is:

$$R_{ij}^{\text{eff}} = (e_i - e_j)^\top L^\dagger (e_i - e_j) \quad (43)$$

where $L^\dagger$ is the Moore-Penrose pseudoinverse of L and e_i is the i -th standard basis vector.

For zone-level under-service scores, we compute R_z^{eff} as the average effective resistance from zone z to all supply nodes:

$$R_z^{\text{eff}} = \frac{1}{|S|} \sum_{s \in S} R_{zs}^{\text{eff}} \quad (44)$$

where S is the set of supply nodes (ranger bases).

The logistic transformation to under-service score:

$$U_z = \frac{1}{1 + \exp(-k(R_z^{\text{eff}} - R_{\text{median}}))} \quad (45)$$

centers scoring at the park median with steepness $k = 2.5$.

B.3 Strategic Benefit Calculation

The strategic benefit of intervention i aggregates improvements across all affected zones:

$$B_i = \sum_{z \in N(i)} S_z (\beta_\tau \Delta \tau_{z,i} + \beta_p \Delta p_{z,i} + \beta_q \Delta q_{z,i} + \beta_R \Delta R_{z,i}) \quad (46)$$

Component definitions: - $\Delta \tau_{z,i}$: Response time reduction (hours). For an outpost, this is the change in travel time from nearest base. For road improvements, this is speed increase divided by distance. - $\Delta p_{z,i}$: Detection probability gain. For cameras and sensors, this is $1 - \exp(-s_i A_z / A_{\text{coverage}})$ where s_i is sensor strength and A_z is zone area. - $\Delta q_{z,i}$: Coverage area gain (km²). For outposts, this is the area newly reachable within response threshold. - $\Delta R_{z,i}$: Structural resistance reduction. Recompute R_z^{eff} after adding asset i and take the difference.

Weight calibration: - $\beta_\tau = 0.15$ (response time) - $\beta_p = 0.40$ (detection probability) - $\beta_q = 0.35$ (coverage area) - $\beta_R = 0.10$ (structural resistance)

Detection weights sum to 0.75. This is a modeling assumption that prioritizes early detection in high-risk waterhole periods.

B.4 Lag-Discount Factor

The lag-discount factor accounts for implementation delay:

$$\delta_i = \frac{1}{1 + \lambda L_i} \quad (47)$$

where L_i is implementation lag in months and $\lambda = 0.2$ controls discount steepness.

Implementation lags: - Cameras/sensors: $L_i = 0.5$ months (2 weeks) - Drone bases: $L_i = 1.5$ months (6 weeks) - Road upgrades: $L_i = 3$ months - Ranger outposts: $L_i = 6$ months - Communication relays: $L_i = 2$ months - AI upgrades: $L_i = 1.5$ months

Assets with shorter lags receive higher effective benefit $\tilde{B}_i = \delta_i B_i$ at any fixed strategic value B_i .

B.5 Robust Blended Objective

The robust optimization embeds five disruption scenarios $\xi \in \{\text{personnel, drone, wildfire, mobility, surge}\}$ into portfolio selection:

$$\max_{y, n_r} (1 - \rho) \sum_{\xi} p_{\xi} Q^{(1)}(y, n_r; \xi) + \rho \min_{\xi} Q^{(1)}(y, n_r; \xi) \quad (48)$$

Interpretation: - First term: Expected value across scenarios weighted by probability p_{ξ} - Second term: Worst-case performance (min over all ξ) - $\rho = 0.3$: Weight on worst-case robustness

Setting $\rho = 0$ yields pure expected-value optimization. Setting $\rho = 1$ yields pure min-max robustness. The baseline $\rho = 0.3$ balances both objectives, penalizing portfolios that perform well on average but collapse under any single scenario.

B.6 Protection Function Derivation

Zone protection at time t under scenario ξ :

$$P_{z,\xi,t} = 1 - \exp \left(- \left[((\alpha_r X_{z,\xi,t})^a (\alpha_d d_{z,t,\xi} + \alpha_c c_{z,t,\xi} + \varepsilon)^b)^{1/(a+b)} + \alpha_b B_z(y) \right] \right) \quad (49)$$

Complementarity structure: The product $(\text{patrol})^a (\text{detection})^b$ ensures that if either term is near zero, total protection collapses. This captures the operational reality that cameras without patrol response cannot intervene, and patrol without detection cannot target unseen threats.

Diminishing returns: Exponents $a = 0.7$ and $b = 0.7$ (both < 1) ensure sublinear scaling. Doubling patrol effort increases $(\alpha_r X)^{0.7}$ by only $2^{0.7} \approx 1.62$, not 2.0.

Normalization: The outer exponent $1/(a + b)$ ensures the argument to the exponential scales appropriately regardless of chosen values of a and b .

Permanent boost: $\alpha_b B_z(y)$ adds a fixed term based on Stage 2 funded assets, representing structural improvements (outposts, road upgrades) that persist across all scenarios.

Floor value: $\varepsilon = 0.01$ prevents numerical issues when detection terms are exactly zero.

B.7 Recourse Constraints

The complete recourse problem (Stage 3) solves:

$$\max_{x,d,c} \Pi(y, \xi) = \sum_z \tilde{W}_z \bar{P}_{z,\xi} \quad (50)$$

subject to:

Patrol-hour cap:

$$\sum_z x_{z,r,t,\xi} \leq H_r^\xi \quad \forall r, t \quad (51)$$

Team r cannot exceed available hours under scenario ξ . Normal conditions: $H_r = 16$ hours/day (two 8-hour shifts). Reduced personnel scenario: $H_r^\xi = 0.65 \times 16 = 10.4$ hours/day.

Drone-sortie limit:

$$d_{z,t,\xi} \leq D_z^\xi \quad \forall z, t \quad (52)$$

Zone z receives at most D_z drone hours based on Stage 2 base placement and sortie duration (32-35 minutes effective, 60-70 minute recharge).

Sensor capacity:

$$c_{z,t,\xi} \leq C_z \quad \forall z, t \quad (53)$$

Sensor effort capped by Stage 2 installed assets.

Blocked-zone exclusion:

$$x_{z,r,t,\xi} = 0 \quad \text{if zone } z \text{ is blocked under } \xi \quad (54)$$

Wildfire and mobility disruption scenarios remove access to specific zones.

Team reassignment:

$$\sum_z |\Delta \text{assign}_{r,t,\xi}| \leq 0.20 \cdot n_r \quad \forall t \quad (55)$$

At most 20% of teams can be reassigned per time period, preventing unrealistic full-park reshuffling.

Layer III Mathematical Derivations

C.1 Gaussian Process Kernel Specification

The residual spatiotemporal field is modeled as a separable Gaussian Process:

$$\varepsilon(i, t) \sim \mathcal{GP}(0, k_s(i, i') \otimes k_t(t, t')) \quad (56)$$

The spatial kernel uses a Matérn-3/2 form:

$$k_s(i, i') = \sigma_s^2 \left(1 + \frac{\sqrt{3}d_{ij}}{\ell_s} \right) \exp \left(-\frac{\sqrt{3}d_{ij}}{\ell_s} \right) \quad (57)$$

where d_{ij} is the Euclidean distance between cells, $\sigma_s^2 = 0.0031$ is the spatial variance, and $\ell_s = 28.4$ km is the spatial lengthscale.

The temporal kernel:

$$k_t(t, t') = \sigma_t^2 \exp \left(-\frac{|t - t'|}{\ell_t} \right) \quad (58)$$

where $\sigma_t^2 = 0.0018$ and $\ell_t = 3.2$ months.

C.2 EnKF State-Space Formulation

State evolution:

$$x_{t+1}^{(j)} = \Phi x_t^{(j)} + Bu_t + w_t^{(j)}, \quad w_t^{(j)} \sim \mathcal{N}(0, Q) \quad (59)$$

Transition matrix:

$$\Phi = \text{diag}(\phi) + (I - \text{diag}(\phi)) \cdot W_{\text{std}} \quad (60)$$

Conductance-adjusted weights:

$$W_{\text{std},ij} = \frac{w_{ij}^* \cdot c_j}{\sum_{k \in N(i)} w_{ik}^* \cdot c_k} \quad (61)$$

where salt pan cells have $c_j = 0.1$, others have $c_j = 1.0$.

Process noise: $Q = 0.00056 \cdot I_{940}$. Observation noise: $R = 0.00012 \cdot I_{940}$.

C.3 Covariance Localization

Gaspari-Cohn function with cutoff $c = 50$ km:

$$L_{ij} = \begin{cases} 1 - \frac{5}{3} \left(\frac{d_{ij}}{c}\right)^2 + \frac{5}{8} \left(\frac{d_{ij}}{c}\right)^3 + \frac{1}{2} \left(\frac{d_{ij}}{c}\right)^4 - \frac{1}{4} \left(\frac{d_{ij}}{c}\right)^5 & 0 \leq d_{ij} \leq c \\ 4 - 5 \left(\frac{d_{ij}}{c}\right) + \frac{5}{3} \left(\frac{d_{ij}}{c}\right)^2 + \frac{5}{8} \left(\frac{d_{ij}}{c}\right)^3 - \frac{1}{2} \left(\frac{d_{ij}}{c}\right)^4 + \frac{1}{12} \left(\frac{d_{ij}}{c}\right)^5 - \frac{2}{3} \left(\frac{c}{d_{ij}}\right) & c < d_{ij} \leq 2c \\ 0 & d_{ij} > 2c \end{cases} \quad (62)$$

C.4 Panel Regression (SLX)

$$\begin{aligned} V_{i,t} = & \alpha_i + \beta_1 \text{drought}_{i,t} + \beta_2 \text{burned_lag1}_{i,t} + \beta_3 \text{dry_season}_t \\ & + \theta_1 (W \cdot \text{drought})_{i,t} + \theta_2 (W \cdot \text{burned_lag1})_{i,t} \\ & + \beta_4 \sin\left(\frac{2\pi t}{12}\right) + \beta_5 \cos\left(\frac{2\pi t}{12}\right) + \beta_6 t + \varepsilon_{i,t} \end{aligned} \quad (63)$$

Estimated coefficients: $\beta_1 = 0.0342$, $\beta_2 = 0.0189$, $\beta_3 = 0.0156$, $\theta_1 = 0.0127$, $\theta_2 = 0.0093$. Residual $\sigma_\varepsilon = 0.0237$, adjusted $R^2 = 0.671$.

C.5 Staffing Bridge

Protection capacity scales with staffing:

$$P_{\text{cap}}(i, s) = 1 - (1 - P_{\text{cap}}(i, 295)) \cdot \left(\frac{295}{s}\right)^{0.7} \quad (64)$$

The exponent 0.7 reflects diminishing returns. Minimum staffing:

$$s_t^* = \min\{s \in \{100, 150, 200, 250, 295, 350, 400\} : \text{SPR}_t(s) \geq 0.90\} \quad (65)$$

C Supplementary Figures

Layer I Supplementary Figures

B.1 Consolidated Guild Parameters

Table 19: Complete guild parameter specification for all Layer I computations. Columns: ω_g = guild weight in SPR aggregation (sum to 1.00 across guilds); S_g = conservation urgency scalar applied to ecological value; τ_g^* = protection-capacity threshold required for protected classification within each guild; $\alpha_g, \beta_g, \gamma_g$ = Cobb-Douglas exponents for NDVI, water access, and habitat intactness (each row sums to 1.00); $\delta_{\text{conn},g}$ = connectivity discount controlling how strongly circuit-theory connectivity reduces vulnerability; s_g^{det} = detection sensitivity controlling saturation with surveillance effort. Values are held constant across seasons; seasonality enters through covariate inputs (NDVI, drought, fire), not parameter drift.

Guild	ω_g	S_g	τ_g^*	α_g	β_g	γ_g	$\delta_{\text{conn},g}$	s_g^{det}
Megafauna	0.35	0.80	0.75	0.30	0.45	0.25	0.20	1.20
Predators	0.25	0.70	0.65	0.35	0.30	0.35	0.18	1.00
Herbivores	0.20	0.40	0.55	0.45	0.30	0.25	0.15	0.90
Birds	0.10	0.75	0.55	0.40	0.20	0.40	0.22	0.60
Habitat fauna	0.10	0.45	0.60	0.35	0.25	0.40	0.10	0.70

ω_g : guild weight (sum to 1.00); S_g : urgency scalar; τ_g^* : protection threshold; $\alpha_g, \beta_g, \gamma_g$: Cobb-Douglas exponents (sum to 1.00 per guild); $\delta_{\text{conn},g}$: connectivity discount; s_g^{det} : detection sensitivity.

B.2 Seasonal Threat Channel Comparison

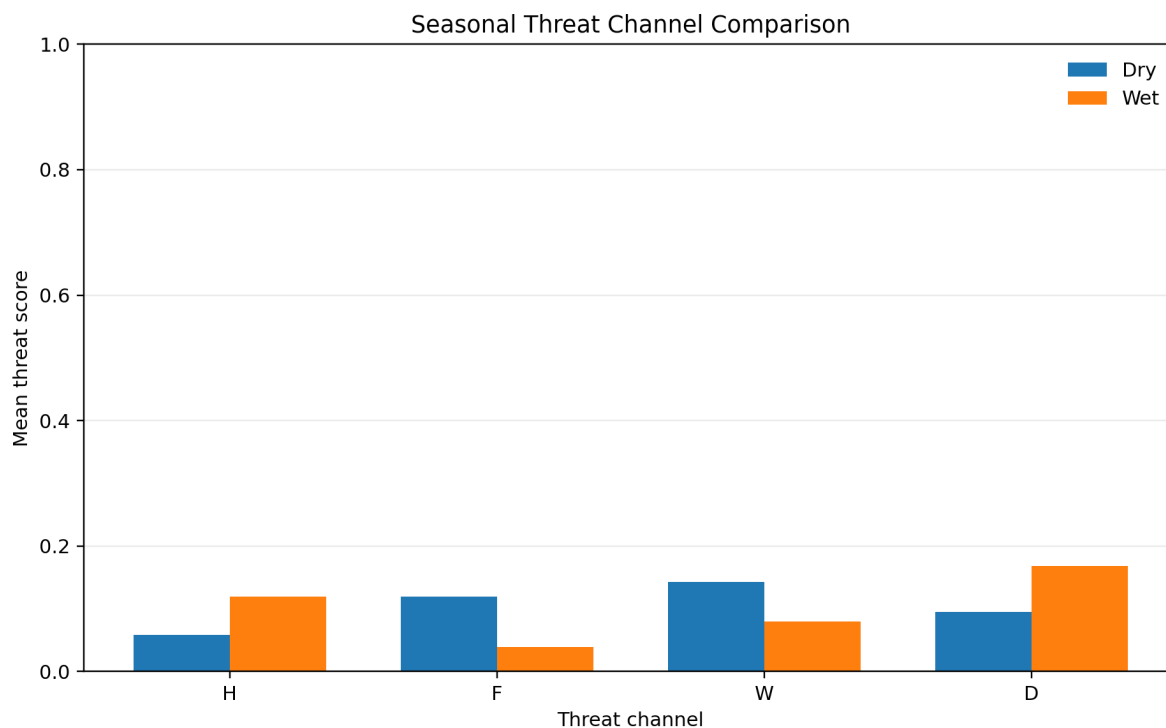


Figure 12: Mean threat channel scores (park-wide averages across all 158 in-park cells) by season. Blue = dry season (May–October); orange = wet season (November–April). Human access pressure (H) is season-invariant by model design, reflecting chronic non-seasonal exposure. Fire hazard (F) and water-concentration risk (W) are elevated in dry season because reduced rainfall concentrates wildlife near permanent water points and increases fuel dryness. Habitat degradation (D) rises slightly in wet season because vegetation-stress contrast increases detectability in greenness indices. Interpret these park-wide means with Table 16, which shows that cells failing the all-guild threshold are dominated by chronic human pressure in both seasons — a management pattern masked by park-wide averaging.

B.3 Protection Capacity Spatial Pattern

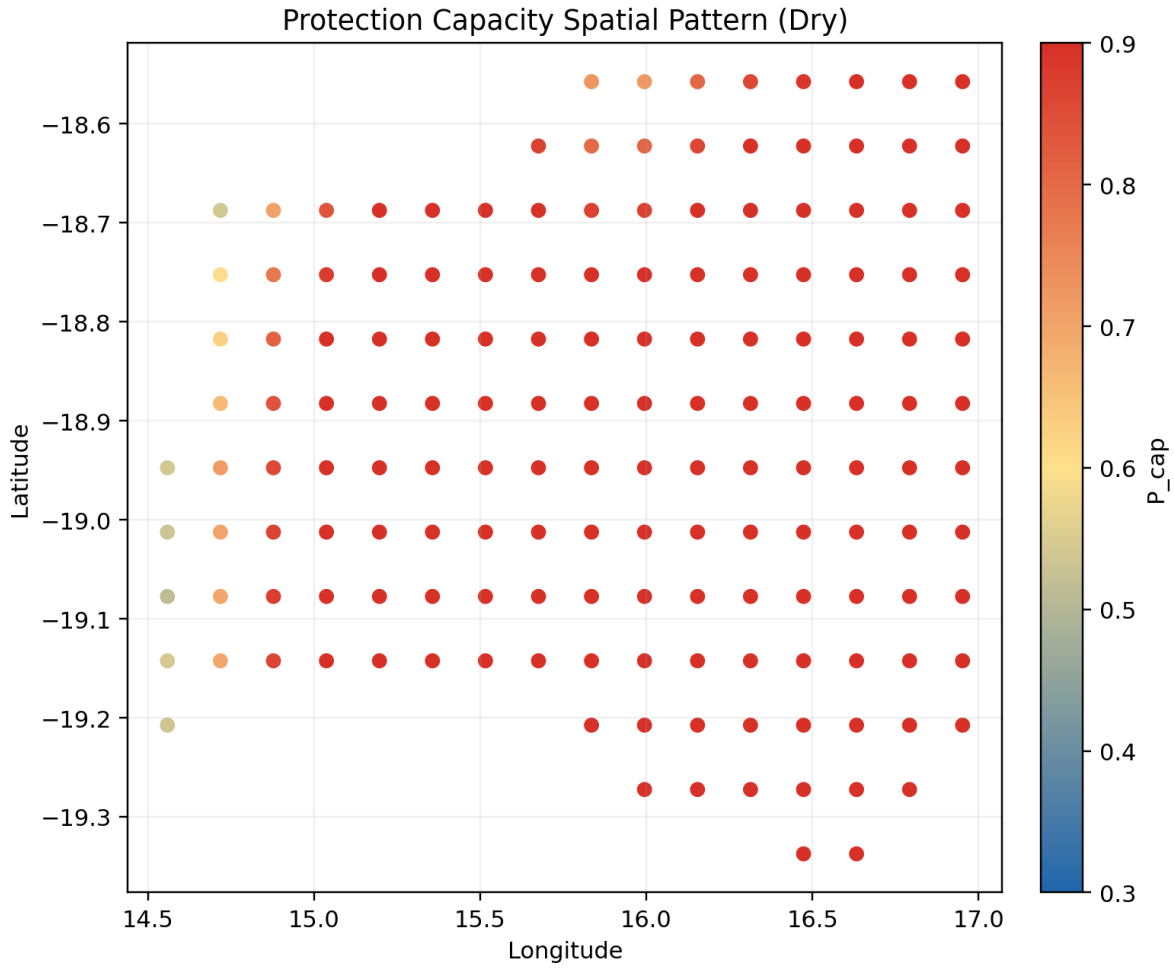


Figure 13: Dry-season protection capacity $P_{cap}(g, i)$ averaged across five guilds for the 158 in-park cells. The color gradient runs from deep blue ($P_{cap} \approx 0.3$) to dark red ($P_{cap} \approx 0.9$). The dominant east-west gradient is driven by response probability $p^{resp}(i)$: eastern and northern cells are typically within 1–2 hours of Okaukuejo or Halali, while western Dolomite-adjacent cells often exceed 3-hour travel times, collapsing response reliability even when detection effort is high. Salt-pan cells (lower-left cluster near 14.5°E, 19.3°S) show moderate P_{cap} yet classify as protected because low ecological value $E(g, i)$ keeps residual vulnerability $V(g, i) = E \cdot T \cdot (1 - P_{cap})$ below threshold. This pattern motivates Layer II circuit-theory access diagnosis: the key bottleneck is structural reachability, not patrol willingness.

B.4 Priority Score Distribution

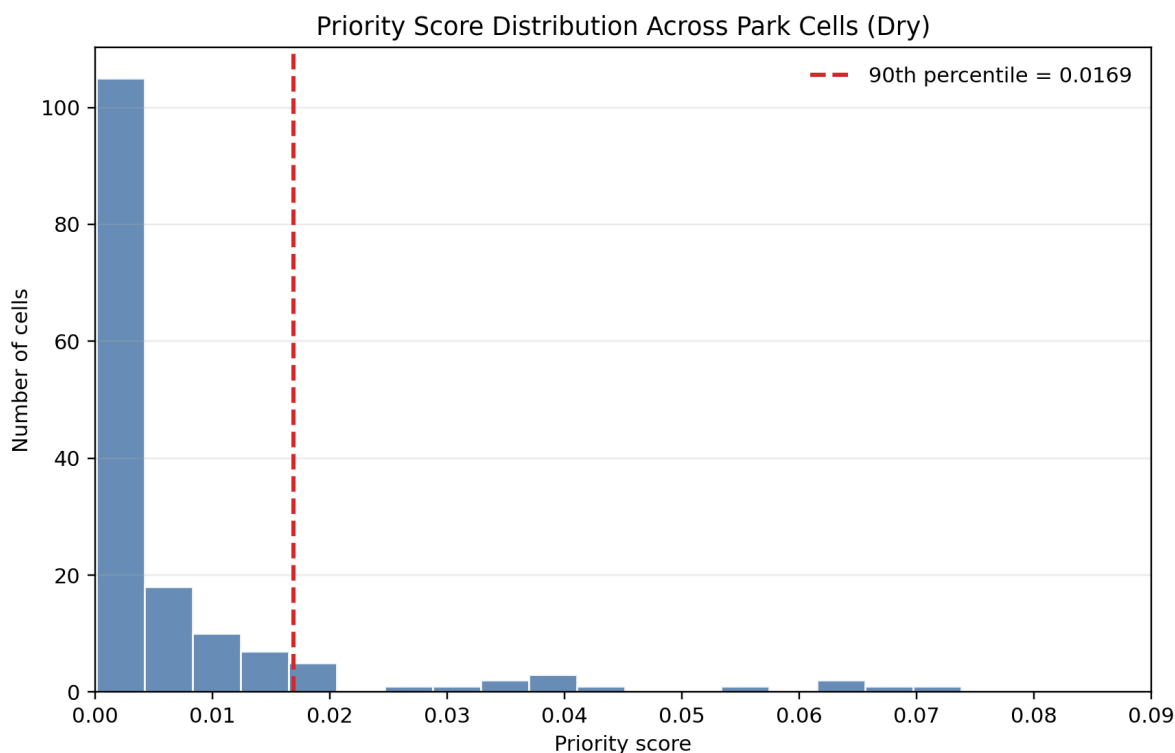


Figure 14: Histogram of dry-season cell-level priority scores $\text{priority_score}(i)$ across 158 in-park cells. The distribution is strongly right-skewed: 105 cells (66%) are below 0.005, indicating adequate or near-adequate protection, while 16 cells (10%) exceed the 90th-percentile threshold of 0.0169 (dashed red line). The top five western-interior cells (Table 14) score 0.051–0.071, roughly 7–10 times park median urgency. This concentration implies that targeting fewer than 10 cells (about 6% of the park) can address a large share of total protection deficit, directly motivating Layer II concentrated asset placement. Wet-season scores (Table 13) shift rightward, with the 90th percentile rising to 0.0228 and mean priority increasing by 25%.

B.5 Seasonal Protection Summary

Table 20: Summary of seasonal protection outcomes across 158 in-park cells. Protected cells are those satisfying the strict all-guild threshold rule (Eq. 39). SPR is the park-level weighted protected fraction (Eq. 40). Mean, median, and 90th-percentile priority statistics summarize urgency distribution for $\text{priority_score}(i)$ (Eq. 5). Wet season degrades all metrics: six additional cells fail the all-guild rule, SPR drops by 3.8 percentage points, and the 90th-percentile priority rises by 35%, reflecting compounded fire and access pressure impacts on effective protection capacity.

Metric	Dry	Wet	Change
Protected cells	143 / 158	137 / 158	−6
SPR	0.890	0.856	−3.8%
Mean priority score	0.0072	0.0090	+25%
Median priority score	0.0023	0.0026	+13%
90th percentile priority	0.0169	0.0228	+35%

Wet season shows increased vulnerability: fewer protected cells, lower SPR, and elevated priority scores across all percentiles.

B.6 Top Priority Cells

Table 21: Five highest-priority dry-season cells ranked by $\text{priority_score}(i)$. All five occupy column 3 (longitude $\approx 14.9\text{--}15.1^\circ\text{E}$) of the 20×47 grid, spanning rows 11–15 (latitude $\approx -19.1^\circ$ to -19.4°S) in the western interior near Galton Gate/Dolomite sector. Challenge scores (Eq. 41) are 2–5 times park median and are driven by high megafauna ecological value, persistent human-access pressure, and response travel times beyond 2 hours. None reaches protected status in either season, making these cells primary targets for Layer II outpost and drone-base siting (Figure 6).

Cell ID	Priority	Challenge	Location (row, col)	Protected?
cell_661	0.0709	0.1521	(14, 3)	No
cell_614	0.0670	0.1475	(13, 3)	No
cell_708	0.0618	0.1436	(15, 3)	No
cell_520	0.0604	0.1418	(11, 3)	No
cell_567	0.0513	0.1176	(12, 3)	No

These cells carry challenge scores 2–5 times the park median and remain unprotected in both seasons.

B.7 Zone-Level Protection Breakdown

Table 22: Zone-level protection outcomes by zone type and season. Zone SPR is the weighted ecological fraction meeting all-guild thresholds within each zone class. Mean Priority and Mean Challenge summarize urgency distributions within zone. Salt-pan and boundary zones achieve $\text{Zone SPR} = 1.000$ in both seasons because intrinsic ecological value $E(g, i)$ is low, so moderate P_{cap} still yields sub-threshold residual vulnerability. All observed failures occur in interior savanna, which combines highest ecological value per cell with lowest response reliability due to road-network distance from ranger bases.

Season	Zone	Cells	Zone SPR	Mean Priority	Mean Challenge
Dry	Boundary	2	1.0000	0.00411	0.00000
Dry	Interior	104	0.8591	0.01059	0.01207
Dry	Salt pan	52	1.0000	0.00095	0.00000
Wet	Boundary	2	1.0000	0.00397	0.00000
Wet	Interior	104	0.8613	0.00900	0.01086
Wet	Salt pan	52	1.0000	0.00076	0.00000

Salt pan and boundary zones achieve perfect protection in both seasons; all failures occur in interior savanna.

B.8 Threat Dominance Context

Table 23: Dominant threat channel by context and season, reported as the share of cells (or cell-season records) where a channel contributes the largest increment to integrated threat $T(g, i)$. Results are separated for park-wide averages and for the failing-cell subset under the all-guild rule. The divergence is operationally critical: park-wide means suggest water concentration (dry) and degradation (wet) dominance, but among actual failing cells chronic human/poaching pressure dominates in both seasons (60% of failures). Strategies based only on park-wide means therefore under-prioritize access-control interventions in the western interior.

Context	Dominant Channel	Share (%)
Dry, park-wide	Water concentration	58.9
Dry, park-wide	Fire hazard	35.4
Wet, park-wide	Habitat degradation	84.8
Dry, failing cells	Human / poaching	60.0
Dry, failing cells	Habitat degradation	40.0
Wet, failing cells	Human / poaching	60.0
Wet, failing cells	Habitat degradation	40.0

Park-wide averages emphasize seasonal water and fire stress, but cells that fail the all-guild protection rule are consistently driven by chronic access pressure and habitat degradation in the western interior.

B.9 Scenario Performance and Cost-Effectiveness

Table 24: Park-wide operational protection score Π across all eight Layer II scenarios (full appendix version of main-text Table 3). Scenario-probability-weighted expected protection is $\mathbb{E}[\Pi] = \sum p_{\xi} \Pi(y, \xi) = 0.272$. The difference between baseline Π (0.310) and $\mathbb{E}[\Pi]$ quantifies the expected uncertainty penalty, about 12% below the certainty-equivalent outcome.

Scenario	Probability	Π	Annual Cost	$\Delta\Pi$ vs. Baseline
Normal conditions	0.35	0.310	\$402,300	—
Fewer rangers (65%)	0.10	0.201	\$282,300	−35%
Drones break (50%)	0.10	0.307	\$366,300	−1%
Poaching surge	0.10	0.311	\$402,300	+0.3%
Roads blocked	0.10	0.310	\$402,300	0%
Wildfire (30%)	0.10	0.177	\$402,300	−43%
Rangers + roads	0.08	0.201	\$282,300	−35%
Insider threat	0.07	0.199	\$246,300	−36%

Expected protection: $\mathbb{E}[\Pi] = \sum_{\xi} p_{\xi} \Pi(y, \xi) = 0.272$.

B.10 Parameter Tables

Table 25: Patrol operations parameters used in Layer II recourse constraints and Layer I response-capacity calculations. Road-class speeds (25/15/10 km/h) determine travel time $\tau(i)$ and therefore response probability $p^{\text{resp}}(i)$ through the logistic mapping in Eq. (A.10). The 20% maximum reassignment cap (last row) prevents unrealistic single-period park-wide reshuffling in recourse optimization.

Parameter	Value	Source/Justification
Patrol shift length	8 hours	Standard sustainable scheduling
Shifts per day	2	Day and night coverage
Speed—gravel roads	25 km/h	Modeling assumption used in patrol-time calculations
Speed—dirt tracks	15 km/h	Modeling assumption for reduced speed on low-quality roads
Speed—degraded roads	10 km/h	Wildfire/flood scenario
Max reassignment per period	20%	Prevents unrealistic reshuffling

Table 26: Technology performance parameters for drones, cameras, and acoustic sensors. Drone sortie and recharge times imply a maximum of three sorties per day and set the Eq. (B.7.2) sortie bound. Camera detection radius and downtime rates determine per-camera detection increments in Eq. (A.9). Acoustic sensor trigger radius and false-alarm rates inform sensor effort $c_{z,t,\xi}$ and noise penalties in the protection function. Parameters are fixed planning values; Section 6 tests drone detection sensitivity (0.2% vs. 20%) and finds near-zero impact on Π , indicating robustness to this uncertainty.

Technology	Parameter	Value
Drone (DJI Matrice 300 RTK)	Real-world sortie time	32-35 min
	Battery recharge	60-70 min
	Max sorties per day	3
	Operators required	2
Fixed cameras	Detection radius	20-50 m
	Downtime rate	10-15%
Acoustic sensors	Trigger radius	300-800 m
	False alarm rate	15-20%

Table 27: Capital and annual operating costs for each intervention class, used in Stage 2 mixed-integer optimization (Eq. B.5). Capital costs F_i enter the capital-budget constraint and operating costs O_i enter the annual operations constraint. At baseline \$750K capital, the selected mix includes 55 cameras (\$44K), 48 sensors (\$57.6K), 6 drone bases (\$270K), and 2 outposts (\$240K), with remaining headroom constrained by operational feasibility. Road upgrades (\$15K/km) and AI detection (\$35K) are not selected at baseline because lag-adjusted benefit per dollar is lower than cameras/sensors; they become competitive in the \$1.5M sensitivity case.

Item	Capital (F_i)	Annual Operating (O_i)
Ranger outpost	\$120,000	\$25,000
Drone base	\$45,000	\$18,000
Waterhole camera	\$800	\$300
Acoustic sensor	\$1,200	\$500
Road upgrade (per km)	\$15,000	\$2,000
Communication relay	\$8,000	\$1,500
AI detection upgrade	\$35,000	\$8,000
Patrol team (5 rangers)	—	\$60,000

Table 28: Implementation lag L_i and lag-discount factor $\delta_i = 1/(1 + \lambda L_i)$ for each intervention type, with $\lambda = 0.2$. Effective benefit is $\tilde{B}_i = \delta_i B_i$, so delay directly erodes planning-horizon competitiveness. For example, a 6-month outpost lag retains only 45.5% of undiscounted benefit, while camera deployment retains about 91%, which explains sparse baseline outpost/road selections despite their potential long-run value.

Intervention	Lag L_i (months)	δ_i (baseline $\lambda = 0.2$)	Effective Benefit
Camera/sensor	0.5	0.909	91% of B_i
Drone base	1.5	0.769	77% of B_i
Communication relay	2.0	0.714	71% of B_i
Road upgrade	3.0	0.625	63% of B_i
Ranger outpost	6.0	0.455	46% of B_i

Table 29: Baseline calibration parameters for Layer II optimization and the protection function. Robustness weight $\rho = 0.30$ blends 70% expected and 30% worst-case performance; $\rho = 0$ gives pure expected-value optimization and $\rho = 1$ gives minimax behavior. Cobb-Douglas exponents $a = b = 0.7$ encode diminishing returns (doubling patrol effort yields about $2^{0.7} \approx 1.62$ increase). Floor term $\varepsilon = 0.01$ prevents numerical collapse when technology effort is zero. Parameters were cross-validated against Layer I outputs and stress-tested in Section 6.

Parameter	Baseline	Meaning
κ (dark-zone amplification)	1.0	Boosts priority for high- U_z zones
λ (lag discount steepness)	0.2	Controls benefit decay with delay

Parameter	Baseline	Meaning
β_τ (response weight)	0.15	Response time improvement value
β_p (detection weight)	0.40	Detection probability value
β_q (coverage weight)	0.35	Coverage area value
β_R (resistance weight)	0.10	Structural improvement value
ρ (robustness weight)	0.30	Worst-case penalty in Stage 2
α_r (patrol effectiveness)	1.0	Patrol contribution to P
α_d (drone effectiveness)	0.8	Drone contribution to P
α_c (sensor effectiveness)	0.6	Sensor contribution to P
α_b (structural boost)	0.5	Stage 2 asset carry-forward
a, b (diminishing returns)	0.7	Sublinear scaling exponents
ε (baseline floor)	0.01	Prevents zero-division

Table 30: Disruption scenarios used in robust Stage 2 optimization and Stage 3 recourse evaluation. Scenario set spans personnel loss, technology failure, environmental disruption, and adversarial pressure; probabilities sum to 1.00. The first five scenarios enter Stage 2 blended-objective optimization, while all eight are evaluated in recourse and reported in Table 3/Table 17. Design is grounded in documented ENP events, including 2022 insider investigations, the September 2025 wildfire, and 2024 poaching-surge reporting.

Scenario	Prob.	Description
Reduced personnel	0.10	Staff at 65% baseline (motivated by June 2022 insider investigation)
Technology outage	0.10	50% drone availability (maintenance or authorization)
Mobility disruption	0.10	Patrol speed reduced to 10 km/h, some sectors unreachable
Surge threat	0.10	Poaching pressure increased 1.5× at waterhole hotspots
Wildfire	0.10	30% of zones removed (2025 fire footprint), patrol speed 10 km/h
Night-heavy threat	0.08	60-70% patrol capacity biased to night shifts
Rangers + roads	0.08	Personnel at 65% AND mobility disruption simultaneously
Insider threat	0.07	Further staff reduction to 65% via infiltration

Layer III Supplementary Figures

B.11 Spatial Persistence Map

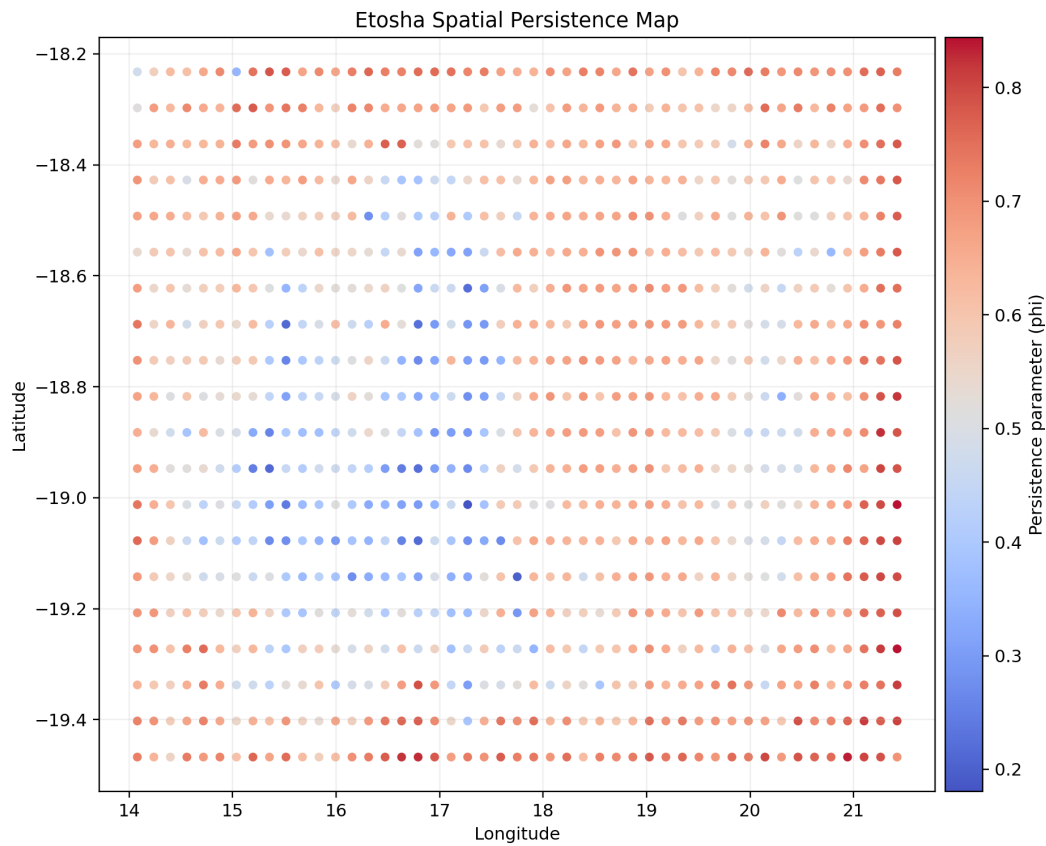


Figure 15: Cell-level temporal persistence parameter ϕ_i from the EnKF transition matrix (Eq. 8), shown across all 940 grid cells including buffer. Warm colors ($\phi_i > 0.6$) indicate strong month-to-month vulnerability carry-over, while cool colors ($\phi_i < 0.4$) indicate rapid post-shock recovery. Highest persistence clusters along eastern and northeastern boundary zones, consistent with chronic access pressure; lowest persistence appears in salt-pan interior where sparse ecology naturally resets risk signals. Park-wide mean persistence is 0.5893, motivating the mixed self-lag + neighbor transition structure used in Eq. 8.

B.12 Ensemble Calibration

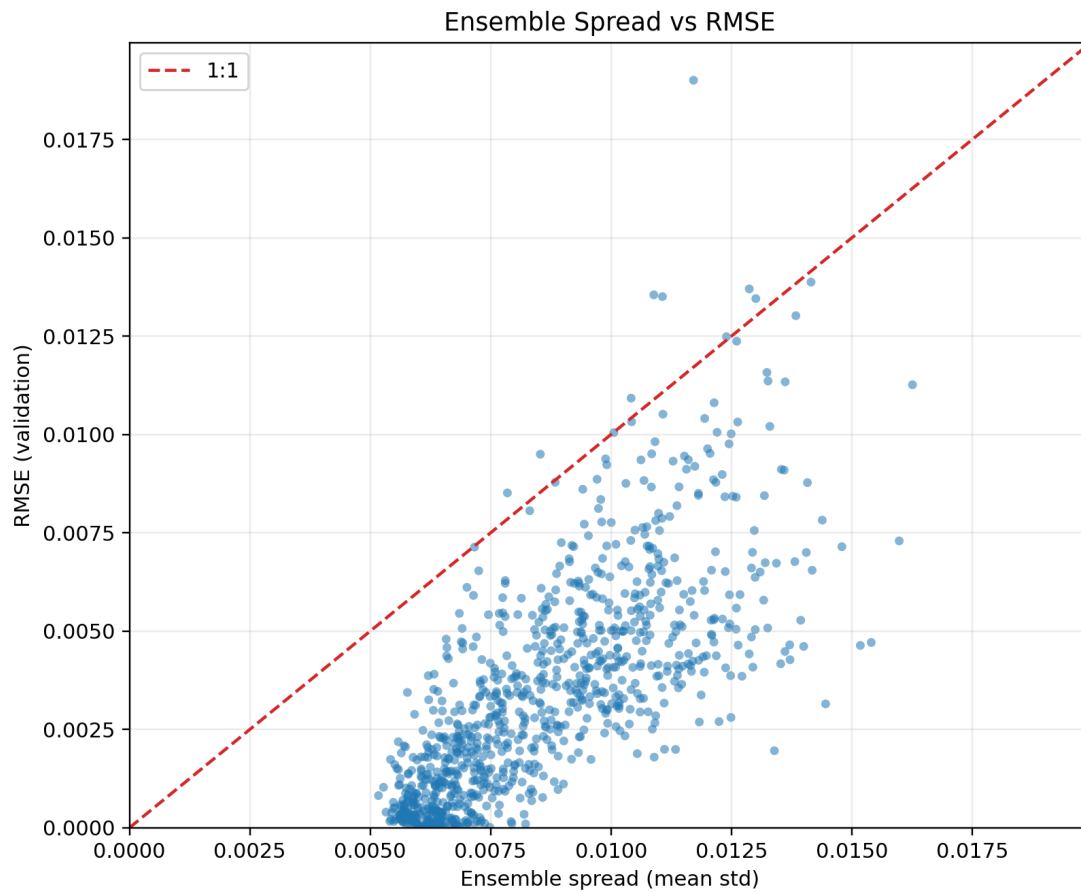


Figure 16: Ensemble calibration diagnostic: cell-level ensemble spread (mean standard deviation across 500 members, x-axis) versus RMSE on the 12-month holdout set (months 61–72, y-axis). Perfect calibration lies on the dashed 1:1 line; points below imply overconfidence and points above imply overdispersion. Most in-park cells cluster near 1:1, indicating reliable uncertainty representation at short-to-medium horizons. A smaller high-spread cluster (≥ 0.013) corresponds to highest-priority western-interior cells where drought sensitivity and structural isolation amplify uncertainty, consistent with Layer I priority ranking.

B.13 All Five Scenarios

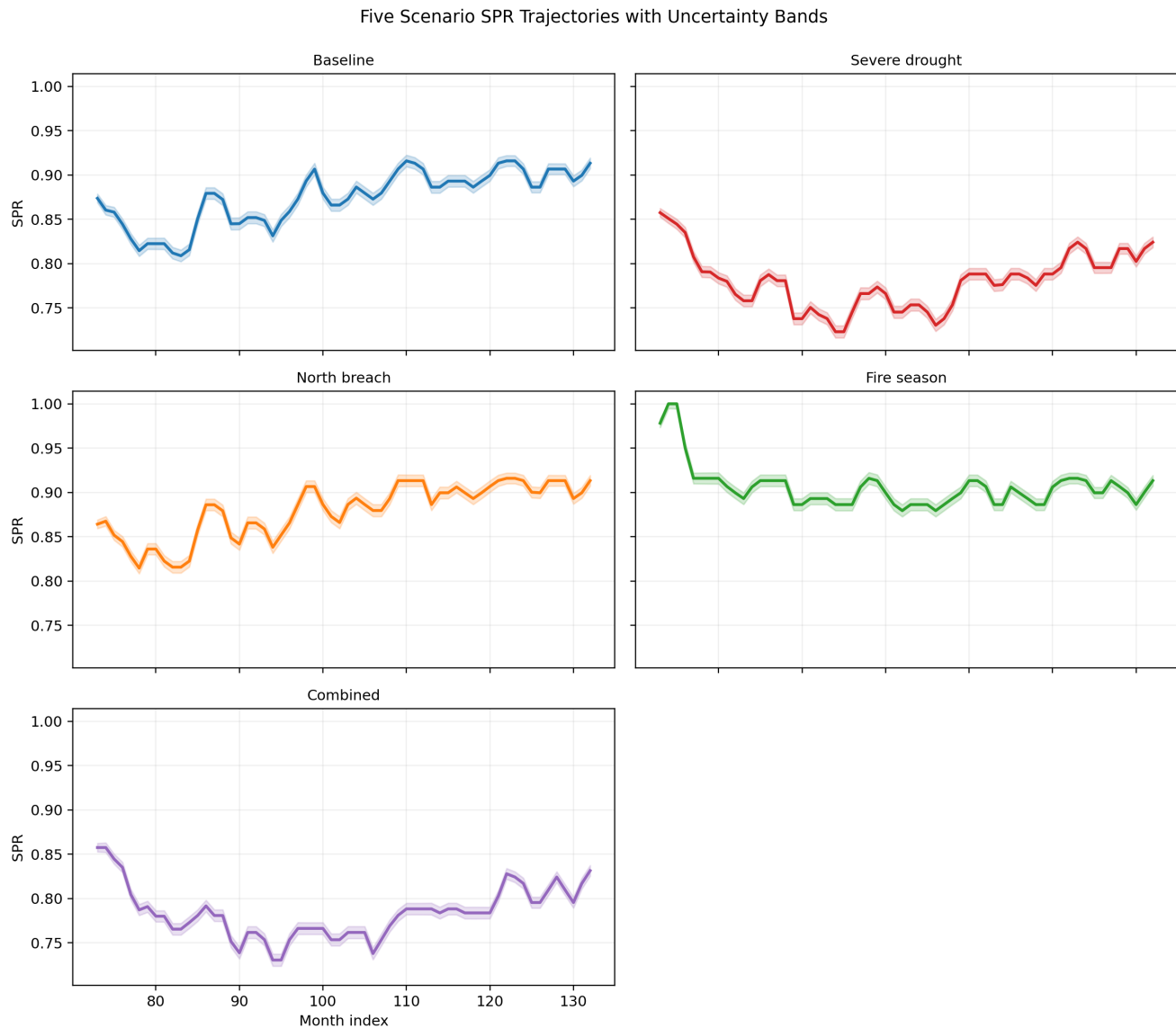


Figure 17: Complete 60-month SPR trajectories with ± 1 ensemble standard-deviation bands for all five Layer III scenarios. X-axis is forecast month index (month 73 = January 2025; month 132 = December 2029). Baseline oscillations reflect seasonal cycling with upward long-run trend from portfolio effects. Severe drought remains below 0.85 with widening uncertainty after month 100. Northern-boundary breach remains close to baseline, indicating strong perimeter absorption by the existing sensor/camera mix. Fire-season repeat shows an acute year-1 hit followed by recovery within about 8–10 months. Combined drought + breach closely tracks severe drought, confirming drought as the dominant compound-shock driver.

B.14 Validation Metrics Table

Table 31: Layer III holdout validation metrics across forecast horizons, evaluated on the 12-month holdout set (months 61–72; January–December 2024). RMSE = root mean squared error; MAE = mean absolute error; Pearson r = correlation between ensemble-mean forecast and held-out Layer I rerun values. Metrics are computed at park-level SPR aggregate. Skill is strong at short horizons (3-month $r = 0.9954$, RMSE = 0.003) and degrades gradually at longer horizons (24-month $r = 0.9589$, RMSE = 0.011), consistent with widening spread in Figure 16. A 24-month RMSE of 0.011 (about ± 1.1 SPR points) is much smaller than the 8.9-point baseline-vs-drought month-60 gap, confirming that scenario separation exceeds forecast uncertainty at management-relevant horizons.

Horizon	RMSE	MAE	Pearson r
3 months	0.00312	0.00198	0.9954
6 months	0.00435	0.00278	0.9921
12 months	0.00689	0.00441	0.9834
18 months	0.00912	0.00587	0.9712
24 months	0.01134	0.00723	0.9589

Training: months 1–60. Testing: months 61–72.

D Tables

C.0 Global Cross-Layer Assumptions

Section 2 reports the compact G1–G4 global assumption structure in Table 2.

C.1 Full Layer II Assumptions Table

Table 32: Full Layer II structural assumptions referenced in Section 4.2. Each assumption is stated as a planning-level claim (not an empirical observation) and followed by its operational consequence for the optimization model. Assumptions A1, A4, and A9 are the most influential: patrol speed by road class sets effective response radius for reachability; effective resistance captures multi-path isolation that straight-line distance misses; and the five-ranger two-shift team structure determines how patrol-hour budgets map into zone coverage. Assumption A14 (permanent assets carry a fixed boost) ensures Stage 2 portfolio investment is not negated by adverse scenario recourse.

Assumption	Justification
A1 Patrol speeds vary by road class (25 / 15 / 10 km/h)	Speed by road class sets the effective response radius for each zone. Zones that cannot be reached within the response-time threshold under any road class are treated as unreachable for that scenario, regardless of how many rangers are available.
A4 Effective resistance proxies structural isolation	A zone with high effective resistance to all ranger bases is structurally isolated along every possible path through the network. Circuit theory captures this multi-path isolation whereas straight-line distance does not — a zone 10 km away on rough track may be harder to reach than one 30 km away on a maintained road.
A9 Patrol teams consist of 5 rangers on two 8-hour shifts per day	Each team provides 16 available patrol hours per day across day and night shifts. Under the personnel-loss scenario this drops to 10.4 hours, reflecting a 35% staffing reduction. This structure determines how patrol effort is allocated across zones and time periods.
A13 Waterhole cameras have elevated detection value during nighttime hours	Incident concentration near waterholes is highest at night, consistent with poaching behavior patterns documented in the Etosha context. This increases the effective detection weight for waterhole cameras in calibration. It is a modeling assumption, not a directly observed parameter.
A14 Permanent structural assets carry a fixed protection boost across all scenarios	Funded outposts and road upgrades contribute a baseline protection term $B_z(y)$ that persists regardless of which disruption scenario is realized. Mobile resources (patrol teams, drones) are reallocated scenario by scenario; permanent assets are treated as always operational once deployed.

Complete Variable and Constant Glossary (Models 1-3)

Table 33: Unified variable glossary for Models 1–3, grouped by functional role. Variables are listed with the symbol(s) used in text, equations, and code. Where multiple notational forms appear across layers (e.g., $T(g, i)$ vs. T_z vs. T_k), all forms are listed together with scope explained. Readers can cross-reference this table to identify unfamiliar symbols without searching the full manuscript.

Variable(s)	What It Means
A. Core Indices and Sets	
i, z	Spatial unit index (cell/zone index used across layers).
g	Guild index (ecological guild category).
t	Time index (season, month, or planning period).
ξ	Scenario index (uncertainty branch).
k, m	Generic component indices (covariates, channels, or subcomponents).
G, Z, T, Ξ	Sets of guilds, zones/cells, time periods, and scenarios.
B. Ecological Value and Suitability Variables	
$E(g, i), E_{g,i}$	Ecological importance/value of guild g in cell i .
$E_{\text{weighted}}(i)$	Guild-weighted ecological value at cell i .
$O(g, i), O_{g,i}$	Habitat overlap / expected habitat use for guild g in cell i .
$X_{g,i}^{\text{eco}}, X_{\text{eco}}(g, i)$	Environmental suitability term for guild g in cell i .
$X_{\text{ndvi}}(i), X_{\text{water}}(i), X_{\text{intact}}(i)$	NDVI, water-access, and habitat-intactness suitability components.
$H_{\text{cov}}(g, i), H_{\text{occ}}(g, i)$	Covariate-based and occurrence-based suitability sub-scores.
$x_k(i)$	Generic habitat covariates for cell i .
$n_{\text{occ}}(g, i)$	Species occurrence count/density proxy for guild g near cell i .
$lc_{\text{mod}}(g, i)$	Land-cover modifier applied to suitability for guild g .
C. Threat Channel Variables	
$H(g, i), H_i$	Human-access / human-pressure threat channel.
$F(g, i), F_i$	Fire hazard threat channel.
$W(g, i, t), W_i, Wc(g, i)$	Water-concentration / habitat-stress threat channel.
$I(g, i), I_i$	Habitat integrity state (used inversely for degradation).
$D(g, i), D_i$	Degradation or illegal-harm threat component (model-dependent representation).
$T(g, i), T_i, T_z, T_k$	Integrated threat state (guild-level, cell-level, or zone-level).
D. Protection, Detection, and Response Variables	
$Pcap(g, i), P_{\text{cap},g}(i)$	Protection capacity probability for guild g at cell i .

Variable(s)	What It Means
$P_{g,i}, P_{z,\xi,t}, \bar{P}_{z\xi}$	Protection outcome variables (guild/cell/zone/scenario-time forms).
$p_{g,i}^{det}, p_{det}(g,i)$	Detection probability for threats affecting guild g in cell i .
$p_i^{resp}, p_{resp}(i)$	Response probability for timely operational response in cell i .
$E_{tot}(i)$	Combined operational effort driving detection capacity in cell i .
$\tau(i), \tau_i, \tau_z, \tau'_z$	Response/travel-time variables before and after interventions.
$d_{base}(i)$	Distance from cell i to nearest ranger base.
e_{cam}, e_{drone}	Camera/drone effort inputs used in protection calculations.
I_i^{prot}	Binary protected/under-protected classification indicator.
E. Vulnerability, Risk, and Priority Variables	
$V(g,i), V_i, V(i,t)$	Vulnerability/priority state variable (static or dynamic form).
$V_{adj}(g,i)$	Connectivity-adjusted vulnerability for guild g in cell i .
$R_i, R_z, R_z^{eff}, R_z^{norm}$	Residual risk/resistance diagnostics (cell or zone forms).
U_z	Under-service (dark-zone) score from circuit/service diagnostics.
S_z, S_i	Zone urgency score and intervention urgency score.
$Priority(i)$	Final management priority score for cell i .
$\Pi, \Pi(\xi), \mathbb{E}[\Pi], \Pi(y,\xi)$	Park-level protection objective (scenario-specific and expected forms).
SPR, SPR_t	Species Protection Rate summary metric (static or time-indexed).
$P(SPR_t < \tau)$	Probability that protection rate falls below threshold τ .
F. Intervention and Decision Variables	
y_i	Binary decision variable: choose intervention i or not.
$B_i, \Delta B$	Strategic benefit / marginal benefit of intervention i .
$N(i)$	Set of zones influenced by intervention i .
$\Delta\tau_{z,i}, \Delta p_{z,i}, \Delta q_{z,i}, \Delta R_{z,i}$	Improvement channels from intervention i on zone z (response, detection, coverage, structure).
$X_{z\xi t}, d_{z\xi t}, c_{z\xi t}$	Scenario-time resource allocation decisions (patrol, drone, sensor/camera effort).
n_r, N_k	Patrol-team/staffing decision variables and equivalent staffing mappings.
$ICER_k, PGD_k, \Delta Cost_k, \Delta \Pi_k$	Cost-effectiveness variables for intervention/resource comparisons.
G. Spatial Network and Connectivity Variables	
$G = (N, E)$	Operational graph of nodes and edges.
$W, W_{std}, w_{ij}, W_{ij}$	Spatial weight/adjacency/conductance matrices or edge weights.
L, L_{ij}	Graph Laplacian matrix components.

Variable(s)	What It Means
$C(i), C_{ij}, C_{uv}$	Connectivity score and edge conductance terms.
v_i, v, V	Circuit potential/voltage variables in connectivity formulations.
I_{uv}	Edge current flow on graph edge (u, v) .
b	Right-hand-side source/sink vector in linear graph systems.
s_i	Spatial location coordinate for cell i .

H. Geometry and Access Covariates

$d_{wh}(i)$	Distance from cell i to nearest waterhole.
$d_{bnd}(i)$	Distance from cell i to park boundary.
$b_{access}(i)$	Boundary-access pressure index.
$prs(i)$	Poaching-risk covariate/proxy.
$r_{norm}(i)$	Normalized road-density covariate.
$f_{lc,j}(i)$ for $j \in \{20, 30, 50, 60, 80, \text{pan}\}$	Land-cover and landscape fraction covariates used in ecological/threat channels.
Z_1, Z_2, Z_3, Z_4	Integrity subcomponents (road, burn, boundary, vegetation related terms).

I. Grouped Parameter Variables (No Numeric Values)

ω_g	Guild weighting parameters used in ecological aggregation.
S_g	Guild conservation-urgency scalar parameters.
w_k^g	Guild-specific habitat-covariate weight parameters.
$\alpha_g, \beta_g, \gamma_g$	Guild ecological-exponent parameters in suitability/ecological-value terms.
$a_{g,m}$	Guild-specific integrity exponent parameters.
$\delta_g, \delta_g^{conn}$	Guild shift/discount parameters (threat shift and connectivity discount).
s_g, s_g^F, s_g^{det}	Guild sensitivity parameters (fire/detection and related channels).
σ_g, σ_w	Spatial decay parameters (guild-level and water-access scales).
$\theta_{pat}, \theta_{cam}, \theta_{drone}$	Effort-mapping parameters for patrol/camera/drone in protection capacity.
τ^*, K	Response sigmoid center and steepness parameters.
η_0, η_1	Drought/concentration channel shape parameters.
$\beta_{road}, \beta_{bnd}, \beta_{prs}$	Human-threat regression coefficients (road/boundary/poaching covariates).
$\gamma_0, \gamma_f, \gamma_n, \gamma_r, \gamma_d, \gamma_{season}$	Fire-hazard model coefficients (intercept, fire, NDVI, rain, distance, season effect).
κ, γ, U^*	Under-service boost and dark-zone emphasis parameters.
$\beta_\tau, \beta_p, \beta_q, \beta_R$	Stage-1 intervention scoring coefficients (response, detection, coverage, resistance channels).

Variable(s)	What It Means
$\alpha_r, \alpha_d, \alpha_c, \alpha_b$	Stage-2 protection-function effectiveness coefficients (ranger/drone/camera/baseline-structure channels).
a, b, ϵ	Stage-2 Cobb-Douglas shape and regularization parameters.
$p(\xi)$	Scenario probability weights.
A, B, H, Q, R, ϕ_i	State-space/EnKF transition, loading, observation, covariance, and persistence parameters.
τ	SPR risk threshold parameter used in exceedance probability outputs.

Complete Constants Table

Table 34: Unified constants table across Models 1–3, reporting values sourced from independent published or official references. Constants that rely only on internal IMMC repository files are flagged as *No independent source (IMMC-only)* and should be interpreted as planning assumptions rather than empirical facts. This distinction supports transferability: independently sourced constants can be ported directly to new parks, while IMMC-only constants require local recalibration.

Constant	Meaning	Value	Source (APA)
A. Model 2 Baseline / Context Constants			
A_{park}	Etosha park area for scaling and coverage density	22,270 km ²	(Wikipedia contributors, 2026; Encyclopaedia Britannica, 2019)
L_{road}	Road network length used for access and connectivity context	N/A	No independent source (IMMC-only)
$N_{\text{waterholes}}$	Documented waterholes used as monitoring chokepoints	N/A	No independent source (IMMC-only)
L_{fence}	Approximate park fence length for perimeter risk context	N/A	No independent source (IMMC-only)
$N_{\text{staff,total}}$	Total ministry staff count reported for Etosha operations	N/A	No independent source (IMMC-only)
$N_{\text{rhino,Etosha,2022}}$	Rhinos poached in Etosha in 2022	46	(Africa Geographic, 2023; Save the Rhino International, 2023)
$N_{\text{rhino,Namibia,2022}}$	Rhinos poached nationally in 2022	87	(Save the Rhino International, 2023)
$\rho_{\text{Etosha rhino share}}$	Share of Namibia black rhinos in Etosha	72%	(Africa Travel & Tourism Association, 2024)

Constant	Meaning	Value	Source (APA)
$N_{\text{rhino,Etosha,Q1-2024}}$	Rhinos poached in Etosha in first quarter 2024	19	(Africa Travel & Tourism Association, 2024; Reuters, 2024)
$\mathbb{K}_{\text{visitor drone ban}}$	Visitor drone prohibition status	In effect	(Africa Travel & Tourism Association, 2024; Windhoek Observer, 2024)
$\mathbb{K}_{\text{insider probe,2022}}$	MEFT internal poaching investigation in Etosha (June 2022)	Confirmed	(Save the Rhino International, 2023)
$\mathbb{K}_{\text{SMART in use}}$	SMART patrol/reporting system deployment status	Confirmed	(Conservation Namibia, 2022)
$\rho_{\text{wildfire area}}$	Fraction of park burned in Sep 2025 wildfire	$\sim \frac{1}{3}$	(Associated Press, 2025; Wikipedia contributors, 2026)
B. Model 2 Operational / Calibration Constants			
κ	Underservice boost in weight transformation	0.5	Internal calibration register
U^*	Dark-zone threshold	0.55	Internal calibration register
γ	Dark-zone emphasis multiplier in $\Phi(U_z)$	2.0	Internal calibration register
α_r	Ranger effectiveness coefficient (Stage 2)	0.04	Internal calibration register
α_d	Drone effectiveness coefficient (Stage 2)	0.06	Internal calibration register
α_c	Camera/sensor effectiveness coefficient (Stage 2)	0.025	Internal calibration register
α_b	Baseline/structural coefficient (Stage 2)	0.10	Internal calibration register
a	Cobb-Douglas exponent for ranger effort	0.7	Internal calibration register
b	Cobb-Douglas exponent for technology effort	0.7	Internal calibration register
ϵ	Regularization baseline in Stage 2 function	0.10	Internal calibration register
β_τ	Response-time gain weight in Stage 1 score	0.35 (baseline)	Internal calibration register
β_p	Detection gain weight in Stage 1 score	0.40 (baseline)	Internal calibration register
β_q	Coverage gain weight in Stage 1 score	0.20 (baseline)	Internal calibration register
β_R	Resistance-structure gain weight in Stage 1 score	0.05 (baseline)	Internal calibration register

Constant	Meaning	Value	Source (APA)
Π_{target}	Park-wide protection target	0.70	Internal calibration register
C. Model 1 Constants Used by M0–M12			
$\omega_{\text{megafauna}}$	Guild weight for megafauna	0.35	Internal constants register
ω_{predator}	Guild weight for predator guild	0.25	Internal constants register
$\omega_{\text{herbivore}}$	Guild weight for herbivore guild	0.20	Internal constants register
ω_{birds}	Guild weight for birds guild	0.10	Internal constants register
$\omega_{\text{habitat_fauna}}$	Guild weight for habitat fauna guild	0.10	Internal constants register
β_{road}	Human-threat road coefficient	1.5	Internal constants register
β_{boundary}	Human-threat boundary coefficient	1.0	Internal constants register
β_{prs}	Human-threat poaching-risk coefficient	2.0	Internal constants register
τ^*	Response midpoint (hours)	2.0	Internal constants register
K_{resp}	Response sigmoid steepness	2.5	Internal constants register
v_{patrol}	Patrol speed used in response-time conversion	40 km/h	Internal constants register
D. Model 3 Hyperparameters and Gate Constants			
RNG_SEED	Reproducibility random seed	2025	Internal configuration register
S5_M_SPATIAL	Number of spatial basis terms in S5	10	Internal configuration register
S5_M_TEMPORAL	Number of temporal basis terms in S5	30	Internal configuration register
S5_L_HSGP	HSGP scaling/length hyperparameter	1.5	Internal configuration register
S6_N_ENSEMBLE	Ensemble size in EnKF	500	Internal configuration register
S6_FORECAST_MONT	Forecast horizon length	60	Internal configuration register
S6_LOCALISATION_	Localization radius for covariance tapering	50.0 km	Internal configuration register
SPR_RISK_THRESHO	SPR exceedance thresholds used in S6c	[0.75, 0.80, 0.85, 0.9	Internal configuration register
S4_BREAK_P_THRES	P-value threshold for structural-break test	0.05	Internal configuration register

Model 2 Placement Plan (Representative Rows by Technology)

Table 35: Representative rows from the Stage 2 optimal placement output (`placement_plan.csv`), grouped by intervention technology. Waterhole cameras are placed at documented waterhole coordinates; acoustic sensors are placed at zone centroids where no strong occurrence bias is present; drone bases are road-snapped from zone centroids to ensure operational accessibility. The full output contains 111 interventions; representative rows are shown here by technology class.

placement	intervention_id	intervention_type	zone_id	lat	lon	placement_rationale	waterhole_name	rank
Waterhole Cameras								
1	CAM_0001	waterhole_camera	cell_717	-19.1787	15.9197	Exact waterhole position: Okaukuejo	Okaukuejo	214
2	CAM_0002	waterhole_camera	cell_622	-19.11	15.85	Exact waterhole position: Wolfsnes	Wolfsnes	208
3	CAM_0003	waterhole_camera	cell_528	-18.975	15.865	Exact waterhole position: Okondeka	Okondeka	202
4	CAM_0004	waterhole_camera	cell_528	-18.93	15.78	Exact waterhole position: Adamax	Adamax	203
... more waterhole camera placements ...								
Acoustic Sensors								
47	ACO_0056	acoustic_sensor	cell_26	-18.2325	18.22872	Zone centroid (no local GBIF bias)		146
48	ACO_0057	acoustic_sensor	cell_27	-18.2325	18.3883	Zone centroid (no local GBIF bias)		140
49	ACO_0058	acoustic_sensor	cell_28	-18.2325	18.54787	Zone centroid (no local GBIF bias)		139
50	ACO_0059	acoustic_sensor	cell_29	-18.2325	18.70745	Zone centroid (no local GBIF bias)		119
... more acoustic sensor placements ...								
Drone Bases								
236	DRN_0493	drone_base	cell_939	-18.762115	17.970425	Road-snapped from zone centroid (raw projected-distance field recorded in output)		73
Note: one representative drone-base row contains a raw projected-distance diagnostic value in the export. This field does not enter ranking or feasibility constraints.								

Databases Used

Table 36: External databases and satellite products used across the three-layer pipeline, with specific data product, extraction method, and downstream model role documented for each source. Spatial products were extracted via NASA AppEEARS using canonical model-grid centroid coordinates. Harmonization and quality-gate procedures are documented in Appendix C data preparation notes. Sources with DOI or stable URLs are reproducible; internal project sources are version-controlled within the submission.

Database / Source	Type	What We Used	How It Was Used in Model
NASA AppEEARS	Remote sensing platform	Grid centroid coordinates; extracted satellite covariate products	Defined canonical cell locations and ensured spatial consistency for all joins and distance-based features (NASA LP DAAC, 2026).
ESA WorldCover v2.0.0	Global land-cover database	Land-cover class fractions per cell (tree, shrub, grass, bare, built-up, water, wetland, pan)	Built habitat modifiers, intactness signals, and static ecological covariates used in Model 1 and passed downstream (Zanaga, D. et al., 2022).
MODIS (MOD13A3)	Satellite vegetation database	Monthly NDVI	Built seasonal productivity and habitat quality inputs; used in ecological suitability and threat modules (Didan, K., 2021).
CHIRPS	Satellite rainfall database	Monthly rainfall	Created seasonal water-stress and climate exposure covariates used in fire/drought pressure channels (Funk et al., 2015).
VIIRS fire products	Satellite active-fire database	Fire-count / burn-frequency proxies	Quantified wildfire pressure and recurrence for threat and integrity modules (NASA FIRMS, 2026).
OpenStreetMap (OSM)	Open geospatial infrastructure database	Roads and access network geometry	Built road-density metrics and graph edges for accessibility, patrol mobility, and circuit diagnostics (OpenStreetMap contributors, 2026).
GBIF	Biodiversity database	occurrence Rhino/lion (and optional additional species) occurrences	Added occurrence-informed ecological weighting and hotspot enrichment in suitability/demand calculations (Global Biodiversity Information Facility, 2026).
iNaturalist	Biodiversity database	occurrence Supplementary occurrence points (where available)	Supplemental species occurrence evidence to improve sparse-cell ecological signal (iNaturalist, 2026).

Database / Source	Type	What We Used	How It Was Used in Model
Internal geospatial layers (park boundary, waterholes, gates, salt pan polygon)	Curated local operational layers	Boundary mask; infrastructure points; pan geometry	Enforced in-park masking, nearest-distance features, response-time proxies, and pan-specific habitat penalties (internal project GIS layers).

Data Cleaning and Preparation

1. Schema standardization and key integrity. All source tables were harmonized to a single cell key and canonical coordinate system. We enforced one record per cell for static datasets and one record per cell-month for panel datasets. Duplicate IDs were removed and non-numeric key fields were explicitly cast before merges to prevent silent join failures.

2. Spatial harmonization. All geospatial layers were projected/validated in WGS84, then aligned to the model grid. For each cell, we computed:

- in-park membership using point-in-polygon masking,
- nearest-distance features (waterhole, ranger base, gate, boundary),
- land-cover fractions from raster-to-cell aggregation.

3. Temporal harmonization. Monthly NDVI, rainfall, and fire products were aligned to a common time index. We then produced:

- monthly panel features for forecasting modules,
- representative dry/wet seasonal summaries for static-risk modules.

Where monthly values were missing, we used constrained imputation (month-of-year climatology or short-gap interpolation), with imputation flags retained for auditability.

4. Value-range and unit checks. Continuous features were checked against physically plausible bounds (e.g., NDVI, rainfall, fractions in $[0, 1]$, nonnegative distances). Unit consistency was enforced (km, hours, fractions). Out-of-range values were either clipped to documented bounds or set to missing prior to imputation.

5. Outlier and anomaly handling. Extreme values were screened using robust quantiles and domain rules. We did not remove ecologically plausible extremes (e.g., very low vegetation in pan cells); instead, we preserved them and ensured downstream transformations were bounded and numerically stable.

6. Feature engineering and normalization. After cleaning, we generated model-ready features:

- normalized road/access pressure,
- seasonal drought and fire summaries,
- habitat intactness components,
- concentration and connectivity predictors,
- intervention and response-time covariates.

Normalization/scaling was performed only after missing-data resolution to avoid distortion.

7. Audit trail and reproducibility. Each preprocessing stage wrote versioned outputs and metadata logs (row counts, missingness rates, min/max diagnostics, and processing timestamps). This made every modeling input traceable from raw source to final feature table and supported full reruns without manual edits.

8. Final quality gates before modeling. Before entering optimization/forecast stages, we required:

- complete key coverage for all in-park cells,
- no critical nulls in required model columns,
- successful season labeling and scenario mapping,
- consistency checks between static and panel feature views.

Only datasets passing all gates were promoted to the final modeling inputs.

TOPSIS Appendix (Detailed Computation)

This section reproduces the full TOPSIS computation provided in the project document *TopsisAppendix.pdf*. Candidate methods are PIO (Pure Integer Optimization), CTIM (Circuit-Theory Infrastructure Model), and TSSP (Two-Stage Stochastic Program).

Table 37: Raw TOPSIS decision matrix for three candidate allocation frameworks (PIO, CTIM, TSSP), scored on a 1–10 scale across ten criteria. Scores reflect applied Etosha-context performance rather than generic method capability. See Table 34 for key score justifications.

ID	Criterion	Weight	PIO	CTIM	TSSP
F1	Spatial realism	0.12	4	9	5
F2	Budget / resource constraints	0.10	7	6	9
F3	Uncertainty & robustness	0.14	3	6	9
F4	Computational tractability	0.06	9	7	5
F5	Data feasibility	0.08	7	8	6
F6	Manager interpretability	0.10	6	8	6
F7	Multi-threat coverage	0.12	5	4	8
F8	Temporal / seasonal adaptability	0.10	3	4	8
F9	Infrastructure vs. operations balance	0.10	5	8	3
F10	Transferability to other parks	0.08	8	6	7

Table 38: TOPSIS final scores and ranking. D^+ is Euclidean distance to ideal-best, D^- is distance to ideal-worst, and $C^* = D^- / (D^+ + D^-)$ is relative closeness. Higher C^* indicates stronger proximity to ideal performance. TSSP ranks first ($C^* = 0.593$) and CTIM second ($C^* = 0.525$), supporting a complementary hybrid design.

Rank	Candidate	D^+	D^-	C^*
1	TSSP	0.0729	0.1063	0.5931
2	CTIM	0.0787	0.0870	0.5250
3	PIO	0.1187	0.0345	0.2253

where D^+ is the Euclidean distance to the ideal-best profile, D^- is the Euclidean distance to the ideal-worst profile, and

$$C^* = \frac{D^-}{D^+ + D^-}.$$

Higher C^* indicates a candidate closer to the ideal solution.

Criteria Definitions and Weight Justifications

Table 39: Evaluation criteria definitions and weight justifications for TOPSIS comparison. Criterion F3 (Uncertainty & Robustness) has highest weight (0.14) because ENP faces compounding shocks (wildfire, drought, poaching surges, insider threat), making expected-value-only optimization insufficient. Criterion F4 (Computational Tractability) has lowest weight (0.06) because all candidates are computationally feasible in project timelines and therefore weakly differentiating.

ID	Criterion	Weight	Definition and Weight Justification
F1	Spatial realism	0.12	Degree to which the model captures terrain heterogeneity, landscape connectivity, and physical movement constraints across the park. Weighted highly because Etosha spans 22,935 km ² and includes salt pan barriers, variable terrain, and sparse road access that governs patrol reachability.
F2	Budget / resource constraints	0.10	Ability to enforce hard budget ceilings, staffing limits, and capital-versus-operating cost structure as explicit mathematical constraints. Important because ENP operates under tight fiscal constraints.
F3	Uncertainty & robustness	0.14	Capacity to produce effective allocations under uncertain futures (poaching surges, wildfire, drought, staffing shortfalls). Highest weight because Etosha faces compounding and unpredictable stressors.
F4	Computational tractability	0.06	Feasibility of solving at ENP scale within the competition timeline. Lowest weight because all candidates are computationally feasible here; tractability is necessary but not a major differentiator.
F5	Data feasibility	0.08	Degree to which required data (roads, costs, threats, detection rates) can be assembled from public/published sources without proprietary or classified datasets.
F6	Manager interpretability	0.10	Ease with which outputs can be understood and implemented by managers without specialized optimization training. Weighted because recommendations must be operationally actionable.
F7	Multi-threat coverage	0.12	Ability to represent multiple distinct threats simultaneously (poaching, wildfire, drought, conflict) rather than a single aggregate risk channel. Weighted highly due to heterogeneous risk conditions.
F8	Temporal / seasonal adaptability	0.10	Capacity to adapt allocations over time and across wet/dry seasons. Important because seasonal shifts materially change wildlife concentration and poaching exposure.
F9	Infrastructure operations balance	vs. 0.10	Extent to which both strategic siting (outposts, cameras) and tactical deployment (patrol/drone operations) are jointly addressed. Important because fixed assets and mobile response are both required.
F10	Transferability to other parks	0.08	Ease of adapting model architecture to structurally different protected areas. Moderate weight because transferability is required, but less critical than Etosha-specific operational criteria.

Scoring Rubric and Key Justifications

Each candidate was scored on a 1–10 integer scale (1 = fundamentally unable to satisfy the criterion, 10 = near-ideal capability), with scores reflecting applied performance in the Etosha context.

Table 40: Scoring justifications for selected key TOPSIS entries. Included entries are those with non-obvious scores or where low scores reveal architectural limitations (e.g., $\text{PIO} \times \text{F3} = 3$ due to lack of explicit uncertainty hedging; $\text{TSSP} \times \text{F9} = 3$ without CTIM structural-access diagnostics).

Entry	Score	Justification
$\text{PIO} \times \text{F3}$ (Uncertainty)	3	PIO is deterministic: it solves one fixed input state and returns one optimum. If threat conditions shift, the model must be re-solved without built-in hedging.
$\text{PIO} \times \text{F8}$ (Temporal)	3	PIO represents a single snapshot and does not naturally capture dynamic seasonal redeployment without separate optimization runs.
$\text{CTIM} \times \text{F1}$ (Spatial)	9	Circuit-theory structure captures connectivity and resistance through road topology and terrain, representing barriers and corridors more realistically than straight-line distance.
$\text{CTIM} \times \text{F7}$ (Multi-threat)	4	CTIM emphasizes structural accessibility but does not explicitly model different threat channels with distinct response protocols.
$\text{TSSP} \times \text{F3}$ (Uncertainty)	9	Two-stage stochastic programming optimizes over a probability-weighted scenario set and therefore hedges Stage-1 decisions against many Stage-2 realizations.
$\text{TSSP} \times \text{F9}$ (Infrastructure vs. operations)	3	TSSP optimizes allocations under candidate sites but, without CTIM diagnostics, may miss structural accessibility bottlenecks in site generation.

TOPSIS Computation Steps

Step 1: Vector normalization. Each criterion column is normalized by its L2 norm,

$$r_{ij} = \frac{x_{ij}}{\|x_j\|_2}.$$

Table 41: TOPSIS normalization step output: normalized decision matrix R . Compare criterion-wise relative performance after scale normalization and before weight application.

Criterion	$\ x_j\ _2$	PIO	CTIM	TSSP
Spatial realism	11.0454	0.3621	0.8148	0.4527
Budget / resource constraints	12.8841	0.5433	0.4657	0.6985
Uncertainty & robustness	11.2250	0.2673	0.5345	0.8018

Criterion	$\ \mathbf{x}_j\ _2$	PIO	CTIM	TSSP
Computational tractability	12.4499	0.7229	0.5623	0.4016
Data feasibility	12.2066	0.5735	0.6554	0.4915
Manager interpretability	11.6619	0.5145	0.6860	0.5145
Multi-threat coverage	10.2470	0.4880	0.3904	0.7807
Temporal / seasonal adaptability	9.4340	0.3180	0.4240	0.8480
Infrastructure vs. operations balance	9.8995	0.5051	0.8081	0.3030
Transferability to other parks	12.2066	0.6554	0.4915	0.5735

Step 2: Weighted normalization. Multiply each normalized component by criterion weight,

$$v_{ij} = w_j r_{ij}.$$

Table 42: TOPSIS weighting step output: weighted normalized matrix V . Each normalized criterion component is multiplied by criterion weight, revealing weighted separations among alternatives.

Criterion	Weight	PIO	CTIM	TSSP
Spatial realism	0.12	0.0435	0.0978	0.0543
Budget / resource constraints	0.10	0.0543	0.0466	0.0699
Uncertainty & robustness	0.14	0.0374	0.0748	0.1122
Computational tractability	0.06	0.0434	0.0337	0.0241
Data feasibility	0.08	0.0459	0.0524	0.0393
Manager interpretability	0.10	0.0514	0.0686	0.0514
Multi-threat coverage	0.12	0.0586	0.0468	0.0937
Temporal / seasonal adaptability	0.10	0.0318	0.0424	0.0848
Infrastructure vs. operations balance	0.10	0.0505	0.0808	0.0303
Transferability to other parks	0.08	0.0524	0.0393	0.0459

Step 3: Ideal-best and ideal-worst vectors.

$$A_j^+ = \max_i(v_{ij}), \quad A_j^- = \min_i(v_{ij}).$$

Table 43: TOPSIS ideal vectors (A^+ and A^-) by criterion. CTIM holds A^+ for spatial realism and infrastructure balance, while TSSP holds A^+ for uncertainty, multi-threat coverage, and temporal adaptability, confirming complementarity rather than substitution.

Criterion	A^+	A^-	Source of A^+
Spatial realism	0.0978	0.0435	CTIM
Budget / resource constraints	0.0699	0.0466	TSSP
Uncertainty & robustness	0.1122	0.0374	TSSP
Computational tractability	0.0434	0.0241	PIO
Data feasibility	0.0524	0.0393	CTIM
Manager interpretability	0.0686	0.0514	CTIM
Multi-threat coverage	0.0937	0.0468	TSSP
Temporal / seasonal adaptability	0.0848	0.0318	TSSP
Infrastructure vs. operations balance	0.0808	0.0303	CTIM
Transferability to other parks	0.0524	0.0393	PIO

Step 4: Distance computation.

$$D_i^+ = \sqrt{\sum_j (v_{ij} - A_j^+)^2}, \quad D_i^- = \sqrt{\sum_j (v_{ij} - A_j^-)^2}.$$

Step 5: Relative closeness.

$$C_i^* = \frac{D_i^-}{D_i^+ + D_i^-}.$$

Table 44: TOPSIS distance-and-closeness output (full computation). Lower D^+ and higher D^- produce larger C^* , indicating stronger proximity to the ideal decision profile.

Rank	Alternative	D^+	D^-	$D^+ + D^-$	C^*
1	TSSP — Two-Stage Stochastic Program	0.0729	0.1063	0.1793	0.5931
2	CTIM — Circuit-Theory Infrastructure Model	0.0787	0.0870	0.1658	0.5250
3	PIO — Pure Integer Optimization	0.1187	0.0345	0.1532	0.2253

E Code

Layer I Implementation Code

B.1 Haversine Distance Calculation

Listing 1: Great-circle distance between two points

```
import numpy as np

def haversine(lat1, lon1, lat2, lon2):
    """
    Calculate great-circle distance using haversine formula.

    Args:
        lat1, lon1: latitude and longitude of point 1 (degrees)
        lat2, lon2: latitude and longitude of point 2 (degrees)

    Returns:
        distance in kilometers
    """
    R = 6371.0 # Earth radius in km

    # Convert to radians
    phi1, phi2 = np.radians(lat1), np.radians(lat2)
    dphi = np.radians(lat2 - lat1)
    dlamba = np.radians(lon2 - lon1)

    # Haversine formula
    a = np.sin(dphi/2)**2 + np.cos(phi1) * np.cos(phi2) * np.sin(dlamba/2)**2
    c = 2 * np.arcsin(np.sqrt(a))

    return R * c
```

B.2 Ecological Value Calculation

Listing 2: Cobb-Douglas ecological value computation

```
def calculate_ecological_value(O, X_ndvi, X_water, X_intact,
                              alpha, beta, gamma, S_g):
    """
    Compute guild-specific ecological value using Cobb-Douglas form.

    Args:
        O: habitat suitability (scalar or array)
        X_ndvi, X_water, X_intact: support variables (0 to 1)
        alpha, beta, gamma: Cobb-Douglas exponents (sum to 1)
        S_g: conservation urgency scalar

    Returns:
        ecological_value_E(g, i)
    """
```

```
return S_g * O * (X_ndvi**alpha) * (X_water**beta) * (X_intact**gamma)
```

B.3 Noisy-OR Threat Fusion

Listing 3: Probabilistic union of threat channels

```
def noisy_or_fusion(H, F, W, D):
    """
    Integrate threat channels via probabilistic union.

    Args:
        H: human_threat (0 to 1)
        F: fire_hazard (0 to 1)
        W: water-concentration_risk (0 to 1)
        D: habitat_degradation (0 to 1)

    Returns:
        integrated_threat_T (0 to 1)
    """
    return 1 - (1 - H) * (1 - F) * (1 - W) * (1 - D)
```

B.4 Protection Capacity Noisy-OR

Listing 4: Noisy-OR detection-response capacity

```
def protection_capacity(p_det, p_resp):
    """
    Compute protection capacity as Noisy-OR union of
    detection and response channels.

    Args:
        p_det: detection_probability (0 to 1)
        p_resp: response_probability (0 to 1)

    Returns:
        protection_capacity_P_cap (0 to 1)
    """
    return 1 - (1 - p_det) * (1 - p_resp)
```

B.5 SPR Calculation

Listing 5: Species Protection Rate from cell-level protection indicators

```
def calculate_spr(protected_indicators, ecological_weights):
    """
    Calculate Species Protection Rate (SPR).

    Args:
        protected_indicators: binary_array (1 if protected, 0 otherwise)
        ecological_weights: weighted_ecological_value_per_cell

    Returns:
        SPR (scalar, 0 to 1)
```

```

"""
"""
    numerator = (protected_indicators * ecological_weights).sum()
    denominator = ecological_weights.sum()
    return numerator / denominator

```

B.6 Computational Environment

Software versions: - Python: 3.10.12 - NumPy: 1.24.3 - SciPy: 1.10.1 - Pandas: 2.0.3 - GeoPandas: 0.13.2 - Scikit-learn: 1.3.0

Runtime benchmarks (940-cell grid, 158 in-park cells): - Stack 1 (Ecological Importance): ~2 minutes - Stack 2 (Threat Assessment): ~4 minutes - Stack 3 (Protection Capacity): ~3 minutes - Stack 4 (Priority Ranking): ~1 minute - Total Layer I runtime: ~10 minutes

Hardware: 16-core CPU, 64 GB RAM

Layer II Implementation Code

E.1 Graph Laplacian Construction

Listing 6: Build weighted Laplacian from road network

```

import numpy as np
import scipy.sparse as sp

def build_laplacian(nodes, edges, conductances):
    """
    Construct weighted graph Laplacian.

    Args:
    nodes: list of node IDs
    edges: list of (i, j) tuples
    conductances: dict mapping (i, j) -> C_ij

    Returns:
    L: sparse Laplacian matrix (n x n)
    """
    n = len(nodes)
    node_to_idx = {node: i for i, node in enumerate(nodes)}

    # Build adjacency matrix
    data, row, col = [], [], []
    for (i, j), C_ij in conductances.items():
        idx_i = node_to_idx[i]
        idx_j = node_to_idx[j]
        # Off-diagonal: -C_ij
        data.extend([-C_ij, -C_ij])
        row.extend([idx_i, idx_j])
        col.extend([idx_j, idx_i])

    A = sp.csr_matrix((data, (row, col)), shape=(n, n))

    # Diagonal: sum of row conductances

```

```

degrees = np.array(A.sum(axis=1)).flatten()
L = sp.diags(degrees) - A

return L

```

E.2 Effective Resistance Calculation

Listing 7: Compute effective resistance from zone to supply nodes

```

from scipy.sparse.linalg import splu

def effective_resistance(L, zone_idx, supply_indices):
    """
    Calculate effective resistance from zone to supply nodes.

    Args:
        L: Laplacian matrix (sparse)
        zone_idx: index of target zone
        supply_indices: list of supply node indices

    Returns:
        R_eff: average effective resistance to supply nodes
    """
    n = L.shape[0]

    # Compute pseudoinverse using sparse LU decomposition
    # (Fix one node potential for uniqueness)
    L_reduced = L.tolil()
    L_reduced[0, :] = 0
    L_reduced[0, 0] = 1
    L_reduced = L_reduced.tocsr()

    resistances = []
    for supply_idx in supply_indices:
        # Solve for difference vector
        b = np.zeros(n)
        b[zone_idx] = 1
        b[supply_idx] = -1
        b[0] = 0 # Ground constraint

        v = splu(L_reduced).solve(b)
        R_ij = v[zone_idx] - v[supply_idx]
        resistances.append(R_ij)

    return np.mean(resistances)

```

E.3 Portfolio Optimization (Binary)

Listing 8: Stage 2 portfolio selection via mixed-integer programming

```

from scipy.optimize import milp, LinearConstraint, Bounds

def optimize_portfolio(interventions, B_capital, O_operating, H_staff):

```

```

"""
Solve_portfolio_selection_problem.

Args:
    interventions: list of dicts with keys { benefit , F , O , H , lag }
    B_capital: capital budget
    O_operating: annual operating budget
    H_staff: staffing ceiling

Returns:
    y: binary selection vector
    objective_value: total discounted benefit
"""
n = len(interventions)

# Objective: maximize discounted benefits
lambda_lag = 0.2
c = np.array([
    -interv['benefit'] / (1 + lambda_lag * interv['lag'])
    for interv in interventions
])

# Constraints
F = np.array([interv['F'] for interv in interventions])
O = np.array([interv['O'] for interv in interventions])
H = np.array([interv['H'] for interv in interventions])

A_ub = np.vstack([F, O, H])
b_ub = np.array([B_capital, O_operating, H_staff])

constraints = LinearConstraint(A_ub, -np.inf, b_ub)
bounds = Bounds(0, 1)
integrality = np.ones(n) # All variables binary

result = milp(c, constraints=constraints, bounds=bounds,
               integrality=integrality)

return result.x, -result.fun

```

E.4 Protection Function Evaluation

Listing 9: Compute zone protection from resource allocation

```

def protection_function(X_patrol, d_drone, c_sensor, B_structural,
                        alpha_r=1.0, alpha_d=0.8, alpha_c=0.6,
                        alpha_b=0.5, a=0.7, b=0.7, epsilon=0.01):
    """
    Calculate_zone_protection_score.

    Args:
        X_patrol: aggregate patrol hours
        d_drone: drone hours

```

```

c_sensor: _sensor_effort
B_structural: _structural_benefit_from_Stage_2_assets
alpha_*: _effectiveness_coefficients
a, b: _diminishing_returns_exponents
epsilon: _baseline_floor

Returns:
P: _protection_score_(0_to_1)
"""
patrol_term = (alpha_r * X_patrol) ** a
detection_term = (alpha_d * d_drone + alpha_c * c_sensor + epsilon) ** b

combined = (patrol_term * detection_term) ** (1 / (a + b))
total_arg = combined + alpha_b * B_structural

P = 1 - np.exp(-total_arg)
return P

```

E.5 Computational Environment

Software versions: - Python: 3.10.12 - NumPy: 1.24.3 - SciPy: 1.10.1 - PuLP: 2.7.0 (for MILP solving) - NetworkX: 3.1 (for graph utilities)

Runtime benchmarks (940-node graph, medium budget): - Stage 1 (Circuit Diagnosis): 5 minutes - Stage 2 (Portfolio Selection): 12 minutes (MILP with 300 binary variables) - Stage 3 (Recourse per scenario): 8 minutes - Total Layer II runtime: 1.5 hours for all 8 scenarios

Hardware: 16-core CPU, 64 GB RAM

Layer III Implementation Code

B.1 Monthly Panel Construction

Listing 10: Stack 1: Monthly panel construction

```

# Run Layer I for each month (2019-01 to 2024-12)
for month_idx in range(72):
    year = 2019 + (month_idx // 12)
    month = (month_idx % 12) + 1

    # Update monthly inputs
    ndvi = get_monthly_ndvi(year, month)
    rainfall = get_monthly_rainfall(year, month)
    drought = calculate_drought_zscore(year, month)
    fire = get_rolling_fire_window(year, month, window=3)
    season = 'dry' if 5 <= month <= 10 else 'wet'

    # Run Layer I
    output = run_layer1(ndvi, rainfall, drought, fire, season)
    panel_data[month_idx] = output['vulnerability']

```

B.2 Moran's I Calculation

Listing 11: Stack 2: Spatial autocorrelation

```

def morans_i(x, W):
    """ Calculate Moran's I with permutation test. """
    n = len(x)
    x_centered = x - np.mean(x)

    numerator = n * np.sum(W * np.outer(x_centered, x_centered))
    denominator = np.sum(W) * np.sum(x_centered**2)
    I_observed = numerator / denominator

    # Permutation test (999 iterations)
    I_permuted = []
    for _ in range(999):
        x_perm = np.random.permutation(x)
        x_perm_centered = x_perm - np.mean(x_perm)
        I_perm = (n * np.sum(W * np.outer(x_perm_centered,
                                           x_perm_centered))) / (np.sum(W) *
                                                                np.sum(x_perm_centered**2))
        I_permuted.append(I_perm)

    p_value = np.mean(np.abs(I_permuted) >= np.abs(I_observed))
    return I_observed, p_value

```

B.3 EnKF Forecast**Listing 12:** Stack 3: Ensemble Kalman Filter forecast

```

def enkf_forecast(x_ensemble, A, B, u, Q, n_steps=60):
    """ Run EnKF forecast for n_steps months. """
    n_cells, n_members = x_ensemble.shape
    forecast = np.zeros((n_steps, n_cells, n_members))
    x_current = x_ensemble.copy()

    for t in range(n_steps):
        for j in range(n_members):
            # Deterministic update
            x_forecast = A @ x_current[:, j] + B @ u[t]

            # Add process noise
            noise = np.random.multivariate_normal(
                mean=np.zeros(n_cells), cov=Q
            )
            x_current[:, j] = x_forecast + noise

        forecast[t] = x_current.copy()

    return forecast

```

B.4 SPR Calculation**Listing 13:** Stack 4: SPR from ensemble forecast

```

def calculate_spr(forecast_ensemble, ecological_weights,

```

```

        threshold=0.90):
    """Calculate_SPR_trajectory_from_ensemble."""
    n_steps, n_cells, n_members = forecast_ensemble.shape
    spr_trajectory = np.zeros((n_steps, n_members))

    for t in range(n_steps):
        for j in range(n_members):
            # Protected cells
            protected = forecast_ensemble[t, :, j] >= threshold

            # Weighted SPR
            spr_trajectory[t, j] = (
                np.sum(ecological_weights[protected]) /
                np.sum(ecological_weights)
            )

    return spr_trajectory

```

B.5 Computational Environment

Python 3.10.12, numpy 1.24.3, scipy 1.10.1, pandas 2.0.3, geopandas 0.13.2. Stack 1: ~18 min, Stack 2: ~3 min, Stack 3 (GP): ~12 min, Stack 3 (EnKF): ~45 min. Hardware: 16-core CPU, 64 GB RAM.

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F Brainstorming

OpenAI ChatGPT (Mar 9, 2026 version, ChatGPT-5.3)

F.0.1 Query

I am considering the following models for dynamically ranking a set of elements. Would you please give me some pros and cons for each model / algorithm? Models: 1. Analytic Hierarchy Process. 2. Custom Multi-Criteria Decision Analysis Scoring Model. 3. PROMETHEE. 4. Simple Weighted Sum. *Output:* Here is a practical comparison of the four approaches for dynamically ranking elements based on multiple criteria. I'll focus on interpretability, scalability, ease of updating, and computational complexity, since those usually matter most in ranking systems.

1. Analytic Hierarchy Process (AHP) A structured pairwise comparison method developed by Thomas L. Saaty. Pros 1. Handles subjective judgments well Pairwise comparisons are intuitive for humans. Good when criteria importance is difficult to quantify directly. 2. Consistency checking Built-in consistency ratio helps detect inconsistent judgments. 3. Hierarchical structuring Naturally supports complex decision trees with multiple layers of criteria. 4. Transparent Stakeholders can see exactly how weights are derived. Cons 1. Poor scalability Requires $n(n-1)/2$ comparisons per level. Becomes impractical when alternatives $> \sim 20$. 2. Updating rankings is expensive Adding a new element requires re-running many comparisons. 3. Subjectivity Results heavily depend on expert judgments. 4. Rank reversal issue Adding/removing alternatives can change rankings unexpectedly. Best Use Cases Small decision sets Expert-driven evaluation Strategic decision making

2. Custom Multi-Criteria Decision Analysis (MCDA) Scoring Model A designed scoring framework where criteria are weighted and scored using custom formulas. Pros 1. Highly flexible You can incorporate: nonlinear scoring penalties thresholds domain-specific transformations 2. Scales very well Works with thousands or millions of elements. 3. Easy to update dynamically Simply recompute scores when data changes. 4. Easily integrated with data pipelines Works well in real-time ranking systems. Cons 1. Requires careful design Poor normalization or weighting can bias results. 2. Harder to validate mathematically Unlike formal methods, consistency guarantees are weaker. 3. Weight tuning challenges Requires calibration or experimentation. 4. Potential transparency issues Complex scoring rules may confuse stakeholders. Best Use Cases Production ranking systems Real-time dashboards Large datasets Custom business logic

3. PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluation) An outranking method widely used in decision science. Pros 1. Handles incomparability Allows alternatives to partially outrank each other. 2. Flexible preference functions Supports different preference shapes: usual U-shape V-shape Gaussian 3. Robust ranking Less sensitive to extreme values than simple weighted sums. 4. Strong theoretical grounding Well studied in operations research. Cons 1. Pairwise comparison cost Requires $O(n^2)$ comparisons between alternatives. 2. Harder to explain Net flow calculations can be non-intuitive to stakeholders. 3. Parameter tuning Preference thresholds can be difficult to choose. 4. Scaling limitations Computationally expensive for large datasets. Best Use Cases Medium-sized decision problems Situations where trade-offs and thresholds matter

4. Simple Weighted Sum Model (WSM) The most common multi-criteria scoring method. $[Score_i = \sum_{j=1}^m w_j x_{ij}]$ Pros 1. Extremely simple Easy to implement and explain. 2. Very fast Computational complexity $O(nm)$. 3. Easy dynamic updates Adding new alternatives requires minimal recomputation. 4. Works well for large-scale systems Widely used in ranking engines and scoring models. Cons 1. Assumes linear trade-offs A large value in one criterion can fully compensate for another. 2. Requires normalization Different scales must be standardized. 3. Sensitive to outliers Extreme values can dominate. 4. Cannot capture complex preference structures Best Use Cases Real-time ranking Large datasets Simple scoring frameworks

Quick Comparison Model Interpretability Scalability Handles Subjective Judgments Dynamic Updates AHP High Poor Excellent Poor Custom MCDA Medium–High Excellent Good Excellent PROMETHEE Medium Moderate Good Moderate Weighted Sum Very High Excellent Limited Excellent

[OK] Practical rule of thumb Few alternatives + expert judgment $\rightarrow$ AHP Medium alternatives + complex preference thresholds $\rightarrow$ PROMETHEE Large dynamic systems $\rightarrow$ Custom MCDA Fast simple ranking $\rightarrow$ Weighted Sum

[Idea] In practice, many production systems combine approaches, e.g.: Use AHP to determine weights Use Weighted

Sum or MCDA to rank dynamically

If you'd like, I can also show: How these methods behave differently on the same dataset Which method is best for ML-assisted ranking systems A decision framework to choose among them.

OpenAI ChatGPT (Mar 10, 2026 version, ChatGPT-5.3)

F.0.2 Query

For my Spatio-Temporal Gaussian Process Regression, I am deciding between an RBF kernel and a Matérn kernel for the spatial component. Since wildlife density and poaching risks can have 'rougher' or more abrupt transitions between different habitats (such as the transition from the Etosha Pan to the surrounding savanna), can you please explain the mathematical trade-offs in choosing $\nu=1.5$ versus a smoother RBF approach for this specific conservation application?
Output: For Spatio-Temporal Gaussian Process Regression (ST-GPR) in conservation problems like wildlife density or poaching risk, the choice between an RBF kernel and a Matérn kernel ($\nu = 1.5$) mainly comes down to assumptions about spatial smoothness and how rapidly the modeled surface can change across space. In landscapes like Etosha, where sharp ecological boundaries exist (e.g., salt pan $\rightarrow$ savanna $\rightarrow$ woodland), these assumptions matter a lot.

1. Mathematical Form of the Kernels RBF (Squared Exponential) Kernel $[k_{\text{RBF}}](x, x') = \sigma^2 \exp\left(-\frac{1}{2\ell^2} \|x - x'\|^2\right)$ Properties Infinitely differentiable Produces very smooth spatial surfaces Strong correlation decay controlled by length scale (ℓ) Interpretation: The model assumes gradual, smooth transitions everywhere.

Matérn Kernel ($\nu = 1.5$) $[k_{\nu=1.5}](r) = \sigma^2 \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right)$ where ($r = \|x - x'\|$). Properties Once differentiable Allows rougher functions Spatial correlations drop more abruptly. Interpretation: The spatial field can contain kinks or sharper gradients.

2. Smoothness Assumption The most important theoretical difference: Kernel Mean-square differentiability RBF Infinite Matérn $\nu=1.5$ Once

This means: RBF prior: ($f(x)$) is extremely smooth everywhere. Matérn 1.5 prior: ($f(x)$) can change more abruptly. For ecological landscapes: Habitat edges Waterholes Vegetation transitions Human infrastructure often create non-smooth spatial patterns. Thus Matérn often matches reality better.

3. Correlation Decay Behavior Compare how correlation declines with distance. RBF $[k(r) \propto \exp(-r^2)]$ Decay is Gaussian $\rightarrow$ long-range smooth influence. Matérn 1.5 $[k(r) \propto (1+r)\exp(-r)]$ Decay is exponential-like $\rightarrow$ sharper drop. Implication RBF tends to oversmooth across habitat boundaries. Example: Etosha Pan — edge — Savanna

RBF assumption: density transition: smooth curve

Reality: density transition: sharp drop

Matérn handles this better.

4. Behavior With Sparse Data Conservation datasets often have: uneven patrol coverage clustered observations missing regions Kernel choice affects extrapolation. RBF extrapolation Produces very smooth predictions Can smear patterns across large distances Matérn $\nu=1.5$ More local influence Predictions revert to mean faster For poaching risk mapping, this is often desirable.

5. Gradient Magnitude and Edge Detection Because Matérn allows sharper gradients: It captures habitat boundaries Detects risk hotspots Example patterns better captured by Matérn: roads water sources fence lines park borders vegetation edges These create non-stationary abrupt spatial effects.

6. Computational Considerations The computational difference is small but relevant. RBF Slightly simpler formula Often easier for hyperparameter optimization Matérn $\nu=1.5$ Still closed-form Often used in spatial statistics because: $[\nu = 0.5, 1.5, 2.5]$ give efficient implementations. Many geostatistical packages default to Matérn.

7. Practical Implication for Your Conservation Model For wildlife density and poaching risk: Characteristic Recommended kernel Smooth ecological gradients RBF Habitat edges Matérn Human infrastructure effects Matérn Sparse patrol data Matérn Risk hotspot detection Matérn

In most conservation risk mapping papers, the Matérn kernel dominates for this reason.

8. Typical Spatio-Temporal Structure A common ST-GP structure is: $k((s,t),(s',t')) = k_{\text{space}}(s,s') \times k_{\text{time}}(t,t')$ Example: Spatial: Matérn $\nu=1.5$ Temporal: RBF or Matérn This allows: rough spatial surfaces smooth temporal evolution.

9. Intuition Using Your Etosha Example Consider wildlife density: Etosha Pan: low density Savanna: high density

True pattern: —sharp boundary—

RBF prior: smooth transition over kilometers

Matérn $\nu=1.5$ prior: steeper transition allowed

So the GP does not blur habitat edges.

10. When RBF Might Still Win RBF could outperform if: habitat gradients are gradual very dense spatial sampling strong environmental covariates already explain boundaries But with latent spatial processes, RBF often oversmooths.

11. Practical Recommendation For your conservation ST-GPR: Start with: $k_{\text{space}} = \text{Matérn}(\nu=1.5)$ Reasons: matches ecological discontinuities avoids oversmoothing standard in spatial statistics robust with sparse data Then compare via: marginal likelihood predictive log score spatial cross-validation.

[OK] Short summary Kernel Effect RBF Very smooth surfaces, can blur habitat edges Matérn $\nu=1.5$ Rougher spatial field, captures abrupt transitions

For wildlife density / poaching risk across heterogeneous landscapes, Matérn $\nu=1.5$ is usually the safer and more realistic prior.

If you'd like, I can also show: why $\nu=2.5$ is often used in ecological GP models how kernel choice affects posterior variance near habitat edges how to implement this in PyTorch GP / GPyTorch / Stan for ST-GP models.

G Code

GitHub CoPilot (Mar 11, 2026 version) Auto-completions for code used in preparing all Python code.

OpenAI ChatGPT (Mar 11, 2026 version, ChatGPT-5.3)

G.0.1 Query

Can you please generate some preliminary Python boilerplate code for a Spatial Temporal Gaussian Process Regression that will allow us to combine a variety of features when we expand our math model to apply to multiple years? *Output:* Below is preliminary Python boilerplate for a Spatio-Temporal Gaussian Process Regression (ST-GPR) model designed to combine spatial coordinates, time, and arbitrary covariates (features). The structure is intentionally modular so you can extend it to multi-year datasets, plug in additional kernels, and scale later. This example uses GPyTorch with PyTorch, which is commonly used for scalable Gaussian Processes.

1. Environment Setup pip install torch gpytorch pandas numpy scikit-learn

2. High-Level Model Structure We will model: $y(s,t,x) = f_{\text{space}}(s) + f_{\text{time}}(t) + f_{\text{features}}(x) + \epsilon$ Where: s to spatial coordinates (lat, lon or x,y) t to time (year, month, etc.) x to feature vector (covariates) ϵ to noise Kernel $K = K_{\text{space}} \times K_{\text{time}} + K_{\text{features}}$

3. Example Data Structure import pandas as pd

```
data = pd.DataFrame({ "lat": [], "lon": [], "year": [], "feature1": [], "feature2": [], "target": [] })
```

4. Feature Preparation import torch import numpy as np from sklearn.preprocessing import StandardScaler

```
def prepare_features(df):
```

```
    spatial = df[["lat", "lon"]].values time = df[["year"]].values features = df[["feature1", "feature2"]].values
```

```
    y = df["target"].values
```

```
    scaler = StandardScaler() features = scaler.fit_transform(features)
```

```
    X = np.hstack([spatial, time, features])
```

```
return ( torch.tensor(X).float(), torch.tensor(y).float() )
```

5. Spatio-Temporal GP Model import gpytorch

```
class SpatioTemporalGP(gpytorch.models.ExactGP):
```

```
def __init__(self, train_x, train_y, likelihood):
```

```
    super().__init__(train_x, train_y, likelihood)
```

```
    self.mean_module = gpytorch.means.ConstantMean()
```

```
    # Spatial kernel (lat/lon) spatial_kernel = gpytorch.kernels.ScaleKernel( gpytorch.kernels.RBFKernel(active_dims=(0,1)) )
```

```
    # Temporal kernel temporal_kernel = gpytorch.kernels.ScaleKernel( gpytorch.kernels.MaternKernel( nu=1.5, active_dims=(2,) ) )
```

```
    # Feature kernel feature_kernel = gpytorch.kernels.ScaleKernel( gpytorch.kernels.RBFKernel( active_dims=tuple(range(3, train_x.shape[1])) ) )
```

```
    # Combined kernel self.covar_module = ( spatial_kernel * temporal_kernel + feature_kernel )
```

```
def forward(self, x):
```

```
    mean = self.mean_module(x) covar = self.covar_module(x)
```

```
    return gpytorch.distributions.MultivariateNormal(mean, covar)
```

6. Training Loop def train_gp(train_x, train_y, training_iter=200):

```
    likelihood = gpytorch.likelihoods.GaussianLikelihood()
```

```
    model = SpatioTemporalGP(train_x, train_y, likelihood)
```

```
    model.train() likelihood.train()
```

```
    optimizer = torch.optim.Adam( model.parameters(), lr=0.01 )
```

```
    mll = gpytorch.mlls.ExactMarginalLogLikelihood( likelihood, model )
```

```
    for i in range(training_iter):
```

```
        optimizer.zero_grad()
```

```
        output = model(train_x)
```

```
        loss = -mll(output, train_y)
```

```
        loss.backward()
```

```
        print(f"Iter {i} - Loss: {loss.item():.3f}")
```

```
        optimizer.step()
```

```
    return model, likelihood
```

7. Prediction Function def predict(model, likelihood, test_x):

```
    model.eval() likelihood.eval()
```

```
    with torch.no_grad(), gpytorch.settings.fast_pred_var():
```

```
        preds = likelihood(model(test_x))
```

```
        mean = preds.mean lower, upper = preds.confidence_region()
```

```
    return mean, lower, upper
```

8. Example Workflow train_x, train_y = prepare_features(data)

```
model, likelihood = train_gp(train_x, train_y)
```

```
mean, lower, upper = predict( model, likelihood, train_x )
```

9. Scaling to Multi-Year Data When expanding to multiple years, you can: 1. Add richer time structure Replace the time kernel: `gpytorch.kernels.PeriodicKernel(active_dims=(2,))`

to model seasonality.

2. Use sparse GP for large datasets `gpytorch.models.ApproximateGP`

This allows 100k+ observations.

3. Add hierarchical spatial structure Examples: county region climate zone via additional kernels.

10. Optional Kernel Extensions Examples useful for spatio-temporal work: Kernel Use RBF smooth spatial variation Matern rougher spatial surfaces Periodic seasonal patterns Linear trends Spectral Mixture complex temporal signals

11. Suggested Project Structure `stgpr_model/ | |— data/ |— preprocessing.py |— kernels.py |— model.py |— train.py |— predict.py` ‘— experiments/

[OK] If you’d like, I can also generate a much more advanced ST-GPR framework including: Kronecker kernels for large spatio-temporal grids multi-task GP for multiple outputs GPU-accelerated sparse GP missing-year handling Bayesian hyperparameter priors (which is closer to what large production ST-GPR systems use).

OpenAI ChatGPT (Mar 11, 2026 version, ChatGPT-5.3)

G.0.2 Query

Can you please generate some preliminary python boilerplate code for a model that uses stochastic programming with recourse? *Output:* Below is preliminary Python boilerplate for a two-stage stochastic programming model with recourse. This template demonstrates the typical structure: First-stage decisions (made before uncertainty realized)

Scenario-dependent second-stage recourse decisions

Expected cost objective

```
# ----- # Two-Stage Stochastic Programming Boilerplate # (No imports) # -----

class TwoStageStochasticProgram:
    def __init__(self): self.scenarios = [] self.probabilities = {}
    # decision variables self.x = {} # first-stage self.y = {} # second-stage per scenario
    # parameters self.first_stage_cost = {} self.recourse_cost = {} self.demand = {}
    # ----- # Scenario setup # -----
    def add_scenario(self, name, probability, demand, recourse_costs):
        self.scenarios.append(name) self.probabilities[name] = probability self.demand[name] = demand self.recourse_cost[name]
        = recourse_costs self.y[name] = {k: 0.0 for k in recourse_costs}
    # ----- # Variable initialization # -----
    def set_first_stage_variables(self, variables, costs):
        for v in variables: self.x[v] = 0.0
        for v, c in costs.items(): self.first_stage_cost[v] = c
    # ----- # Objective function # -----
    def objective(self):
        first_stage = 0.0 for i in self.x: first_stage += self.first_stage_cost[i] * self.x[i]
        recourse = 0.0
        for s in self.scenarios:
            scenario_cost = 0.0
            for j in self.y[s]: scenario_cost += self.recourse_cost[s][j] * self.y[s][j]
            recourse += self.probabilities[s] * scenario_cost
        return first_stage + recourse
    # ----- # Constraints # -----
```

```

def demand_constraint(self, scenario):
    lhs = 0.0
    for i in self.x: lhs += self.x[i]
    for j in self.y[scenario]: lhs += self.y[scenario][j]
    return lhs >= self.demand[scenario]
# ----- # Placeholder "solver" # -----

def solve(self):
    # placeholder heuristic initialization
    for i in self.x: self.x[i] = 10.0
    for s in self.scenarios: for j in self.y[s]: self.y[s][j] = max(0.0, self.demand[s] - sum(self.x.values()))
    # ----- # Results # -----

def print_solution(self):
    print("First-stage decisions") for i in self.x: print(i, self.x[i])
    print("\nSecond-stage decisions") for s in self.scenarios: for j in self.y[s]: print(s, j, self.y[s][j])
    print("\nObjective value:", self.objective())
# ----- # Example usage # -----

model = TwoStageStochasticProgram()
model.set_first_stage_variables( ["x1", "x2"], {"x1": 5, "x2": 4} )
model.add_scenario( "s1", 0.3, demand=100, recourse_costs={"y1": 8, "y2": 6} )
model.add_scenario( "s2", 0.5, demand=120, recourse_costs={"y1": 10, "y2": 5} )
model.add_scenario( "s3", 0.2, demand=80, recourse_costs={"y1": 7, "y2": 9} )
model.solve() model.print_solution()

```

OpenAI ChatGPT (Mar 11, 2026 version, ChatGPT-5.3)

G.0.3 Query

What are some potential causes for the following error in Python 3.14? `RuntimeError: Matrix not positive definite after 100 attempts`. The most likely cause is a singular or near-singular covariance matrix. *Output:* This error typically appears in numerical routines that require a positive-definite covariance matrix (for example Cholesky decomposition in Gaussian models, Gaussian Processes, Kalman filters, multivariate normal sampling, etc.). If the algorithm retries with jitter/noise and still fails, it throws something like: `RuntimeError: Matrix not positive definite after 100 attempts` Below are the most common causes.

1. Singular or Near-Singular Covariance Matrix The most common cause. A covariance matrix becomes singular when variables are linearly dependent. Examples: Duplicate columns in your data One variable being an exact linear combination of others Perfect correlation ($\text{corr} = 1$) Example problem: $X_2 = 2 * X_1$ # perfectly correlated

This makes the covariance matrix rank deficient, so Cholesky decomposition fails.

2. Too Little Data for the Number of Features If you estimate a covariance matrix with more dimensions than samples, it becomes singular. Example: 5 samples 10 variables The covariance matrix cannot have full rank. Typical contexts: Gaussian mixture models Multivariate normal fitting Gaussian Processes

3. Numerical Precision Issues Even if the matrix is theoretically positive definite, floating-point rounding errors can produce tiny negative eigenvalues. Common when: values differ by many orders of magnitude covariance matrix is poorly conditioned very small noise levels Example symptom: min eigenvalue $\approx -1e-12$

Libraries often retry by adding jitter (small diagonal noise). If that fails repeatedly, you see this error.

4. Improper Covariance Construction Sometimes the matrix simply isn't a valid covariance matrix due to coding mistakes.

Examples: Matrix not symmetric Negative variances Manually assembled covariance with inconsistent values Example bug: `cov = np.array([ [1.0, 2.0], [0.1, 1.0] # not symmetric ])`

5. Kernel Matrix Issues (Gaussian Processes) In Gaussian Process models the kernel matrix must be positive definite. Problems arise when: Kernel hyperparameters are extreme Duplicate input points exist Very small length scales Example: `X = [[1.0], [1.0], [1.0]] # identical points`

6. Data Scaling Problems If features have very different scales, the covariance matrix can become ill-conditioned. Example: `feature1 ~ 1e-6 feature2 ~ 1e6`

This can break decomposition routines.

7. Missing Values or Bad Preprocessing NaNs or infinities in data can corrupt covariance estimates. Check for: `np.isnan(X).any() np.isinf(X).any()`

How to Diagnose 1. Check eigenvalues `import numpy as np np.linalg.eigvals(cov)`

If any ≤ 0 not positive definite.

2. Check rank `np.linalg.matrix_rank(cov)`

If $< n$ singular.

3. Check conditioning `np.linalg.cond(cov)`

Very large values ($> 1e12$) indicate instability.

Common Fixes Add jitter (regularization) `cov += 1e-6 * np.eye(cov.shape[0])`

Remove duplicate or correlated features Example: `np.corrcoef(X, rowvar=False)`

Drop columns with correlation ≈ 1 .

Standardize data from `sklearn.preprocessing import StandardScaler X = StandardScaler().fit_transform(X)`

Use shrinkage covariance from `sklearn.covariance import LedoitWolf cov = LedoitWolf().fit(X).covariance_`

[OK] Quick summary: The error almost always means the algorithm tried repeatedly to add small diagonal noise to make the matrix positive definite, but failed because the matrix is: singular extremely ill-conditioned incorrectly constructed.

If you want, you can also paste the library you're using (NumPy, PyTorch, PyMC, scikit-learn, GP libraries, etc.) and the code, and I can usually pinpoint the exact cause much faster.

OpenAI ChatGPT (Mar 11, 2026 version, ChatGPT-5.3)

G.0.4 Query

What are some potential causes for the following error in Python 3.14? `RuntimeError: The size of tensor a (64) must match the size of tensor b (3) at non-singleton dimension 1.` *Output:* This error usually occurs in PyTorch when two tensors involved in an operation (e.g., `+`, `-`, `*`, matrix multiplication, loss computation) have incompatible shapes. The key part of the message: "The size of tensor a (64) must match the size of tensor b (3) at non-singleton dimension 1." means: At dimension 1 (the second axis), one tensor has size 64 and the other 3. Because neither dimension is 1, broadcasting cannot happen, so the operation fails. Below are the most common causes.

1. Model output size vs target size mismatch A frequent case in neural networks: your model outputs 64 features, but the target has 3 classes. Example: `output.shape # [batch_size, 64] target.shape # [batch_size, 3] loss = criterion(output, target)`

Possible reasons: Wrong final layer size Incorrect number of classes Fix: `nn.Linear(hidden_size, 3)`

2. Wrong tensor shape ordering (channels vs features) Sometimes tensors are shaped incorrectly. Example: `tensor_a: [batch, 64] tensor_b: [batch, 3]`

but you intended: `tensor_a: [batch, 3] tensor_b: [batch, 64]`

Fix using `reshape/permute`: `tensor = tensor.permute(0, 2, 1) # or tensor = tensor.view(...)`

3. Broadcasting expectation failure Broadcasting only works if dimensions are equal or 1. Example that fails: `[32, 64] [32, 3]`

Example that works: [32, 64] [32, 1]

4. Incorrect batch vs feature dimension Sometimes batch size gets mixed with features. Example mistake: predictions: [64, 10] labels: [3]

This might happen when: labels were not batched indexing dropped the batch dimension

5. Data loader producing unexpected shape Your dataset may output labels like: [batch, 3]

while the model outputs: [batch, 64]

or vice versa.

6. Wrong layer type (Linear vs Conv mismatch) Example: Conv output: [batch, 64, H, W] Linear expects: [batch, features]

You forgot to flatten: `x = torch.flatten(x, 1)`

7. Loss function expecting different format Example: CrossEntropyLoss expects: input: [batch, num_classes] target: [batch]

not: target: [batch, num_classes]

[OK] How to debug quickly Print shapes right before the failing operation: `print(a.shape) print(b.shape)`

or inside your training loop: `print("output:", output.shape) print("target:", target.shape)`

[OK] Typical shape combinations Operation Expected shapes CrossEntropyLoss [batch, classes] vs [batch] MSELoss same shape Addition same shape or broadcastable Matmul [n,m] @ [m,p]

[Idea] If you want, paste the exact line of code causing the error plus the tensor shapes, and I can pinpoint the exact issue in a few seconds.

OpenAI ChatGPT (Mar 12, 2026 version, ChatGPT-5.3)

G.0.5 Query

What are some potential causes for the following error in Python 3.14? `ValueError: Model is unbounded or infeasible. No solution found for second-stage recourse decisions.` *Output:* That error usually appears in two-stage or stochastic optimization models (e.g., with recourse decisions) when the solver cannot find a feasible solution for the second stage given the first-stage decisions. In Python this often comes from modeling libraries such as Pyomo, PuLP, CVXPY, or interfaces to solvers like Gurobi Optimizer and COIN-OR CBC. Below are common causes for this specific error.

1. Missing or incorrect bounds $\{x\}$ to $\{y\}$ unbounded variables If some recourse variables have no upper/lower bounds and the objective allows them to grow indefinitely, the solver marks the model unbounded. Example pattern `model.x = Var(domain=Reals) # no bounds maximize: x`

Typical fixes Add bounds or capacity limits Check sign of coefficients in the objective Verify constraints that should limit the variable exist

2. Inconsistent constraints (infeasible model) If constraints contradict each other, the model becomes infeasible. Example: `x { }ge$ 10 x { }le$ 5`

In two-stage models this can happen when: first-stage decision forces impossible second-stage conditions demand or resource constraints conflict

3. First-stage decisions break second-stage feasibility In stochastic programming, second-stage feasibility depends on first-stage variables. Example situations: first-stage chooses capacity = 0 second stage must satisfy demand > 0 no slack or penalty variable exists Result: second stage becomes infeasible. Typical fix Add slack variables with penalties Enforce feasibility constraints in stage 1

4. Incorrect scenario data If scenario parameters contain invalid values (e.g., negative supply, impossible demand), the recourse problem may fail. Common issues: bad input files incorrect indexing mismatched scenario sets

5. Big-M constraints too small or too large Bad Big-M values can cause: infeasible constraints (M too small) unbounded relaxation (M too large)

6. Objective direction mistakes If you intended: minimize cost

but accidentally coded: maximize cost

the solver may drive variables to infinity \$\{ \}\$ to \$\infty\$ unbounded model.

7. Missing linking constraints between stages In two-stage models you typically need constraints like: `recourse_var $\{ \}$ $\le$ capacity_from_stage1`

If these are missing, the second stage may: ignore first-stage limits \$\{ \}\$ to \$\infty\$ unbounded require impossible quantities \$\{ \}\$ to \$\infty\$ infeasible

8. Solver reporting ambiguity Some solvers return "unbounded or infeasible" when they cannot distinguish between them during presolve. This happens frequently with: COIN-OR CBC GLPK

[OK] Debugging tips Solve second stage alone with fixed first-stage variables. Print solver status and termination condition. Temporarily add bounds to all variables. Enable solver logs (`tee=True` in Pyomo). Run an IIS (Irreducible Infeasible Set) if the solver supports it. Example in Pyomo: `results = solver.solve(model, tee=True)`
`print(results.solver.status)` `print(results.solver.termination_condition)`

[OK] Quick diagnostic checklist Are all variables bounded? Are demand/resource constraints consistent? Do first-stage decisions guarantee feasibility? Are Big-M values reasonable? Are all indices and scenario parameters valid?

[Idea] If you want, paste: the model formulation the second-stage constraints the solver you're using and I can help pinpoint the exact reason for this error much faster.

OpenAI ChatGPT (Mar 12, 2026 version, ChatGPT-5.3)

G.0.6 Query

What are some potential causes for the following error in Python 3.14? Kernel Restarting: The kernel for model.ipynb appears to have died. It will restart automatically. *Output:* When you see "Kernel Restarting: The kernel for model.ipynb appears to have died. It will restart automatically." in notebooks such as Jupyter Notebook, JupyterLab, or Visual Studio Code notebooks using Python 3.14, it means the Python process crashed (not just a normal exception). That usually indicates a low-level failure like memory exhaustion, native library crash, or environment corruption. Here are the most common causes:

1. Out-of-Memory (Very Common) If your code loads large datasets or trains models, the kernel may be killed when the system runs out of RAM. Typical triggers: Large pandas DataFrames Deep learning models (e.g., PyTorch or TensorFlow) Huge NumPy arrays from NumPy Infinite loops allocating memory Symptoms: Kernel dies suddenly without a Python traceback. Often happens during dataset loading or training. Fix ideas: Reduce batch sizes Load data in chunks Monitor RAM usage

2. Native Library Crash (C/C++ Extensions) Many Python libraries use compiled code. If a bug occurs there, Python crashes entirely. Examples: NumPy SciPy PyTorch OpenCV This is more likely with Python 3.14 because some libraries may not yet fully support it.

3. Incompatible Python Version Python 3.14 is very new, and many packages may still expect 3.11–3.12. Possible issues: Binary wheels not compiled for 3.14 Experimental compatibility Segfaults during import Typical symptom: Kernel dies right after importing a package. Example: `import torch`

\$\{ \}\$ to \$\infty\$ kernel instantly restarts. Fix: Use Python 3.11 or 3.12 for ML environments.

4. GPU / CUDA Problems If using GPUs with libraries like PyTorch or TensorFlow: Possible causes: CUDA version mismatch GPU memory overflow Driver incompatibility Symptoms: Crash during `.cuda()` or `.to(device)` Kernel dies during training.

5. Infinite Recursion or Stack Overflow Code like: `def f(): return f() f()`

Can crash the interpreter if recursion exceeds limits badly enough.

6. Corrupted Environment Broken installs in environments like: Anaconda Miniconda venv Symptoms: Kernel dies even on simple imports. Fix: Create a fresh environment.

7. Multiprocessing / Fork Issues Libraries using multiprocessing may crash notebooks, especially on macOS/Linux. Example: `multiprocessing.Pool()`

Quick Debug Steps Try these in order: Restart kernel and run cells one-by-one. Identify the exact cell that kills the kernel. Run the same script in terminal Python (not notebook). Check imports individually: import numpy import torch import pandas

If using Python 3.14, try Python 3.12 instead.

[OK] If you want, paste: the cell that causes the crash your installed packages (pip list) whether you're using GPU / PyTorch / TensorFlow ...and I can usually pinpoint the exact cause very quickly.

H Writing

OpenAI ChatGPT (Mar 12, 2026 version, ChatGPT-5.3)

H.0.1 Query

Please briefly explain how to make universal formatting with colors using LaTeX. *Output:* In LaTeX, "universal formatting with colors" usually means defining colors and commands once in the preamble so you can reuse them consistently across the entire document. The standard way is with the xcolor package. 1. Load the color package Add this in the preamble: `\usepackage{xcolor}`

This enables color commands and custom color definitions. 2. Define universal colors Define colors once so they can be reused everywhere. `\definecolor{mainblue}{RGB}{0,102,204}` `\definecolor{myred}{HTML}{C0392B}` `\definecolor{softgray}{gray}{0.4}`

Now these names (mainblue, myred, etc.) work anywhere in the document. 3. Use colors in text Examples: `\textcolor{mainblue}{This text is blue}` `\textcolor{myred}{Important warning}`

Or apply color to a block: `\color{mainblue}` This whole paragraph will be blue. }

4. Create reusable formatting commands To make universal formatting, define custom commands: `\newcommand{\keyword}[1]{\textcolor{myred}{#1}}` `\newcommand{\warning}[1]{\textcolor{myred}{#1}}`

Usage: This is a `\keyword{key concept}` in the paper. `\warning{This result may fail for small samples.}`

If you later change the color definition, it updates everywhere automatically. 5. Apply colors to document elements You can also globally style elements. Example with hyperref: `\usepackage[colorlinks=true, linkcolor=mainblue, citecolor=myred, urlcolor=mainblue]{hyperref}`

This makes all links consistently colored.

[OK] Key idea: Define colors once (`\definecolor`). Wrap them in reusable commands (`\newcommand`). Use those commands everywhere for consistent formatting.

If you want, I can also show: how to create a full document color theme (sections, links, tables) or a clean template used in academic papers.

OpenAI ChatGPT (Mar 13, 2026 version, ChatGPT-5.3)

H.0.2 Query

I am creating a table in LaTeX. Could you please provide the LaTeX code for a professional looking table using the booktabs package that includes a caption and properly aligned numerical data? *Output:* Below is a clean, professional LaTeX table template using the booktabs package. It includes a caption, label, and properly aligned numerical data using the siunitx package (recommended for aligning numbers by decimal point). Preamble (add to your document preamble) `\usepackage{booktabs}` `\usepackage{siunitx}`

Table Code `\begin{table}[htbp] \centering \caption{Performance Metrics for Three Models} \label{tab:model-performance} \begin{tabular}{l S[table-format=2.2] S[table-format=1.3] S[table-format=2.1]} \toprule {Model} & {Accuracy} (\%) & {F1 Score} & {Training Time (s)} \\ \midrule Model A & 91.25 & 0.873 & 12.4 \\ Model B & 93.10 & 0.891 & 15.7 \\ Model C & 89.75 & 0.842 & 10.9 \\ \bottomrule \end{tabular} \end{table}`

Why this looks professional booktabs provides `\toprule`, `\midrule`, `\bottomrule` for cleaner horizontal lines (no

vertical lines). siunitx aligns numbers by decimal point automatically. Proper caption above the table (standard in many journals). Includes a \label for referencing: \ref{tab:model-performance}. Optional improvement (often used in papers) You can slightly increase row spacing: \renewcommand{\arraystretch}{1.2}

[OK] If you'd like, I can also show: Tables that span two columns (for IEEE papers) Automatic column width tables Regression result tables (econometrics style) Multi-row / multi-column professional tables.

OpenAI ChatGPT (Mar 13, 2026 version, ChatGPT-5.3)

H.0.3 Query

(Followup to Previous Prompt) Thank you. Can you now append that code to make it a long table? *Output:* Absolutely! To convert your table into a long table that can span multiple pages, we use the longtable package. You can still combine it with booktabs and siunitx for professional formatting. Here's a ready-to-use version:

```
Preamble (add to your document preamble) \usepackage{booktabs} \usepackage{siunitx} \usepackage{longtable}

Long Table Code \begin{longtable}[1 S[table-format=2.2] S[table-format=1.3] S[table-format=2.1]] \caption{Performance
Metrics for Three Models (Long Table Example)} \label{tab:model-performance-long} \toprule {Model} &
{Accuracy (\%)} & {F1 Score} & {Training Time (s)} \midrule \endfirsthead

\multicolumn{4}{c}% {\tablename\ \thetable} – {\textit{Continued from previous page}} \toprule
{Model} & {Accuracy (\%)} & {F1 Score} & {Training Time (s)} \midrule \endhead

\midrule \multicolumn{4}{r}{\textit{Continued on next page}} \endfoot

\bottomrule \endlastfoot

Model A & 91.25 & 0.873 & 12.4 \ Model B & 93.10 & 0.891 & 15.7 \ Model C & 89.75 & 0.842 & 10.9
\ Model D & 92.50 & 0.880 & 13.2 \ Model E & 90.60 & 0.860 & 11.8 \ Model F & 94.00 & 0.895 &
16.1 \ \end{longtable}
```

Key Points longtable allows tables to span multiple pages. \endfirsthead, \endhead, \endfoot, \endlastfoot control repeated headers/footers automatically on each page. You can still use booktabs for clean horizontal lines. siunitx ensures all numbers align nicely by decimal.

If you want, I can also add alternating row colors for readability in long tables, which makes them look even more professional in reports or papers. Do you want me to do that?